Chapter 3

Measurement of surface waves

This chapter illustrates the theory and practice of surface wave data acquisition. The measurement is an experimental process involving the generation of surface waves and the observation of their effects in the medium to investigate. The final aim is the estimation of the wave propagation parameters, which are then used for subsurface characterization. In most applications, the focus is on the wave velocity and its dependence on frequency (see also Chapter 4). The attenuation of surface waves and its dependence on frequency is of interest for the estimation of the material damping (see also Chapter 5). In some applications, the polarization (i.e., the ratio between different components of the particle motion) is also analyzed.

The experimental procedures and the equipment have a primary importance in the whole process of surface wave testing. The acquired data have limitations that affect the available and usable information. Most limitations of the measured data cannot be overcome by data processing.

The general principles of seismic data acquisition will be introduced in the chapter, and then the measurement of surface waves will be presented as sampling of a multidimensional signal. In order to explain limitations and trade-off of surface wave acquisition, some basic principles of signal processing will be introduced even if they are a key element of Chapter 4, where they are presented in more detail. This sequence has been chosen to follow the actual workflow of surface wave tests: acquisition—processing—inversion. The correct planning of field activities requires an understanding of subsequent processing. In turn, the latter is influenced by limitations and constrains introduced in acquisition.

Elements of signal processing are discussed from a conceptual point of view to outline the main issues related to acquisition and processing of surface wave data. A formal treatment of signal processing is outside the scope of the present book, and the reader is referred to specific textbooks on the matter (e.g., Bendat and Piersol 2010; Santamarina and Fratta 2010).

Finally, the typical acquisition parameters and procedures will be discussed, and seismic equipment of common use will be described.

3.1 SEISMIC DATA ACQUISITION

3.1.1 Seismic data

Data acquisition is the first step of any seismic characterization method. The acquisition of seismic data is the generation and observation of the effects of the propagation of seismic waves, in time and in space. A seismic source, a set of receivers, and an acquisition system are deployed with an appropriate geometrical configuration to record the wave field. The seismic wave induces vibrations (i.e., motion of the particles from their position of equilibrium). The motion involves, for a point in the medium, a variation of stress and strain in time. Each receiver detects the effect in time (the particle motion or the associated pressure variations) of the propagating wave at a specific location. Particle motion is typically sampled in terms of velocity or acceleration, which are easier to observe than displacements.

The seismic *trace* (e.g., a time signal) is the elementary unit of seismic data. It describes the medium response to a certain source in a given position in space. Seismic traces are usually represented as wiggles or as density plots (Figure 3.1).

The identification of the different wave types in a single trace is not straightforward, even in global seismology applications where propagation paths can be very long and different events are well separated in time.

In engineering and exploration geophysics, multichannel recording is the standard practice. Recording simultaneously with a plurality of receivers at different locations allows the wave propagation to be observed in space and time. The acquired set of multiple traces, often called seismic record or seismic gather or multichannel seismogram, represents the effects of the propagating wave field at different locations. The different events, or wave types, that constitute the wave field can be identified in a record that measures the vibrations in time at different distances. In a simple seismic acquisition, surface waves and body waves with direct, refracted, and reflected paths are propagated. They appear in different regions of the seismograms with different properties. The example of Figure 3.2 shows schematically refracted, reflected, and surface waves in a simple laterally homogeneous model.

Seismic records are analyzed and processed to estimate the properties of different wave types. For instance, the velocity of events can be obtained extracting the travel time at different distances along the propagation path, although the attenuation can be estimated extracting the amplitude variations. In actual practice, it is not possible to perform a single complete acquisition in which, ideally, all wave types are recorded. The wave field is complex even in rather simple media, with multiple wave types and complex propagation paths. Optimal recording of each event has its

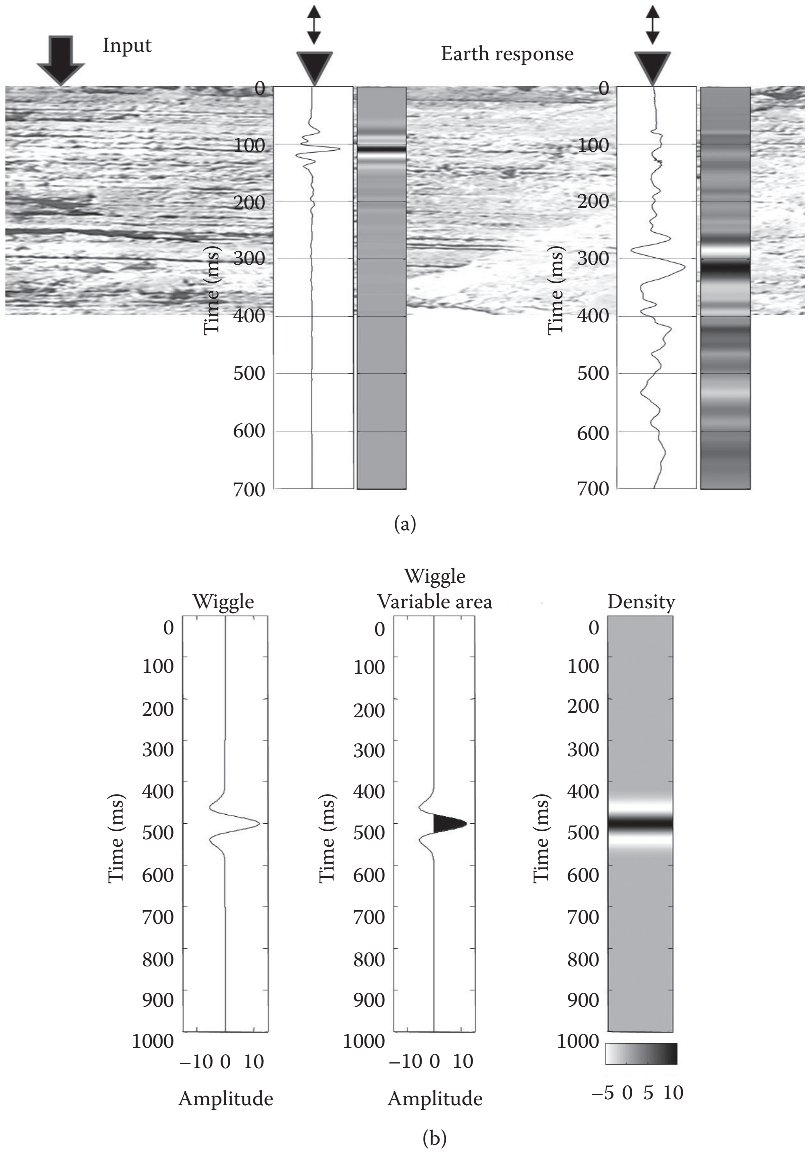

Figure 3.1 Seismic acquisition as generation of waves and observation of their effects: (a) seismic traces describe the effect of the wave at a specific location; (b) each trace can be represented as a wiggle or as a density plot.

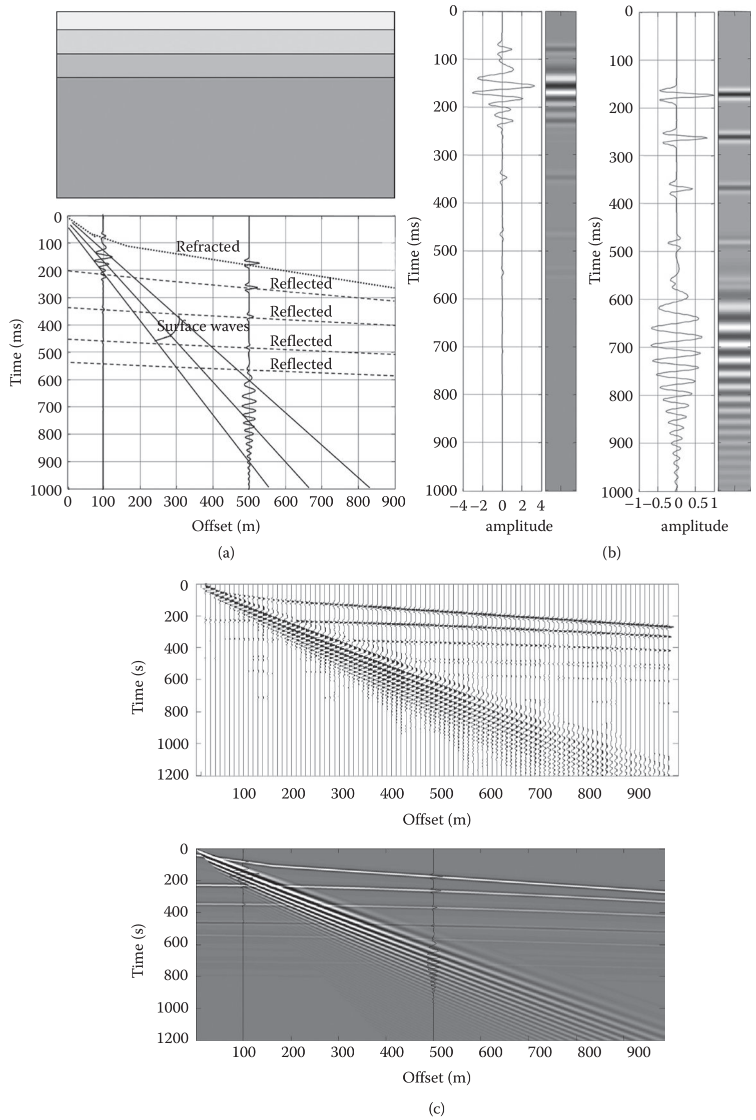

Figure 3.2 Surface seismic experiment: (a) a simple 1D model (top) and the schematic representation of the position of the different wave types in time and offset (bottom); (b) two sample traces are plotted as wiggle and density plots; (c) a wiggle multichannel seismogram (top) and a density grayscale representation of the same seismogram (bottom).

own requirements, for instance, in terms of offsets. The ideal measurement would require an infinite number of ideal receivers. In reality, a limited number of receivers with specific characteristics and limitations are available.

The seismic data acquisition requires a careful design and control, as with any laboratory experiment or field measurement. The survey design shall consider the objective of extracting the *propagation parameters* of some *wave types*, which will be used to characterize the medium under investigation, given a *processing and inversion method*, a desired *resolution*, and an estimated target *depth*, also considering an expected *level of noise*.

In seismic refraction applications, for example, the acquisition aims at extracting the travel times of refracted body waves as a function of the source–detector distance. Even if a seismic source will not generate only refracted waves, they will be identified in recorded data for their kinematic properties because they are associated with the first arrivals in the wave train. The maximum offset and the receiver spacing have to be designed according to the depth of the targets.

In surface wave methods, the acquisition scope is the observation and measurement of different propagation properties of the surface waves: the phase or group velocity, the attenuation, the amplitude, and the polarization, as a function of frequency. Therefore, measuring surface waves requires recording the effects of the surface wave propagation—reducing or controlling the presence of other wave phenomena in the acquired data. The crucial difference with respect to most other seismic acquisitions is that, despite the fact that data are acquired in time and space, the completeness and accuracy of the data depend on the property distribution in the *frequency domain*. The need for evaluating the data in the frequency domain is one of the peculiar aspects of surface wave measurement.

3.1.2 Surface wave acquisition

For most surface wave methods, data analysis is commonly performed in the frequency domain. The characterization of the subsoil is then carried out by solving an inverse problem (see Chapter 6), in which the properties of different wavelengths are mapped into the parameters of the subsurface at different depths. Generating and recording a wide wavelength range of surface waves is therefore crucial in order to collect information about different layers.

The wavelength depends on the frequency and on the velocity of the propagating wave. The latter is a function of the mechanical properties of the subsurface, which are unknown (because they are the objective of the experiment) and cannot be controlled. Therefore, the key parameter is the frequency: the lower the frequency, the longer the wavelength.

Surface waves are a low-velocity event and are often very dispersive in shallow applications. The duration of the wave train, even at relatively short offset, can be large, exceeding some seconds at less than 100 m in distance. It is not possible to generate only surface waves; therefore, due to the needed long duration of records, seismograms acquired to observe surface waves typically contain all other seismic events. Their identification and separation is not straightforward, but surface waves are, in general, high-energy dominant events in active seismic records. In shallow engineering applications, the surface wave acquisition is less challenging than refraction or reflection data acquisition.

Surface wave measurement requires the accurate estimation of the wave propagation properties in the frequency domain, over a wide frequency range, but the data are contaminated by incoherent and coherent noise. The limitations of the space and time sampling affect the accuracy and the bandwidth of the estimated properties, which are the final objectives of the survey design and of the data acquisition, in a way that is not intuitive. The surface wave measurement can be presented as a problem of multidimensional signal sampling. To explain the survey design and the data acquisition, the concepts of signals in time and space and of coherent and incoherent noise have to be introduced. The relationships between different domains (time and frequency, and space and wavenumber) and the corresponding transforms are indeed essential to understand the limitations of the data acquisition.

3.2 THE WAVE FIELD AS A SIGNAL IN TIME AND SPACE

The propagation of the wave field in a continuous medium induces vibrations, which are motion of the particles from their position of equilibrium. The wave field can be completely described with a continuous 4D function of space and time representing, for example, the vector particle velocity. Considering the plane surface waves at the free surface in a 1D medium, the wave field is a continuous vector function of distance x and time t. If we consider a single component of the particle displacement, it is a continuous scalar function of offset and time, $s(x,t)$. In Figure 3.3, a snapshot of a vertical section, strained by a propagating plane Rayleigh wave at constant frequency, is shown at a fixed time instant. The bottom continuous image depicts, in grayscale, the vertical component of the particle velocity at the surface, as a function of offset and time. A slice at constant time represents the particle velocity versus offset at a given time, while a section at constant offset represents the particle velocity of a specific point (such as point A) in time.

We define a *signal* as a physical quantity, measurable over time and space; in this case, as in communication theory, the signal is also a dataset carrying information. The ideal acquisition experiment should generate a

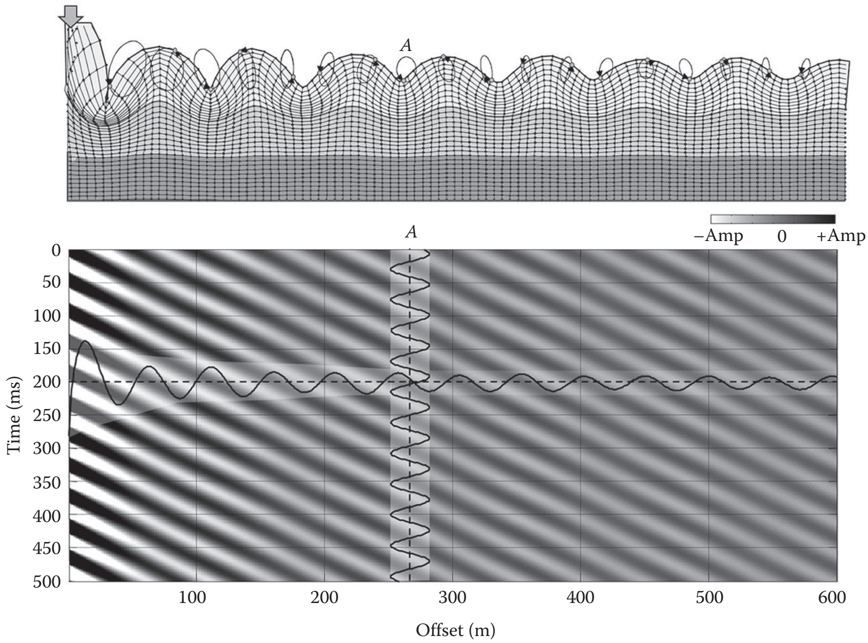

Figure 3.3 The wave field is a continuous function of space and time. Considering a single component of the particle motion of the points along a straight line at the free surface, the wave field is a 2D continuous function of *x* and *t*. A seismic trace is the value at constant *x* (for instance, at point A).

wave field with only surface waves in a noise-free environment, over the required wide frequency band. The ideal signal, that is, the effect of the wave propagation, should be observed continuously and infinitely, and its true value should be recorded without any distortion induced by the acquisition system and equipment.

In reality, this is not possible; the ideal wave field cannot be generated and the signal cannot be ideally recorded. A seismic source will generate a band-limited wave field, with different wave types superimposed on each other in different domains. The presence of lateral variations, of near-field effects, and so on, is not an acquisition limitation, strictly speaking. Yet, it affects the nature of the propagating wave field and has to be considered in the acquisition design. Other nonsource-generated events will superimpose onto the vibration field, constituting incoherent or pseudorandom noise. It is not possible to generate and observe only the desired signal, rather a composition of signal $\overline{s}$ and noise n

$$w(x,t) = \overline{s}(x,t) + n(x,t) \tag{3.1}$$

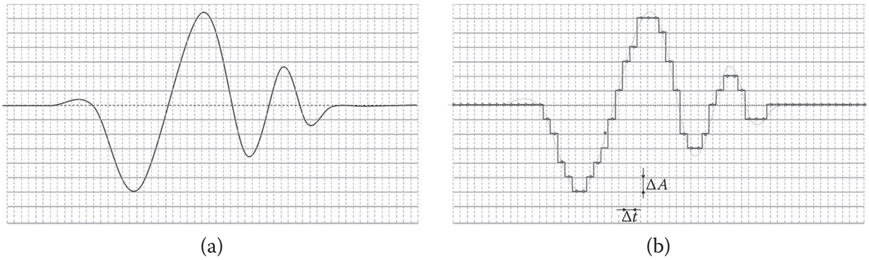

Figure 3.4 Schematic representations of (a) a continuous signal and its digitized version (b) the sampling and quantization transform a continuous real variable into a discrete array with a limited set of possible values.

The wave field is a physical quantity, for instance, the particle velocity, continuous in time, in the continuous medium to be investigated. The measurement of the wave field at a specific location involves the transformation of the physical quantity into an analog electric signal using a transducer, its transmission, conditioning, digitization, and recording. This "measurement chain" transforms the ground motion into a seismic trace. A seismic trace is a discrete signal (i.e., a limited representation of the original signal). Moreover, it is distorted by the imperfections and limitations of the measuring system and is affected by other types of noise.

An important aspect of the digitization is that the physical quantity is not recorded continuously in time and amplitude; sampling and quantization are schematically represented in Figure 3.4. A continuous vibration signal, defined at any time instant as a real-valued number, is transformed into a discrete series of points that can assume a limited discrete set of amplitude values.

The fidelity of the digital data is the similarity of the recorded version of the signal to the true physical quantity. It depends mainly on the density of sampling (i.e., the time interval between samples) and on the resolution of the analog-to-digital conversion, as will be discussed in Section 3.3.6.

In seismic data acquisition, the sampling of the signal *in space* is even more important than the time sampling. The effect of the wave propagation is indeed detected by a limited number of receivers. The wave field w is recorded on a discrete, finite set of points, in time and space. The effects of the sampling in time and space are added onto the physical noise (other events, coherent and incoherent).

Figure 3.5 represents the difference between the ideal experiment and the reality. Ideally, surface waves should be measured continuously in time and space. In reality, the gathered data are limited by the time and space sampling, are affected by the instrument, and are contaminated by noise:

$$\text{data} = \text{sampling}\,(\text{instrument}\,(\overline{s} + n)) \tag{3.2}$$

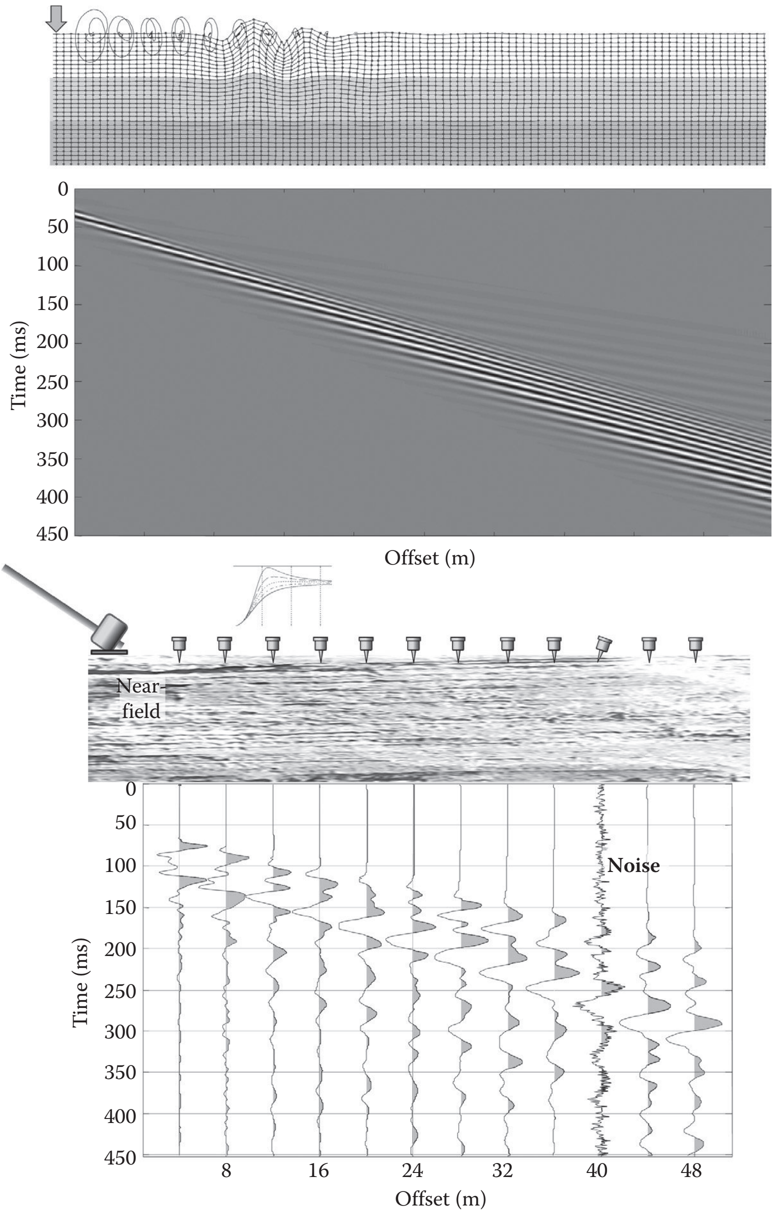

Figure 3.5 An ideal experiment with a continuous wave field consisting only of surface waves and a real acquisition with discrete sampling, different types of noise, and the equipment effects.

3.3 ACQUISITION OF DIGITAL SEISMIC SIGNALS

Before discussing the effects and limitations of sampling and noise, the formal relationships of signal analysis have to be introduced.

3.3.1 Spectral analysis and wave field transforms

The concept of *frequency* is intuitive for cyclic processes; in general, frequency indicates the number of occurrences of an event per unit of time. For a harmonic signal, the frequency is the reciprocal of the time duration of a cycle, called a *period*. Noncyclic signals can be decomposed into the sum of cyclic functions. In Figure 3.6, a signal (e.g., a seismic trace) is represented

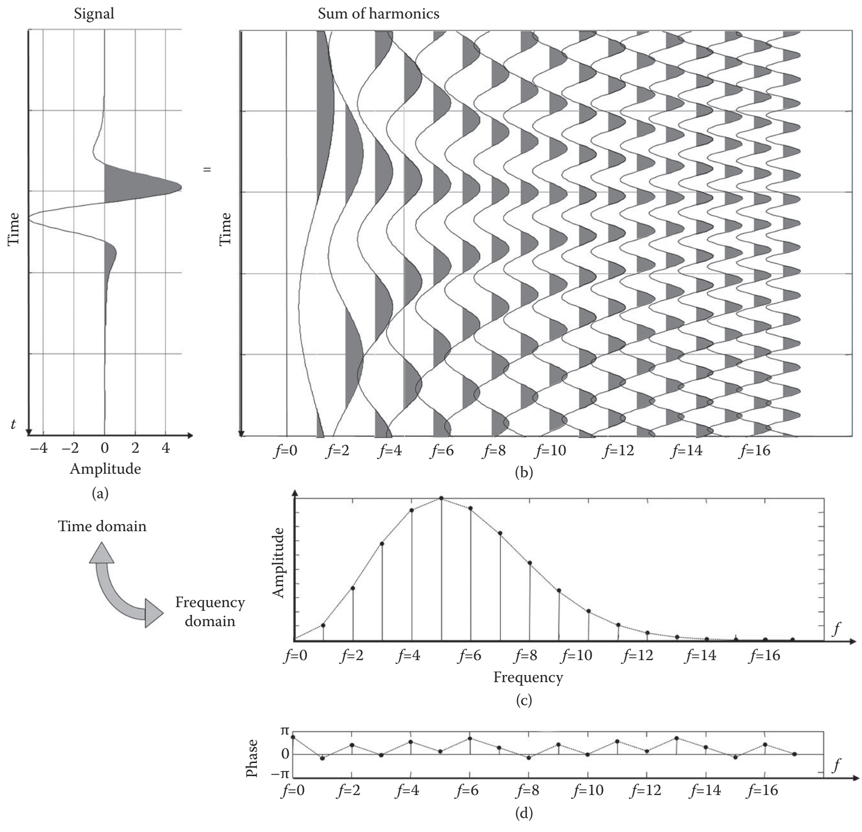

Figure 3.6 (a) A generic nonperiodic signal can be decomposed in (b) the sum of simple cyclic functions. The amplitude and phase of the elementary cyclic signal are the frequency-domain representation of the signal, or its spectrum, consisting of the (c) amplitude and (d) phase.

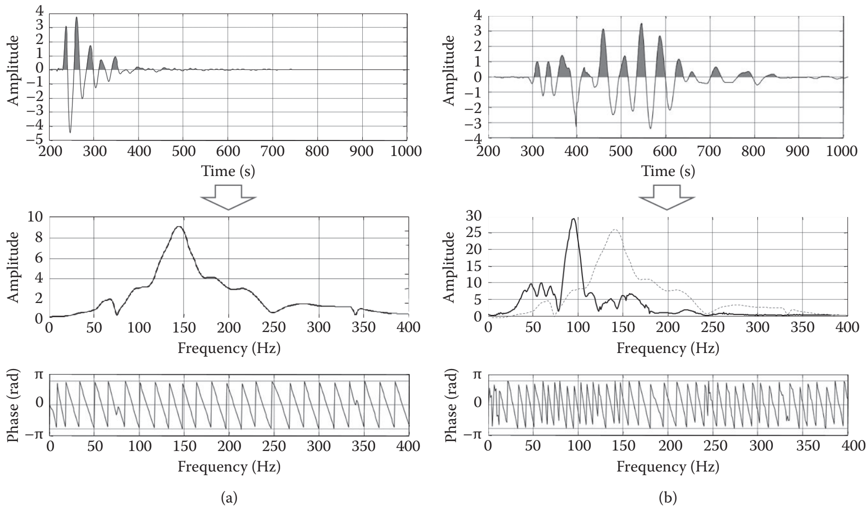

Figure 3.7 Example of spectral analysis of two seismic signals with different frequency content. The amplitude spectrum of the signal (a) is overimposed on the amplitude of the spectrum of the signal (b).

in time domain in Figure 3.6a. It can be exactly obtained by summing the cyclic signals plotted in Figure 3.6b. Each cyclic signal has a different frequency, amplitude, and phase. Each frequency is a multiple of the minimum frequency corresponding to the inverse of the total duration of the signal. The series of amplitudes and phases of these elementary signals (Figure 3.6c and d) represent the trace in frequency domain, also known as the spectrum of the trace.

Often the word *spectrum* is used to indicate the amplitude spectrum of a signal, and the process of decomposing a signal in its frequency components is called *spectral analysis*. The spectral analysis, in practice, consists of finding the amplitude and phase of the elementary cyclic signals. Two real seismic traces are analyzed in Figure 3.7. Their amplitude spectra show different frequency content. The signal on the left is dominated by higher frequency components than the signal on the right.

The mathematical operators used to decompose a signal in its cyclic components are called *transforms*.

3.3.2 Fourier series and Fourier transform

Integral transform are used to transform a function of one variable (e.g., time) into a related function of a different variable (e.g., frequency). The two variables specify two *domains*. The original signal depends on time, and it is addressed as the *time-domain* representation of the function. Its transform depends on frequency and is the *frequency-domain* representation of the function. In theory, the transform is perfectly invertible, and no information is lost in transforming. It is possible to define an inverse operator (i.e., the inverse transform), which takes back the transformed signal to the original signal.

The Fourier transform derives from the Fourier series, which is the decomposition of an arbitrary periodic signal into a sum of harmonics. It can be shown that an infinite, periodic signal of period T can be decomposed into the sum of an infinite number of harmonic (sine or cosine) functions with frequency $f_n = n/T$, with an amplitude A_n and a phase φ_n.

The Fourier transform is the extension to nonperiodic signals, where the period is infinite. The discrete spectrum then becomes a continuous function.

The Fourier transform can be written as

$$G(f) = \int_{-\infty}^{+\infty} g(t) e^{-j2\pi ft} dt \tag{3.3}$$

and it allows going from the time-domain signal g to the frequency-domain signal G. The original independent variable t is eliminated via integration over the infinite interval. The inverse transform operates on the frequency-domain signal G and allows going back, from frequency domain to time domain. The inverse transform can be written as

$$g(t) = \int_{-\infty}^{+\infty} G(f) e^{j2\pi ft} df \tag{3.4}$$

The two representations of the signal are g and G, and the relationship is written as

$$g(t) \xleftrightarrow{F} G(f) \tag{3.5}$$

The original time signal and its frequency spectrum constitute a *Fourier transform pair*. The two sides of the Fourier transform pair are complementary views of the same signal. It is possible to go from one domain to the other without any loss of information.

The same concept applies to signals in different original domains. For instance, the signals for which the independent variable is *space* are transformed into the *wavenumber* domain. The wavenumber k indicates the cycles per unit distance and is the reciprocal of the wavelength. The Fourier pair in this case can be indicated as

$$g(x) \xleftrightarrow{F} G(k) \tag{3.6}$$

In some applications, it is common to use the circular wavenumber, representing the spatial frequency in radians per meter: $\kappa = 2\pi k$. The relationship between wavenumber and circular wavenumber is equivalent to the relationship between frequency f and circular frequency, $\omega = 2\pi \cdot f$.

3.3.2.1 Properties of the Fourier transform

Some basic properties of the Fourier transform are of paramount importance for seismic signal processing. In particular, the following are recalled in this section: linearity, scaling, shifting, and the convolution theorem. The properties are hereafter described for time signals, but they are valid for any Fourier pair. For example, for signals in space, the wavenumber would substitute for the frequency in the following sentences.

The linearity of the transform implies that scaling a function scales its transform pair

$$g(t) \xleftrightarrow{F} G(f) \quad \text{then} \quad ag(t) \xleftrightarrow{F} aG(f) \tag{3.7}$$

and adding two functions corresponds to adding the two spectra

$$\begin{matrix} g(t) \xleftrightarrow{F} G(f) \\ h(t) \xleftrightarrow{F} H(f) \end{matrix} \quad \text{then} \quad g(t) + h(f) \xleftrightarrow{F} G(f) + H(f) \tag{3.8}$$

The scaling property states that a multiplication of the scale factor s to the independent variable (time) changes inversely the frequency axis of the spectrum of the signal. If a signal is shrunk in time, its spectrum is stretched in frequency (and vice versa)

$$g(t) \xleftrightarrow{F} G(f) \quad \text{then} \quad g(t/s) \xleftrightarrow{F} sG(s \cdot f) \tag{3.9}$$

Another relevant property is related to time shift, which induces a phase shift proportional to the time shift and to the frequency; that is, the same time shift corresponds to more cycles for a high-frequency (wavenumber) than for a low-frequency (wavenumber) function

$$g(t) \xleftrightarrow{F} G(f) \quad \text{then} \quad g(t - t_0) \xleftrightarrow{F} e^{-j\omega t_0} G(f) \tag{3.10}$$

Many physical phenomena can be described by their effects on signals. One of the operators that combine two signals to produce a new signal is the convolution, formally denoted by an asterisk. By definition, the convolution between two signals h and g is

$$p = h * g \quad \text{then } p(t) = \int_{-\infty}^{\infty} g(u) h(t - u) du \tag{3.11}$$

The convolution theorem states that the convolution between two signals in time domain is equivalent to the multiplication of their spectra in the frequency domain, and vice versa

$$\begin{array}{lcl} g(t) \xleftrightarrow{F} G(f) & & g(t) * h(f) \xleftrightarrow{F} G(f) \cdot H(f) \\ & \text{then} & \\ h(t) \xleftrightarrow{F} H(f) & & g(t) \cdot h(f) \xleftrightarrow{F} G(f) * H(f) \end{array} \tag{3.12}$$

The aforementioned properties of the Fourier transform are used in the following sections to illustrate the limitations of the sampling and of the acquisition.

3.3.3 Sampling

The data acquisition involves an analog-to-digital conversion, in which the continuous signal is replaced by a discrete series of values at fixed time intervals. The reciprocal of the sampling interval Δt is called sampling frequency F_S. Figure 3.8 illustrates the sampling of a signal. The sampling process implies loss of information.

The process of sampling can be described as a multiplication of the signal y by a sampling *comb* function. The *comb* function consists of an infinite set of regularly spaced impulses, or Dirac's deltas. The spectrum of the sampled function is therefore equal to the spectrum of comb$\cdot y$, which is the convolution of the two spectra, the one of the signal with the one of the comb function

$$F(\text{comb} \cdot y) = F(\text{comb}) * F(y) \tag{3.13}$$

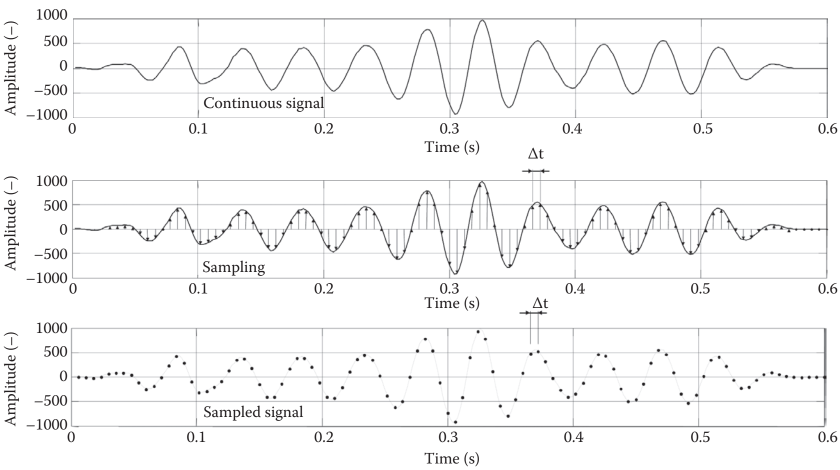

Figure 3.8 Sampling of a continuous signal involves reading the values of the signal at a finite discrete set of points. The sampled signal is defined only at a series of (usually evenly spaced) time instants.

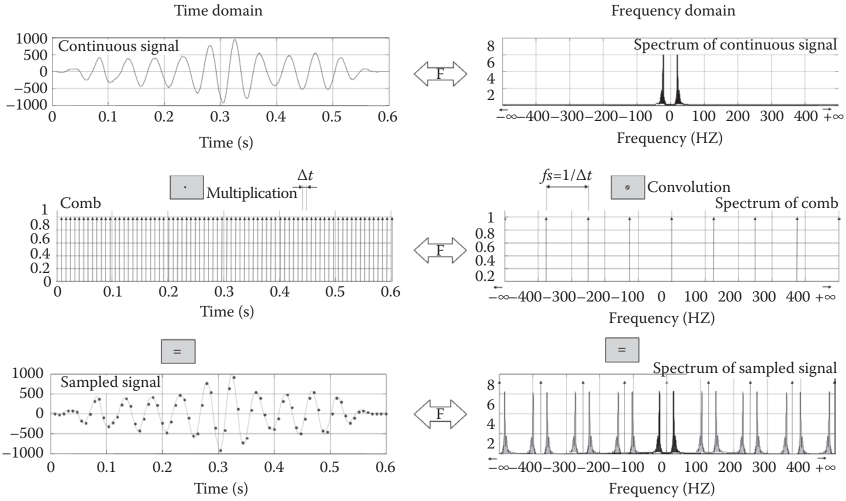

Figure 3.9 Sampling of a continuous function, in time domain (left) and in frequency domain (right); the spectrum of the sampled signal is a repetition of the original spectrum, with a periodicity equal to the inverse of the sampling.

The time-and frequency-domain representations of the sampling are illustrated in Figure 3.9. The top row shows the continuous signal with its symmetric (even) amplitude spectrum. The middle row shows the sampling function, with impulses spaced by Δt, and its amplitude spectrum, with impulses spaced by $f_s = 1/\Delta t$. The comb function is multiplied by the continuous signal, and its spectrum is convolved with the spectrum of the continuous signal. The results are shown in the third row, where the sampled signal is represented on the left and its spectrum is depicted on the right. An infinite set of copies of the original spectrum is obtained, and the spacing between these replicas is equal to the sampling frequency, f_s.

These replicas are separated (i.e., not overlapping each other) when the maximum frequency $f_{\max}$ in the signal is lower than half of the sampling frequency, as in Figure 3.10a (top).

Given a certain sampling frequency, f_S, if the maximum frequency exceeds the limit of $0.5f_S$, the replicas of the spectrum overlap and their values are summed up (as in Figure 3.10b (bottom)).

The Nyquist frequency F_{Nyq} is the maximum observable frequency in sampled data. It is equal to half the sampling frequency.

From an acquisition planning perspective, given a maximum frequency in the signal $f_{\max}$, it is possible to evaluate the sampling frequency needed to sample without loss of information as

$$F_S > 2 \cdot f_{\max} \tag{3.14}$$

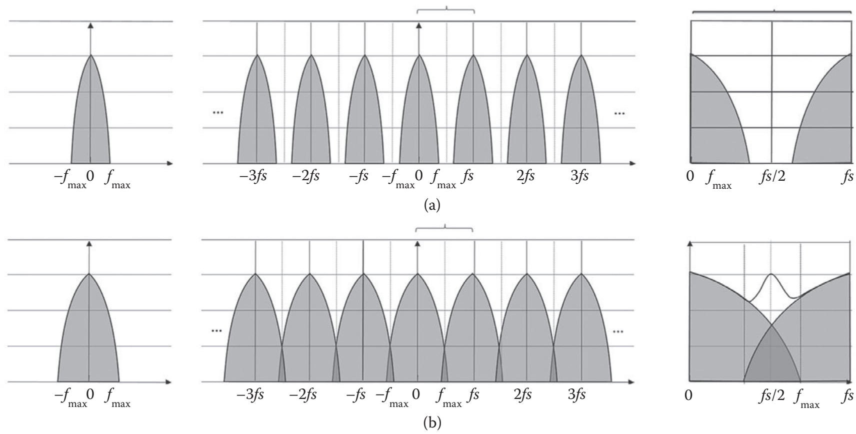

Figure 3.10 The copies of the spectrum do not overlap when the sampling frequency is higher than $2 \cdot f_{max}$.

This criterion is known as the Nyquist–Shannon theorem; no information is lost by regular sampling provided that the sampling frequency is greater than twice the highest frequency in the sampled signal.

3.3.4 Interpolation and aliasing

When the sampling frequency is sufficient, it is possible to recover the spectrum of the continuous signal by multiplying the transform of the sampled function by a boxcar, that is, a box window equal to 1 between $-f_S/2$ and $f_S/2$ and zero outside. This is equivalent to the convolution in time domain with the boxcar spectrum, which is a sinc function. Therefore, the complete sinc function can be used to interpolate a sampled function without any information loss, when the sampling criterion is verified.

If the spectrum has frequencies higher than half of the sampling frequency, the actual spectrum is summed up to its replicas. In this case, the original signal cannot be recovered. Any frequency component above $f_S/2$ is represented as a fictitious component at a lower frequency. This error is known as aliasing. In Figure 3.11, examples of sufficient and insufficient samplings are provided.

Aliasing makes different signals indistinguishable when sampled and creates distortions and artifacts in the reconstruction. A well-known example of distortion is the apparent reverse spinning direction of wheels in movies. When the time sampling is not sufficient to sample the forward moving frequency, related to the distance between the spikes, an apparent negative frequency appears.

To avoid aliasing, the sampling rate has to be increased to satisfy the requirements of the sampling theorem. Alternatively, antialias filters can

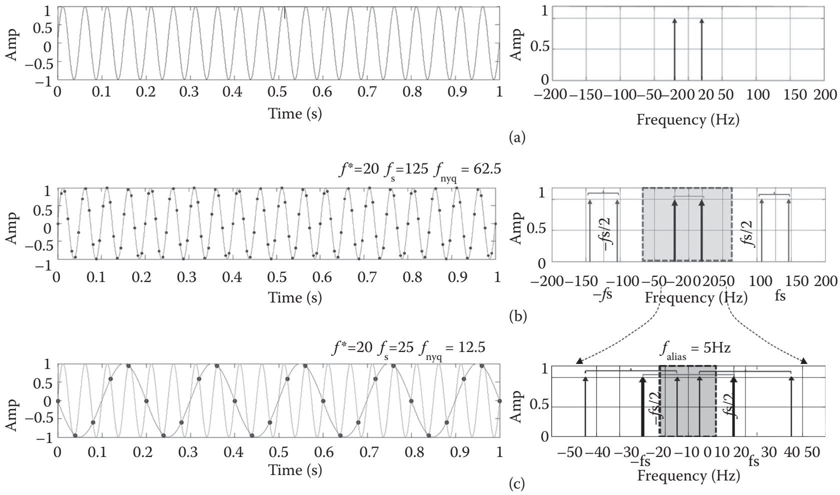

Figure 3.11 Example of aliasing— (a) a continuous signal with the frequency of 20 Hz is plotted with its spectrum. (b) When sampled at 125 Hz with a Nyquist frequency of 62.5 Hz, the first copies of the spectrum appear centered at +125 Hz and −125 Hz, and they fall outside of the range (−62.5 to +62.5), allowing a correct reconstruction of the signal. (c) The signal is sampled at 25 Hz, hence with a usable range (−12.5 to +12.5 Hz). The aliased copies at 5 Hz (25 – 20 Hz) fall in the range, and the signal appears as aliased.

be implemented to remove all components above the highest frequency of interest for the specific application from the signal. These filters restrict the bandwidth of the signal to satisfy the condition of proper sampling by removing frequency components above the Nyquist frequency. The most effective option is represented by electronic analog filters that remove the high-frequency components before the digitization of the signals (hardware filter). Alternatively, the filter can be implemented through digital filtering after oversampling of the signal (software filter). The antialias protection for spatial sampling can be done via spatial oversampling followed by digital filtering or by using arrays of analog receivers that are summed into a single trace.

3.3.5 Windowing

The signals described in the previous sections are still considered indefinite in length, although any real data are limited to a finite time interval. The beginning and end of the acquisition can be described as windowing, or multiplication of an infinite signal by a finite time window. A window function, or tapering function, is a mathematical function that is zero-valued outside a chosen interval. A rectangular window, or boxcar window, is constant inside the interval and zero elsewhere. When the signal *s* is multiplied by the window, the product is zero-valued outside the interval.

We can represent the windowed signal $z(t)$ as follows

$$z(t) = w(t)s(t) \tag{3.15}$$

where $s(t)$ represents the original signal and the function $w(t)$ represents the window function, for example, for the boxcar window

$$w(t) = \begin{cases} 1 - \dfrac{T}{2} < t < \dfrac{T}{2} \\ 0 \quad \text{elsewhere} \end{cases} \tag{3.16}$$

Any window applied to a signal affects its spectrum because of the truncation spectral leakage. For example, windowing of a harmonic signal causes its transform to have nonzero values at frequencies other than the actual frequency of the signal (Figure 3.12). The result of the windowing is a

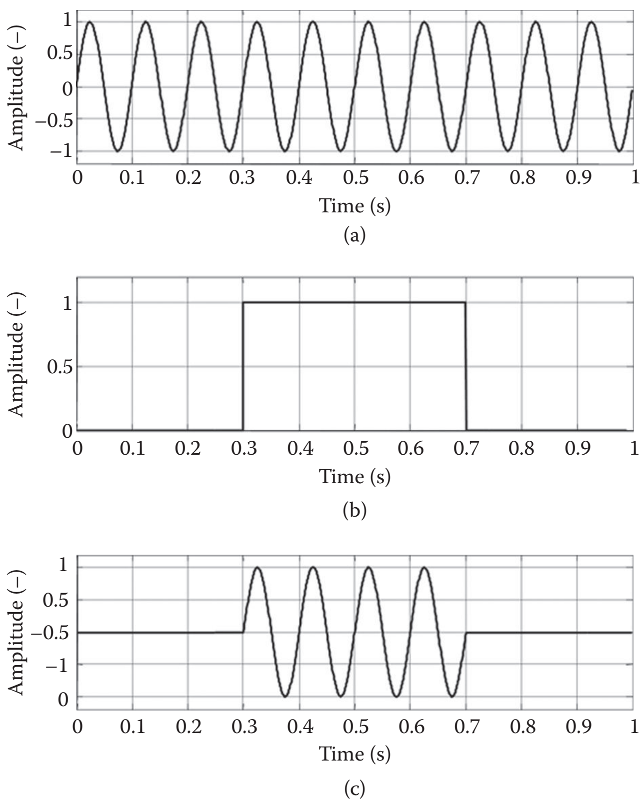

Figure 3.12 Windowing and spectral leakage: (a) harmonic function; (b) boxcar window; (c) windowed signal.

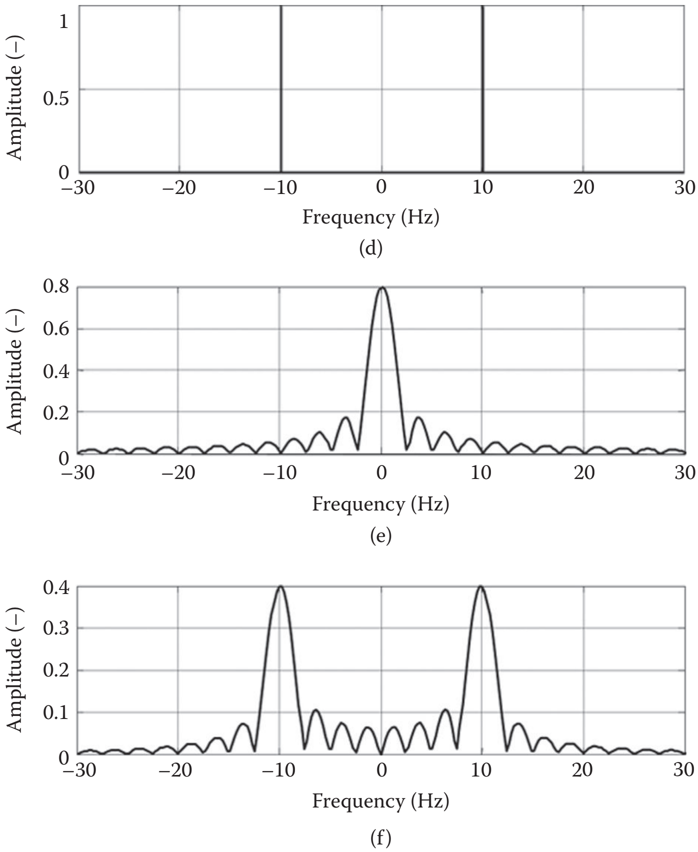

Figure 3.12 (Continued) Windowing and spectral leakage: (d) spectrum of the harmonic function; (e) spectrum of the boxcar window; (f) spectrum of the windowed signal.

truncated signal, zero-valued outside of the window length, and a spectrum replicating the window spectrum. The amplitude of the actual frequency component is altered, and side lobes associated with the spectrum of the box window appear. Windowing a signal corresponds to the multiplication of the signal (a) by the chosen window (c) and is equivalent to the convolution, that is, to the multiplication of the signal spectrum (b) with the window spectrum (d). The spectral leakage spreads the energy out of its real position and affects the resolution of the spectrum (f).

Leakage reduces the spectral resolution and the ability to distinguish multiple events. Figure 3.13 illustrates two examples of ambiguities in the spectral representation of finite signals.

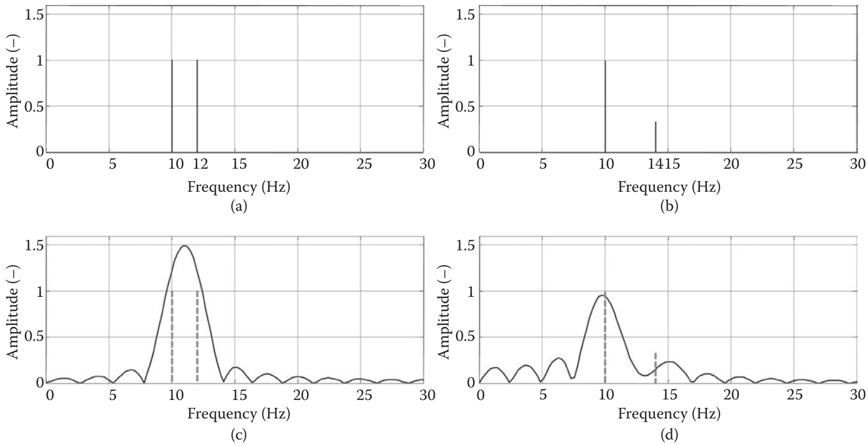

Figure 3.13 Windowing and spectral resolution. The limited acquisition introduces spectral leakage and different events cannot be resolved: (a, b) the actual spectra of two signals composed by the superposition of two harmonic functions; (c, d) the corresponding spectra after convolution with a boxcar window.

Two events, such as two harmonic components of a signal, with similar amplitude and a small frequency difference cannot be properly identified in the spectrum (they cannot be *resolved*) when the main lobes interfere (Figure 3.13a). A stronger event can mask a weaker one with the leakage of its side lobes, also known as ripples (Figure 3.13b).

The jump between 0 and 1 in the values of the boxcar window produces high side lobes. Window functions with a smooth transition between 0 and 1 (e.g., Hanning and Hamming windows) increase the dynamic range at the price of a loss of resolution (Santamarina and Fratta 2010).

The concept of spectral resolution and its dependence on the window size is extremely important for surface wave acquisition because it affects the possibility of separating multiple modes of propagation. For a signal in space, the window is the length of the observation (i.e., the aperture of the array), and this is a key acquisition parameter. The effect of multidimensional windowing is further discussed later in this chapter.

3.3.6 Quantization and analog-to-digital conversion

As mentioned before, the acquisition of digital data implies analog-to-digital conversion, which not only requires sampling but also implies quantization. A real physical signal consists of a variable that can assume, between its maximum and minimum, any real value (i.e., an infinite set of values). Digital data have a finite number of discrete values. The true value has to be rounded, or truncated, to be represented. Usually, an electronic device converts the input analog signal into the digital output, which is coded as a binary number, proportional to the magnitude of the input.

The resolution of the conversion indicates the number of discrete values that can be produced within the input range. Because the output value is usually in binary form, the resolution is expressed in bits and the number of divisions is measured in a power of two. A 4-bit conversion implies 2^4 levels, corresponding to 16 possible levels in the measurement scale, for example, from −8 to +7 if a signed integer coding is used. Typical values of resolution of analog–digital converters (ADCs) in modern seismic acquisition systems are between 16 and 24 bits. The resolution can also be measured in physical units, as the minimum change of input that creates a change in the output level. Usually, input signals are measured in electrical units (because the particle motion is converted into an electric voltage by the transducer) and the resolution can be expressed in volts. The least significant bit (LSB) voltage is the minimum change in input tension corresponding to a change of output. The LSB voltage is equal to the full range divided by the number of divisions.

The quantization error or quantization distortion is the difference, due to the truncation or rounding, between the actual input analog signal and the output digital signal. In a typical case, with a signal of much higher amplitude than the LSB, the quantization noise is not correlated with the signal. In the case of rounding, the distribution is uniform and is between − LSB/2 and + LSB/2, giving a zero mean and a standard deviation of $\frac{LSB}{\sqrt{12}}$, which is the RMS level of this noise. If the signal is uniformly distributed in the full range, then the quantization noise has a ratio equal to the resolution (i.e., 2^Q) corresponding to a quantization noise of $20\log_{10}(2^Q)$. In Figure 3.14, a 4-bit conversion is shown with the associated quantization noise.

The resolution of the conversion affects the accuracy and data integrity. The resolution is called dynamic range, and typical values are from 16 to 24 bits. Some details are given at the end of this chapter when describing data acquisition systems and the AD converter.

3.3.7 Acquisition of 2D signals

Seismic data are multidimensional functions of space and time. For a linear array of receivers, seismic data are two-dimensional (2D) signals (i.e., function of time and distance from the source). It is therefore important to extend the formulation to multidimensional transforms and spectral analysis. Some examples are reported in the following to discuss the limitations of the acquired data.

The two transforms (from time to frequency and from space to wavenumber) can be applied simultaneously to the seismic dataset. The 2D Fourier transform can be written as

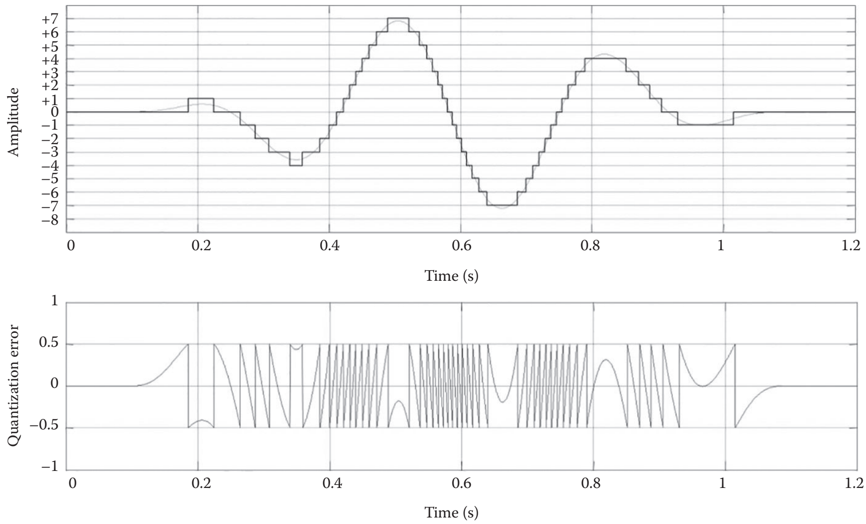

Figure 3.14 Example of signal digital conversion and quantization error. The difference between the real signal and its digitized version represents the inaccuracy introduced by the process. When the signal has large amplitude compared to the unit-amplitude interval, the quantization error is uncorrelated.

$$G(f,k) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(t,x) e^{-j2\pi(kx+ft)}\, dx\, dt \tag{3.17}$$

The original space and time variables are eliminated with a double integration to obtain the representation of the signal as a function of the two transformed variables, frequency and wavenumber. The two paired domains are the time–space domain (*t*–*x*) and the frequency–wavenumber domain (*f*–*k*).

The corresponding inverse transform can be written as

$$g(t,x) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} G(f,k) e^{j2\pi(kx+ft)}\, dk\, df \tag{3.18}$$

The *f*–*k* transform is widely used in seismic data processing, and it is part of the operation called wave field transforms.

The sampling and aliasing issues discussed in the previous sections apply to the 2D domain. In most applications, a regularly spaced array of receivers is deployed to sample the wave field in a regular way in time and offset. In some applications, the spacing between receivers may be not regular, introducing extra complexity in the transforms.

3.3.7.1 Effects of finite sampling

It is useful to discuss the consequences of the windowing in 2D to extend the concept of spectral resolution that was introduced earlier. Considering a unit-amplitude, harmonic plane wave propagating in the x direction with circular wavenumber k_0 and circular frequency ω_0 is

$$s(x,t) = e^{i(\omega_0 t - k_0 x)} \tag{3.19}$$

The frequency–wavenumber spectrum of the signal is

$$S(k,\omega) = 4\pi^2\,\delta(k-k_0)\,\delta(\omega-\omega_0) \tag{3.20}$$

Figure 3.15 shows a graphical representation of this frequency–wavenumber spectrum. Both the circular wavenumber k_0 and circular frequenycy ω_0 of the signal are resolved exactly.

Over space and time, the infinite integrals in Equation 3.17 imply perfect sampling of the signal in both domains. In practice, of course, this is not possible, and the signal is sampled at a limited number of positions and times. Let us initially consider the effect of limited sampling in the time domain. Assume that the signal is sampled with a boxcar window of duration T.

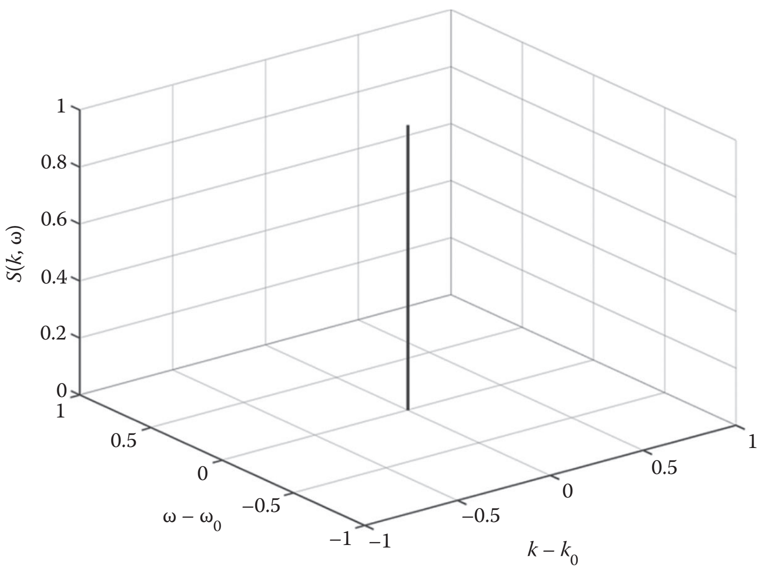

Figure 3.15 Frequency–wavenumber spectrum of a harmonic plane wave.

The frequency–wavenumber spectrum of the sampled signal is

$$\begin{aligned} Z(k,\omega) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} z(x,t)e^{-i(\omega t-kx)}dt\ dx \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} u(t)s(x,t)e^{-i(\omega t-kx)}dt\ dx \\ &= \int_{-\infty}^{\infty}\int_{-\frac{T}{2}}^{\frac{T}{2}} s(x,t)e^{-i(\omega t-kx)}dt\ dx \end{aligned} \tag{3.21}$$

To illustrate the effect of limited temporal sampling, consider the example of a unit-amplitude, harmonic plane wave once again

$$\begin{aligned} Z(k,\omega) &= \int_{-\infty}^{\infty}\int_{-\frac{T}{2}}^{\frac{T}{2}} e^{i(\omega_0 - k_0 x)}\ e^{-i(\omega_t - kx)}\ dt\ dx \\ &= 2\pi|T|\delta(k-k_0)\operatorname{sinc}\left[\frac{T}{2}(\omega-\omega_0)\right] \end{aligned} \tag{3.22}$$

When compared to Equation 3.19, in terms of circular frequency, the Dirac delta function has been replaced by a sinc function that depends on the duration of the sampled signal T. The frequency–wavenumber spectrum is shown in Figure 3.16.

Limited sampling decreases the resolution of the frequency of the signal as indicated by the width of the main lobe centered at $\omega = \omega_0$ and causes energy to "leak" into adjacent side lobes. The frequency resolution defined according to the Rayleigh criterion is equal to one-half of the width of the main lobe, which is $\omega_{\text{Rayleigh}} = 2\pi/T$ for the sampling window $w(t)$. Increasing the sampling duration T improves the frequency resolution. Alternative sampling windows to control side-lobe leakage and frequency resolution are discussed, for instance, in Bendat and Piersol (2010).

Limited sampling in the spatial domain produces similar results. Assume that the signal is sampled within an *aperture* of length D. The sampled signal may now be represented as follows

$$z(x,t) = u(t)v(x)s(x,t) \tag{3.23}$$

where $u(t)$ is defined as before and $v(x)$ is defined as

$$v(x) = \begin{cases} 1 - \dfrac{D}{2} \le x \le \dfrac{D}{2} \\ 0 \quad \text{elsewhere} \end{cases} \tag{3.24}$$

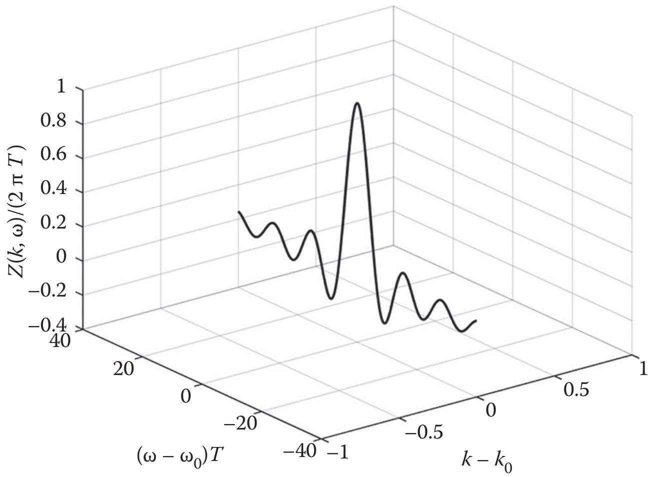

Figure 3.16 Frequency–wavenumber spectrum of a harmonic plane wave with finite temporal sampling.

The frequency–wavenumber spectrum of the sampled signal may be calculated as follows

$$
\begin{aligned}
Z(k,\omega) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} z(x,t)e^{-i(\omega t-kx)}dt\,dx \\
&= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} u(t)v(x)s(x,t)e^{-i(\omega t-kx)}dt\,dx \\
&= \int_{-\frac{D}{2}}^{\frac{D}{2}}\int_{-\frac{T}{2}}^{\frac{T}{2}} s(x,t)e^{-i(\omega t-kx)}dt\,dx
\end{aligned}
\tag{3.25}
$$

Considering once again the example of a unit-amplitude, harmonic plane wave

$$
\begin{aligned}
Z(k,\omega) &= \int_{\frac{-D}{2}}^{\frac{D}{2}}\int_{\frac{-T}{2}}^{\frac{T}{2}} e^{i(\omega_0-k_0x)}e^{-i(\omega t-kx)}dt\,dx \\
&= |D||T|\operatorname{sinc}\left[\frac{D}{2}(k-k_0)\right]\operatorname{sinc}\left[\frac{T}{2}(\omega-\omega_0)\right]
\end{aligned}
\tag{3.26}
$$

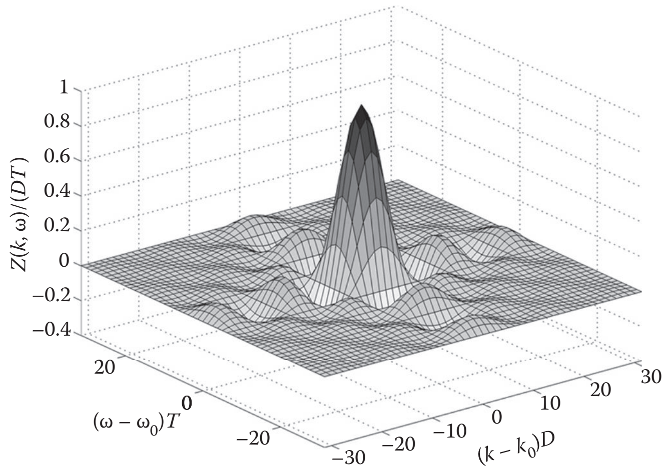

Figure 3.17 Frequency–wavenumber spectrum of a harmonic plane wave with finite temporal and spatial sampling: surface plot.

The frequency–wavenumber spectrum of the temporally and spatially sampled signal is shown in Figure 3.17. The limited resolution and leakage due to finite sampling is present in the wavenumber domain as well.

As before, we can quantify the resolution in the wavenumber domain via the Rayleigh criterion, which is $k_{\mathrm{Rayleigh}} = 2\pi/D$. Increasing the aperture D will improve the resolution of the signal in the wavenumber domain.

Combining the individual functions that define temporal and spatial sampling into a single function

$$w(x,t) = u(t)v(x) \tag{3.27}$$

it is possible to express the frequency–wavenumber spectrum in a compact form through the convolution theorem

$$\begin{aligned} Z(k,\omega) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} w(x,t)s(x,t)e^{-i(\omega t-kx)}dt\,dx \\ &= W(k,\omega) * S(k,\omega) \end{aligned} \tag{3.28}$$

where $*$ is the convolution operator and $W(k,\omega)$ is the Fourier transform of $w(x,t)$

$$W(k,\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} w(x,t)e^{-i(\omega t-kx)}\,dt\,dx \tag{3.29}$$

Due to finite sampling in time and space, the limited resolution has important consequences in the context of multimode surface waves. In the example of Figure 3.18, the signal is composed of the first two Rayleigh wave modes at a given frequency with equal amplitudes. In Figure 3.18a, the difference between the wavenumbers of the two modes is sufficiently large given the spatial sampling aperture D; hence, each mode can be identified correctly. In Figure 3.18b, the difference in wavenumbers is less, but the individual peaks in the frequency–wavenumber spectrum can still be resolved. The difference

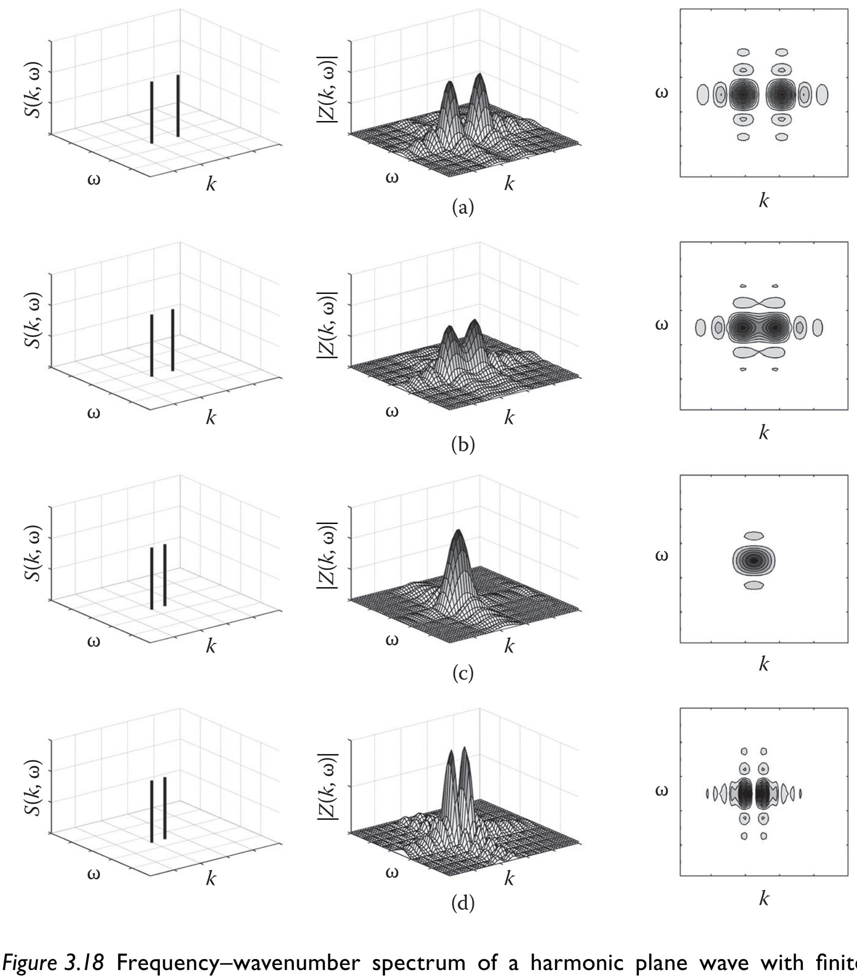

Figure 3.18 Frequency–wavenumber spectrum of a harmonic plane wave with finite temporal and spatial sampling—effect of different spectral resolution related to different apertures. Different distance in wavenumber between the modes are considered for panels a, b, c. In panel d a longer aperture produces a higher spectral resolution and resolves the two modes with the same distance of panel c.

in wavenumbers is further reduced in Figure 3.18c, resulting in a single, “apparent” peak for the combined signal that lies between the “true” peaks associated with the individual modes. Finally, in Figure 3.18d, the difference in wavenumbers is the same as Figure 3.18c, but the spatial sampling aperture is doubled. As a result, the wavenumber resolution is improved and the individual peaks can once again be resolved. In summary, to accurately measure multiple surface wave modes, the spatial sampling aperture has to provide sufficient wavenumber resolution to distinguish between modes.

3.4 ACQUISITION OF SURFACE WAVES

Ideally, the signal to be measured is the effect of pure surface waves, without noise, over the whole frequency band of interest. In reality, the source generates surface waves over a limited frequency band and other wave types are radiated. Moreover, the dataset is affected by noise and by the limitations of the acquisition. The receivers respond without distortions only in a limited frequency range, and the acquisition device records digital traces, with a finite limited sampling and resolution. An example of real shot gather is shown in Figure 3.19. The data show the presence of body waves and of other types of coherent noise. Incoherent noise is also present, as is clearly visible on trace 44. The acquired data are also affected by

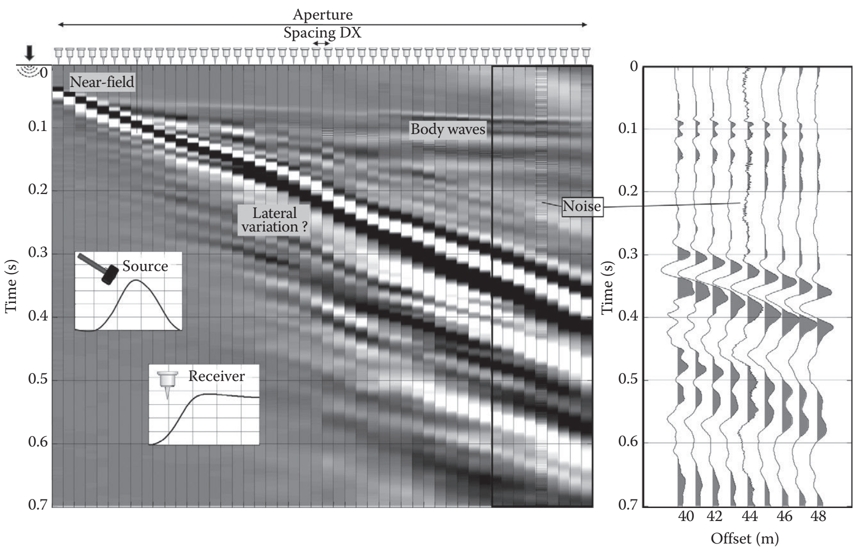

Figure 3.19 Real acquisition with noise, instrument effects, and sampling. Lateral variations appear as changes of dip of the main surface wave event. Body waves are visible as fast early arrivals of P-wave refractions.

the limited source spectrum, the receiver response, and other distortions related to near-field and possibly lateral variations.

These issues cannot be eliminated, but they have to be minimized. The experiment should generate broadband surface waves, with amplitude sufficient to minimize the effect of the incoherent noise. The sampling in time and space should be designed to extract the properties accurately over the desired range of wavelengths, limiting the effects of the coherent noise.

The following issues are discussed in the next sections: coherent and incoherent noise, sampling limitations and survey geometry design, and finally the equipment.

3.4.1 Noise

In general, a *signal* is any function of time and/or space. For seismic tests, a signal is the part of the data carrying meaningful (or useful) information. The unwanted part of the data (i.e., the part that does not carry any useful information) is called *noise*. The latter can mask, distort, and interfere with the signal. The definition of noise is therefore dependent on the objectives of the acquisition.

For instance, in active tests, only the surface waves generated by the controlled seismic source are useful signals. All other recorded events are considered noise. According to this definition, deterministic signals corresponding to an Earth response that cannot be modeled by the assumed Earth model are also considered noise.

Noise is usually classified as *coherent* and *incoherent* noise. Incoherent noise is often referred to as ambient noise, or *random* noise, even if it is statistically not random. Incoherent noise is not deterministically reproducible. It is not a source-generated effect, and it is not correlated with the signal. On the contrary, coherent noise is related to the experiment. It is deterministically reproducible and is correlated to the signal.

The recorded data contain signal and noise: the noise superimposes onto the signal, in different domains, and affects our ability and accuracy in estimating the signal properties. The degree of contamination of the data depends on the relative importance of signal and noise.

The signal-to-noise ratio (SNR) is defined as the ratio between the signal power S and the noise power N. It is often expressed in decibels, as

$$\mathrm{SNR[dB]} = 20\log_{10}\frac{S}{N} \tag{3.30}$$

The data acquisition should be designed and performed with the objective of maximizing the SNR to maximize the accuracy of the estimation of the signal properties.

Different strategies are used to tackle the two types of noise. For the incoherent noise, increasing the power of the signal improves the SNR and reduces the distortions of the noise. The coherent noise, due to its deterministic nature and reproducibility, requires different approaches. For example, the acquisition geometry can be designed to minimize the presence of some coherent events, such as the near-field effect.

3.4.1.1 Incoherent noise

Incoherent noise can be the effect of the background vibrations at the site produced by natural and human sources: traffic, vibrating and moving machines, wind, and movements of surface or ground water. It is often dominated by surface waves, which are, however, incoherent with respect to the acquisition experiment. They are generated by sources, the position and time of activation of which are unknown. Passive surface wave methods are based on analysis of these components of the background noise. Specific signal processing techniques are used to extract the propagation properties without knowing the source position and timing.

Other types of incoherent noise are electric or electronic noise in receivers and cables, and in the acquisition system. This noise may be generated by power lines and other external sources and by imperfections in the recording system.

When acquiring active data, recording the ambient vibrations can be useful for understanding the nature and the level of the incoherent noise. Noise records can be collected by recording data from the receiver spread without activating the seismic source. The noise records often are poorly interpretable in time-offset, but their spectrum discloses valuable information. Understanding the nature of the noise can be useful to evaluate the data during the acquisition stage. For example, a strong single-frequency noise can mask the data in time domain without causing relevant distortion in the useful frequency range. Usually narrow-band noise originates from human activities (e.g., industrial facilities or alternate current in the electrical network). When evaluating the ambient noise properties, long records are required. The spectral analysis of single short windows can have peaks and a bandwidth that do not represent the real site conditions. Figure 3.20 shows an example of the noise spectrum at a quiet site; Figure 3.20a shows the averaged amplitude spectrum of 60 min of passive recording on 48 channels. In Figure 3.20b, the spectrum was computed from two single records 2 s long.

The incoherent noise is summed to the signal and affects all estimates made from the data. Improving the data quality means recording data from which the desired propagation properties can be accurately extracted. Increasing the SNR reduces the uncertainty of the estimation. For instance, the uncertainty on the phase shift between two harmonic traces, with a random noise, is a function of the SNR.

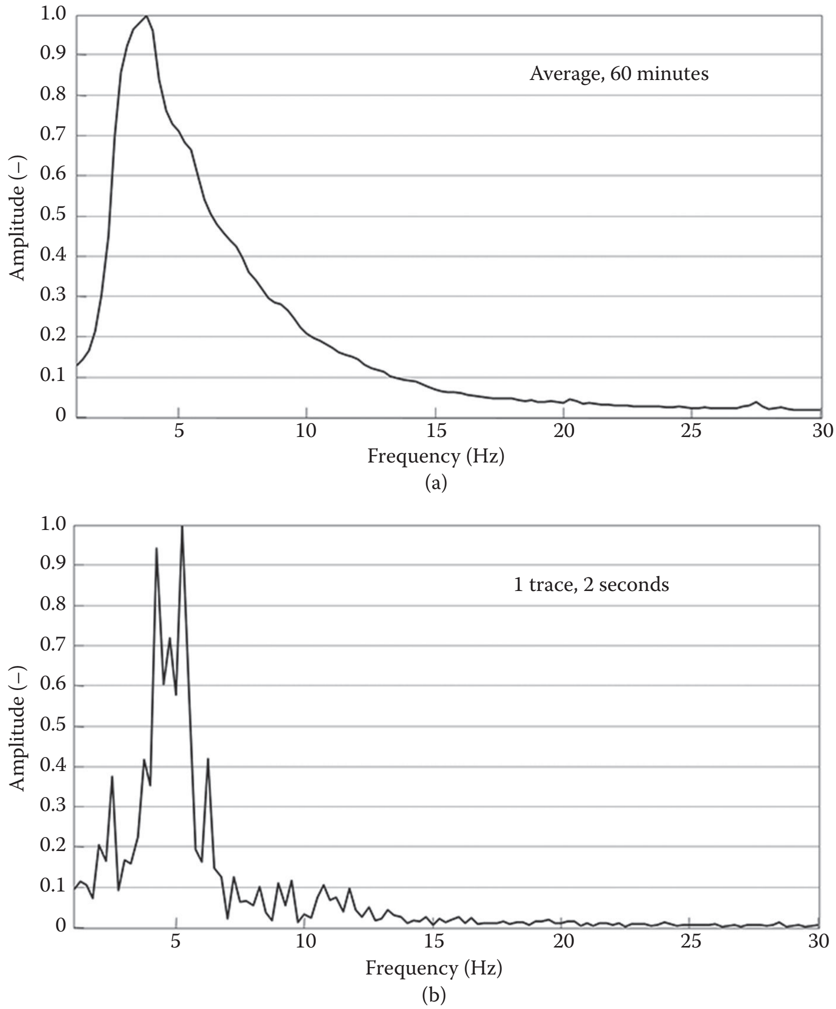

Figure 3.20 Amplitude spectrum of the ambient noise. The spectrum of a single trace, 2 s long, is shown in panel (b). The averaged spectrum is plotted in panel (a).

For a single harmonic trace, the phase distortion of the recorded data $d = s + n$ can be written as (Figure 3.21)

$$= \tan^{-1}\left(\frac{N\sin\quad_n}{N\cos\quad_n + S}\right) = \tan^{-1}\left(\frac{\sin\quad_n}{\cos\quad_n + S/N}\right) \tag{3.31}$$

where φ_n is the phase difference between signal and noise φ_n.

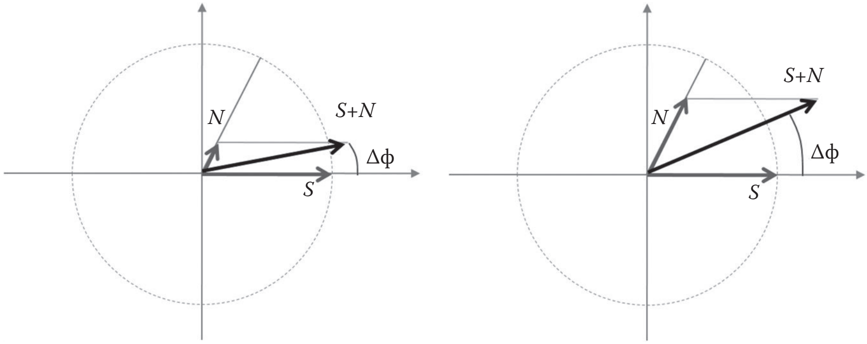

Figure 3.21 The phase distortion due to the presence of noise. For a given phase difference between signal and noise, the phase distortion decreases with the ratio between signal amplitude and noise amplitude.

If the noise has a random phase uniformly distributed in the range $(-\pi, +\pi)$, the phase distortion $(\varphi_{S+N} - \varphi_S)$ has a probability density function $P(\varphi)$

$$P = \frac{1}{2\pi}\left(\frac{\cos \quad \cdot S/N}{\sqrt{1-(S/N)^2 \cdot \sin^2}}\right) \tag{3.32}$$

The distribution is a function of the SNR, and it is symmetric. For SNR >1, it is limited to between $-\sin^{-1}\left(\frac{1}{S/N}\right)$ and $+\sin^{-1}\left(\frac{1}{S/N}\right)$. In Figure 3.22a, the distribution of the phase rotation for different values of the SNR greater than 10 is shown. In Figure 3.22b, the maximum phase distortion is plotted as a function of the SNR.

The phase rotation induced by the noise has a direct effect on the estimation of the velocity. If the phase velocity is estimated from the phase difference between two traces (see Section 4.3), the uncertainty directly depends on the SNR.

In active surface wave data, the SNR is a function of frequency and offset. The incoherent noise has an amplitude spectrum that depends on the site. It is related to the noise sources and to the subsurface response (e.g., see the spectrum of Figure 3.20). If it is not dominated by near sources, it is spatially stationary in terms of amplitude. On the contrary, the signal (the source-generated surface waves) has a frequency- and offset-dependent amplitude spectrum. With the simplest attenuation model, an exponential function of offset with constant damping ratio D and constant velocity V, the signal power in dB takes the form

$$A_{\mathrm{dB}}(f,x) = 20 \cdot \log_{10}\left(A_0(f) \cdot e^{-2\pi f/V \cdot D \cdot x}\right) \tag{3.33}$$

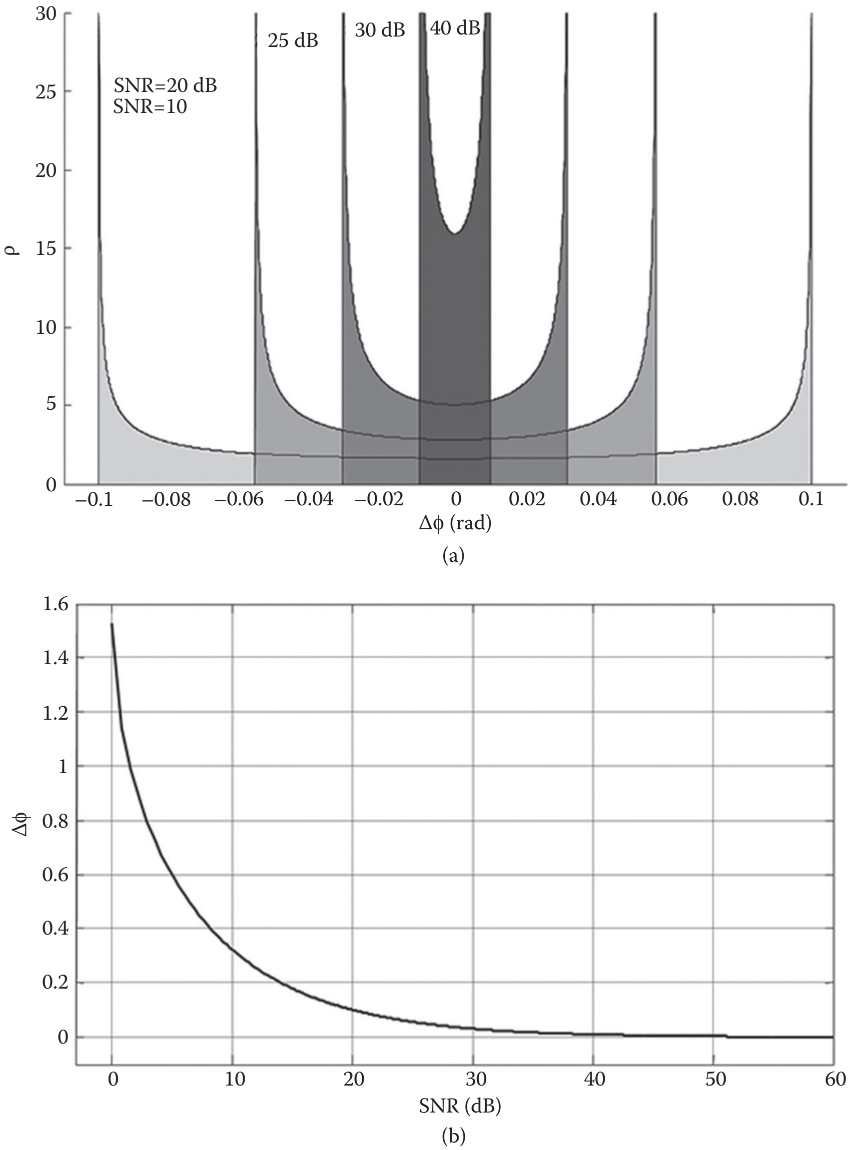

Figure 3.22 Maximum phase distortion induced by noise on a monochromatic signal as a function of the signal-to-noise ratio (SNR): (a) distribution of phase rotation for different values of SNR; (b) maximum phase distortion as a function of SNR.

which gives a set of hyperbolic isoamplitude lines in the *f–x* domain for a constant unit reference amplitude A_0. The line with an attenuation of *c* dB is

$$X = \frac{c \cdot ln(10) \cdot V}{20 \cdot D \cdot 2\pi} \cdot \frac{1}{f} \tag{3.34}$$

as represented in Figure 3.23a. Considering a single frequency, the signal has a linear decay with offset when the amplitude is measured in dB. The total level of signal and noise, however, tends to the noise floor when the SNR is low (as represented in Figure 3.23b).

The SNR distribution versus offset and frequency is important because only data with a sufficient SNR should be used to estimate the signal properties. As discussed earlier, the phase uncertainty depends on the SNR. The far-offset region, for example, will not contribute to the signal estimation in the high-frequency range. By defining a minimum acceptable SNR, a frequency-dependent maximum offset is obtained—for example, the curve −40 dB of Figure 3.23a.

With a more realistic attenuation model, the shape of these graphs would be more complicated, but the conclusions would be similar. Also, in real data, a defined threshold of minimum SNR is reached at different offsets. Figure 3.24 shows the amplitude spectrum of a real gather. Ambient noise has been recorded at the same site. The amplitude *f–x* spectrum of the noise, in Figure 3.24b, is spatially stationary. The amplitude spectrum of the signal, in Figure 3.24c, shows a typical frequency- and offset-dependent attenuation. The amplitude decay of two frequencies is shown in Figure 3.24d.

3.4.1.2 Increasing the SNR for incoherent noise

Increasing the SNR is a primary objective of the acquisition, and it can be achieved either by reducing the level of the noise or by increasing the level of the signal.

Reducing the noise is sometimes possible. It can be done by acquiring data during quiet times (at night, the human noise is lower) or by using better equipment with a lower noise level. A careful execution of the field operations can reduce the incoherent noise. The meticulous deployment and ground coupling of the receivers including burying receivers, avoiding people and vehicle movements, and avoiding vibrations during the acquisition are important practices for minimizing the incoherent noise.

Increasing the signal level can be done by increasing the energy of the source. Possible strategies for increasing the signal level include using more powerful sources or combining different sources for different frequency ranges. An alternative approach is the so-called "vertical stacking" or

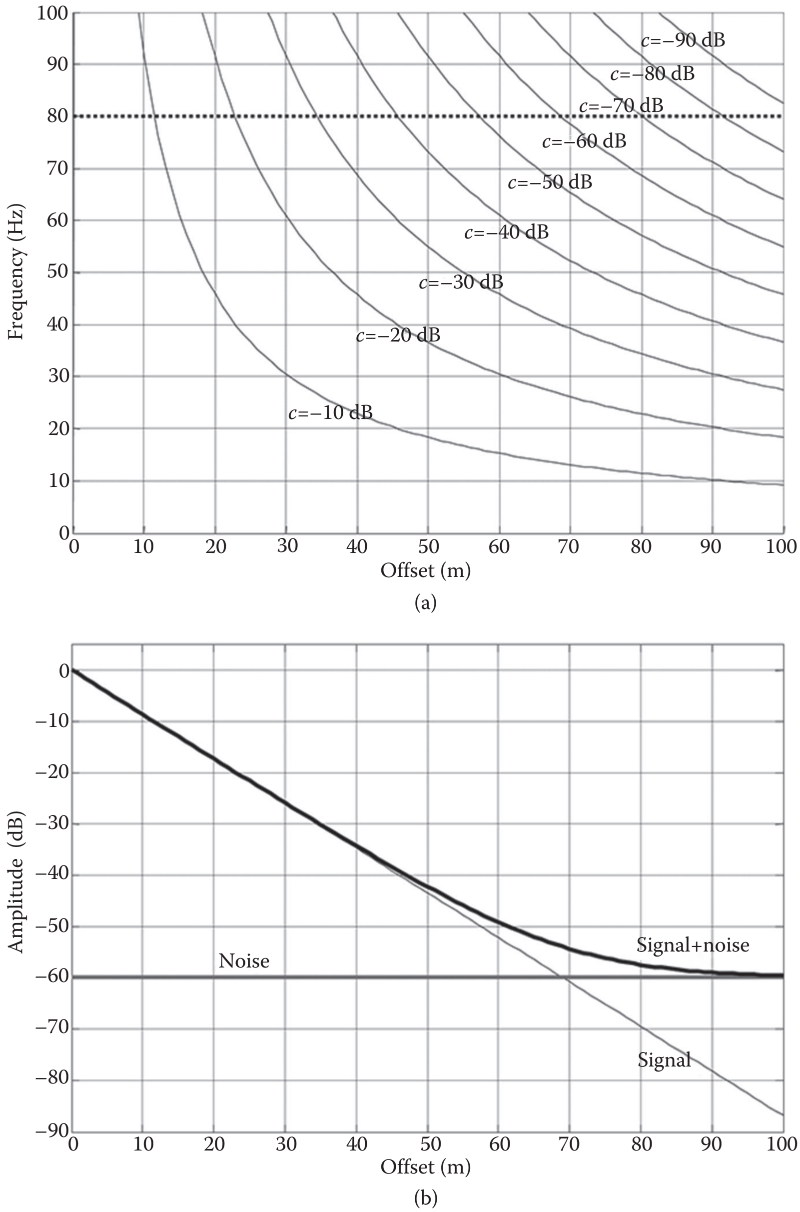

Figure 3.23 (a) The amplitude decay of a signal is plotted. The curves represent the decay of a pure signal with a simple attenuation model (single mode, constant damping, and constant frequency). (b) The amplitude of a recorded frequency with offset is represented schematically as the sum of signal and noise on a dB scale.

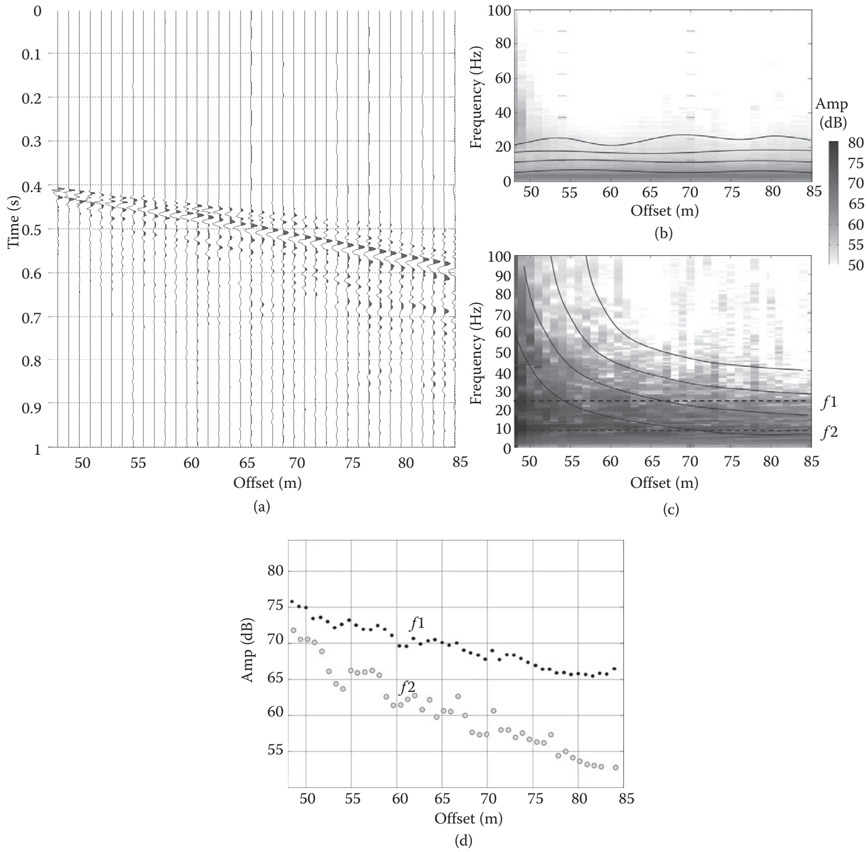

Figure 3.24 Comparison between the signal from an active source and background noise at a real site: (a) shot gather of active-source data; (b) amplitude *f–x* spectrum of the background noise; (c) amplitude *f–x* spectrum of the active-source gather; (d) spatial amplitude decay of two frequency components.

summation of multiple synchronized repetitions of the test. If the incoherent noise is not deterministically reproducible and has at least a random phase component, stacking increases the SNR by the square root of the number of repetitions. An example of vertical stacking is shown in Figure 3.25. The same experiment is repeated 25 times, and the traces are summed.

The signal, when perfectly reproducible among multiple repetitions, increases its amplitude with the number of stacked signals, n. The noise can have stationary amplitude but, if incoherent, has a stochastic phase. If the phase is random, as in a frequency component of the random noise, we can assume a uniform distribution in $(-\pi, +\pi)$; the absolute value of the phase difference between two summed components is linearly distributed in $(-\pi/2, +\pi/2)$, and

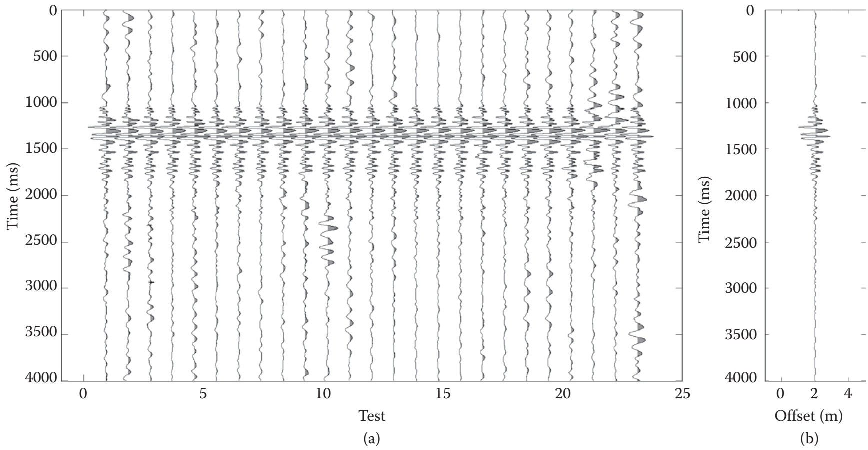

Figure 3.25 The summation of multiple repetitions improves the signal-to-noise ratio because the signal increases more than the noise when the latter is not coherent or deterministically reproducible: (a) ensemble of 25 individual shots; (b) stacked signal.

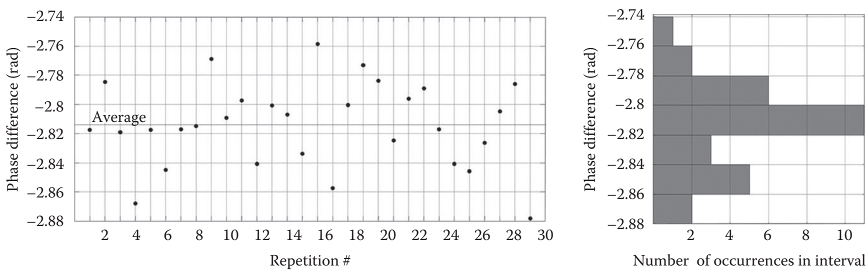

Figure 3.26 The phase difference between two traces estimated for 30 repetitions of the test. The standard deviation in this example is about 1% of the estimated mean value.

this makes the sum lower than its maximum. Considering a constant amplitude, with vertical stacking, the random noise increases as $\sqrt{n}$. The SNR then increases as $n/\sqrt{n} = \sqrt{n}$.

Several repetitions of the experiment are necessary for vertical stacking. If the recorded data are not summed directly in the field but are recorded separately, they provide a way of evaluating the uncertainty of the estimated wave parameters versus the stochastic components of the noise. The phase difference between two traces spaced at 5 m is plotted for 30 repetitions of the test in Figure 3.26. The average phase difference is about 2.8141 radians, and the standard deviation is about 0.0289 radians, or about 1%.

The uncertainty on the phase difference can be used to compute the uncertainty on the wavenumber and propagated to the phase velocity.

It should be noted that such statistical estimations only consider the stochastic uncertainty due to a random component of the incoherent noise. It does not take into account epistemic uncertainty and coherent noise.

3.4.1.3 Coherent noise

Coherent noise consists of deterministically reproducible events. Typically, these are source-generated events that are not associated with the signal of interest. In this case, increasing the power of the source or summing multiple repetitions does not increase the SNR. Indeed, these events increase their power proportionally (in theory) or repeat themselves with the same character. Understanding and identifying the coherent noise and its properties is important to design the strategies to mitigate its effects in the acquisition and processing stages.

The definition of coherent noise is not straightforward. It depends on the objectives of the measurement and on the assumption of the interpretation models. Any part of the data that cannot be explained by the assumed model has to be considered coherent noise. Citing Scales and Snieder (1998), "noise is that part of the data that we choose not to explain." The coherent noise can be a deterministic seismic event propagating in the subsurface and carrying information about some properties of the Earth but not included (maybe for simplicity) in the model that is used to interpret the data. As an example, we can say that in seismic reflection surveying, the surface waves are considered coherent noise. The signal is the reflected wave field.

In most of the cases, surface wave methods assume a one-dimensional (1D) Earth and far-field plane and linear-surface waves. Therefore, when acquiring surface waves, we can consider coherent noise as any other seismic event and, in general, all the effects of the differences between the assumed model and the true Earth response. Whatever produces data that are not explained correctly by the assumed subsoil model is coherent noise. Real data contain then different types of coherent noise: *nonsurface wave events* and *noncompliant surface waves*.

The nonsurface wave events include body waves from direct, refracted, and reflected paths in the subsurface. They also include air blasts (i.e., acoustic waves associated with the source stroke and traveling through the air).

The noncompliant surface wave coherent noise includes all the surface wave energy that does not follow the assumed propagation model for intrinsic and geometric reasons. For example, discontinuities and heterogeneities in the subsurface and in the topography produce diffractions of the surface waves and conversion of body waves into surface waves.

These phenomena, often called scattering, do not propagate directly from the source to the receivers. The associated apparent velocity along the receiver spread can be different from that of the direct surface waves.

The lateral variations of the medium properties affect the wave propagation properties in a deterministic way and distort the extracted velocity. They introduce artifacts in the estimated parameters when a laterally homogenous medium and wave are assumed and can be considered coherent noise. Their presence should be identified and considered in the processing stage.

Surface waves that propagate a short distance from the source cannot be treated as horizontally traveling plane waves. They have different propagation properties and are not correctly interpreted by most inversion approaches. This point has been discussed by many authors and is known as the near-field effect (Stokoe et al. 1994). The near-field effect is deterministic; hence, it can be classified as coherent noise.

Slightly different, from a measurement perspective, is the case of higher modes. The surface wave propagation has a multimodal nature, and the energy associated with higher modes depends on the stratigraphy and on the experimental setup. Especially at inversely dispersive sites, higher modes may dominate the experimental wave field. If the interpretation approach considers only the fundamental mode, or if the acquisition and processing procedures do not allow the identification of multiple events, higher modes have to be considered coherent noise and must be carefully filtered out.

However, in case of higher modes of surface waves, improving the model is recommended; they are not a different, unrelated event, and their properties depend on the same parameters that we try to estimate with the fundamental mode. They can be identified in data and included in the adopted Earth response model; hence, they can be considered as useful signals. Examples of multimodal processing and inversion are reported in the literature (for instance, Gabriels et al. 1987; Maraschini et al. 2010). The advantage is twofold. For one, this adds extra information. Second, upgrading the model allows the acquired data to be correctly interpreted even when the higher modes are dominant and cannot be removed.

3.4.1.4 Body waves

In records, body waves can be P- and S-waves propagating from the source to receivers with different paths. In shallow, small-scale tests, body wave amplitude is often much lower than the surface wave amplitude. They are superimposed onto each other in time-offset, but often they tend to map into different portions of the *f*–*k* spectrum. They can be easily identified and removed or ignored. Strong P-waves from refracted paths can be muted in time if necessary.

In large-scale acquisition, for exploration applications, the guided waves can become dominant events in the far offset. Guided modes are surface waves, leaky modes with a dominating pressure component, and interfering total internal reflections of the supercritical energy in a low-velocity waveguide. Because they are faster and often have a smaller damping ratio, they tend to dominate in the far offset. They are usually separated in terms of velocity and frequency, and they can be analyzed for a joint inversion with the Rayleigh waves or Scholte waves. The joint inversion of Rayleigh and guided waves is discussed in Chapter 8.

3.4.1.5 Air blast

The sound emitted by the seismic source propagates as a pressure wave in the atmosphere, sometimes coupling with the ground, and it is detected by seismic receivers. Even in Vibroseis acquisition, the vibrator sound correlates with the pilot and appears as a high-frequency event in the data.

In active data, it is common to observe a linear, high-frequency event propagating with fairly constant velocity at the speed of sound, which varies from 310 to 360 m/s, for a temperature varying from –30 to +50°C. Indeed, pressure and humidity variations can be neglected in normal conditions, and the speed of sound v_{air} can be written as

$$v_{air} = 331.3 \cdot \sqrt{1 + \frac{T}{273.15}} \; [\mathrm{m/s}] \tag{3.35}$$

where T is the temperature.

The presence of a strong air blast is potentially troublesome in surface wave measurement because it may superimpose on the Rayleigh wave signature. Indeed, the velocity of propagation in air may be very close to Rayleigh wave propagation. The air blast can be observed in time-offset, and it is identified in the *f–k* spectrum as a linear event, with constant velocity and very low attenuation, usually extending to high frequency. It can be muted in time-offset and ignored in the processing stage. In Figure 3.27, the air blast is visible in the shot gather as a high-frequency event preceding the surface wave energy (as annotated).

3.4.1.6 Near-field

Rayleigh waves can be regarded as plane waves only beyond a certain distance from the source (i.e., in the far-field) (Richart et al. 1970). The inner zone is addressed as the zone in the near-field. In the two-station technique (see Section 4.3), the measurements are usually considered as affected by near-field effects if the first receiver is placed at a distance from

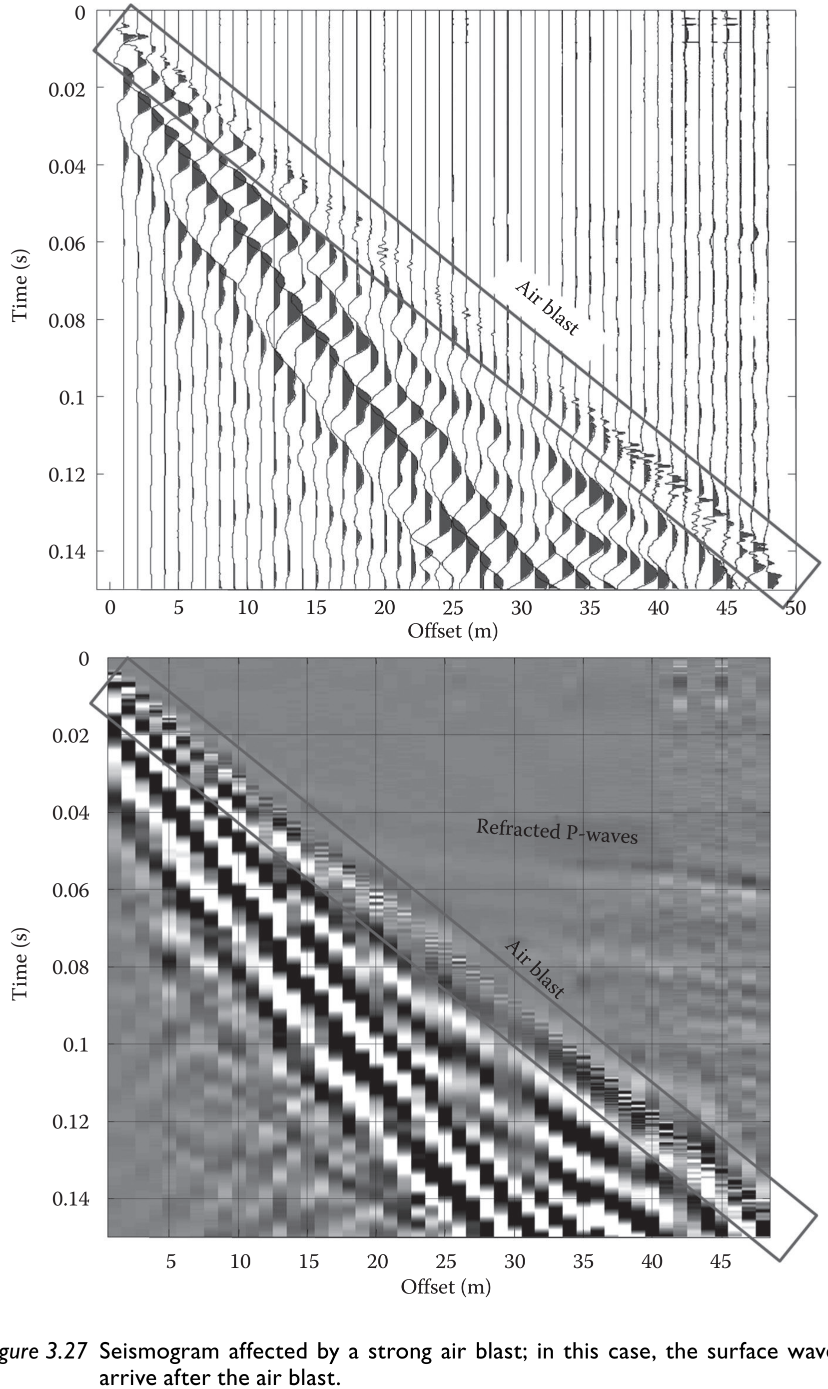

Figure 3.27 Seismogram affected by a strong air blast; in this case, the surface waves arrive after the air blast.

the source less than half the considered wavelength (Stokoe et al. 1994), although some authors suggest larger values for complex stratigraphy (Sànchez-Salinero 1987; Tokimatsu 1995). The near-field is a function of the frequency. In broadband data, the near-field for some frequencies is the far-field for other (higher) frequencies. Some processing techniques allow identifying the zone affected by near-field in which receivers should be muted to get a reliable estimate of Rayleigh wave propagation parameters (see Section 4.4).

3.4.1.7 Lateral variations

The acquired data are usually processed with the aim of extracting a single-space invariant set of propagation properties for the considered site, integrating or averaging over multiple receivers. The assumed propagation is therefore assumed to be laterally homogeneous—linear in time-offset and in phase-offset. This is consistent with the assumed 1D Earth model used for the inversion in most applications (see Chapter 6). The presence of lateral variation in the subsoil produces effects that are not correctly interpreted.

In this context, the effects of lateral variations can be considered as coherent noise. This requirement should be considered when selecting the location of the test, avoiding known lithological boundaries but also trying to avoid the acquisition along the dip direction for expected dipping layers.

In general, it can be observed that the length of the array is an important parameter, as far as the risk of lateral variations is concerned; the longer the array, the higher the chance of significant lateral variations. In addition to different processing techniques, a set of shorter arrays might be preferred for investigating sites where lateral variations are expected. The trade-off is between the lateral resolution and the spectral resolution, which increases with increasing aperture.

Once data are acquired, the identification of lateral variations is an essential step of processing. With the exception of extreme cases, it is not easy to detect the presence of lateral variations from the time-offset data. Processing techniques have been developed to test this basic hypothesis of one-dimensionality using the multichannel data. The lateral variations can be identified by comparison of information extracted from different portions of the array.

3.4.1.8 Higher modes

Surface wave propagation often exhibits multiple modes of propagation. This is common at inversely dispersive sites, where velocity inversions have a strong impact on the energy distribution among modes. Higher modes may also have a relevant role at normally dispersive sites with large impedance contrasts and even with smooth velocity profiles.

Higher modes should be considered as useful information. Their phase velocity, as well as their spatial attenuation and energy distribution, depends on the subsoil properties, and they can be theoretically simulated and inverted. They only become coherent noise when the acquisition, processing, or inversion techniques are not able to deal properly with them.

Therefore, the acquisition should be designed in order to allow the identification of higher modes. Even in cases where the fundamental mode is dominating, the effect of higher modes on the recorded data cannot be neglected. A multichannel acquisition is necessary to properly identify the different modes. However, with some processing technique, the effects of higher modes can be misinterpreted. For instance, jumps of the phase and the oscillating behavior of the amplitude with the offset can be due to the superposition of different modes. The example in Figure 3.28 refers to a site where a higher mode has significant energy in the relevant frequency band.

The shot gather in time-offset and the *f–k* spectrum show the presence of two modes, with similar energy. The seismogram acquired with an impulsive source and 24 vertical geophones is shown in Figure 3.28a. The presence of two events with different velocities can be recognized in time-offset. The two modes can also be separated in the *f–k* spectrum (Figure 3.28b). In this case, the spectral resolution allows the separation of the two modes and hence the identification of the properties of both. The effects of mode superposition are clearly illustrated with a representation of the 20 Hz component of the seismogram (Figure 3.29). Phase rotations and minima of the amplitude are observed at the offsets where the two modes are in phase opposition and where they interfere destructively.

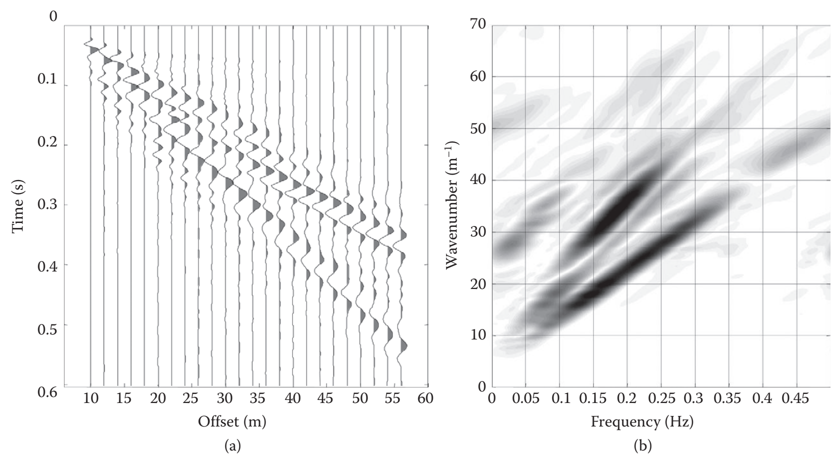

Figure 3.28 Seismogram and *f–k* spectrum for a case with highly energetic higher modes. The aperture is sufficient to separate the events: (a) seismic gather; (b) *f–k* spectrum.

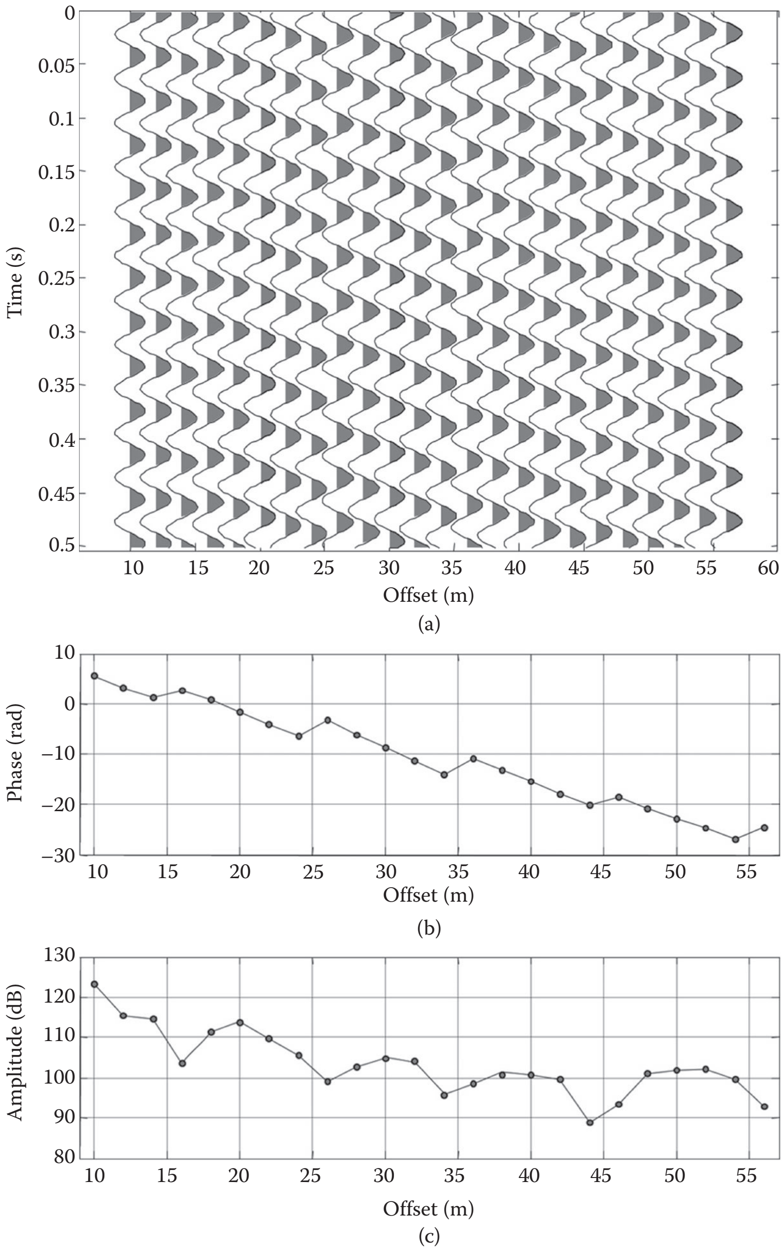

Figure 3.29 Single-frequency component extracted from the seismic record of Figure 3.28: (a) filtered time records; (b) phase versus offset; (c) amplitude versus offset. The amplitude decay shows the interference among multiple modes, and phase rotations are visible as well.

3.4.2 Sampling

The effects of the wave propagation at a specific location are recorded in time as seismic traces, with a fixed time step and a finite length. A finite number of receivers is deployed, and this implies a discrete spatial sampling, with a certain distance between receivers and a total length of the receiver array. The wave field is then sampled using a 2D discrete window in time and space. This induces limitations in terms of accuracy and bandwidth.

The limitations and consequences of the sampling are presented and discussed in time-offset and frequency–wavenumber domains in the following. Yet, the discussion applies independently on the processing method (see Chapter 4) because these limitations are intrinsic. The effects of the sampling are important in any situation and become critical when multiple modes are present.

3.4.2.1 Spatial and temporal discrete and finite sampling

As discussed in Chapter 2, the wave field associated with surface wave propagation in a layered medium can be written in terms of mode superposition, with a discrete set of modal curves. The eigenvalues describe the kinematics: for each frequency f, a set of wavenumbers $k_1\ (f)$, $k_2\ (f)$, ... , $k_n\ (f)$ and phase velocities $v_1 = f/k_1$, $v_2 = f/k_2$, ... , $v_n = f/k_n$ can be found.

The energy propagates only in correspondence with modal wavenumbers. This description of the wave field is valid only in the far-field, neglecting lateral variations, noise, attenuation, and so on, and it allows describing the effects of the sampling on the observed kinematics. The *spatial* sampling is particularly important and directly affects the accuracy of the estimated velocity, the ability to resolve mode, and the ability to identify secondary events with lower amplitudes.

When the wave field is sampled, 2D window w is introduced, and the effects are observed on a discrete set of points

$$r_{obs} = s(x,t) \cdot w(x,t) \tag{3.36}$$

Because the multiplication in time-offset domain is a convolution in frequency–wavenumber domain of the signal spectrum with the spectrum of the 2D window function, we can write

$$R_{obs} = s(f,k) * w(f,k) \tag{3.37}$$

The 2D boxcar window and its spectrum are depicted in Figure 3.30.

A single event—ideally homogeneous, continuous, and infinite—would appear as a spike in f–k. When sampled, it will appear as a spectrum of the acquisition window (Figure 3.31). The effects in k domain are typically more severe than the effects in f domain because the number of sampling

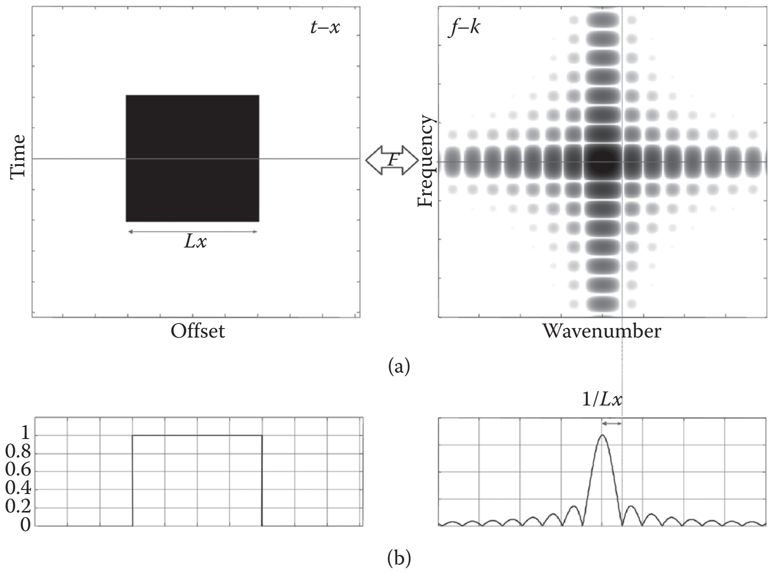

Figure 3.30 Time-offset boxcar window: (a) *t–x* and *f–k* domains; (b) slides at a constant time and at a constant frequency.

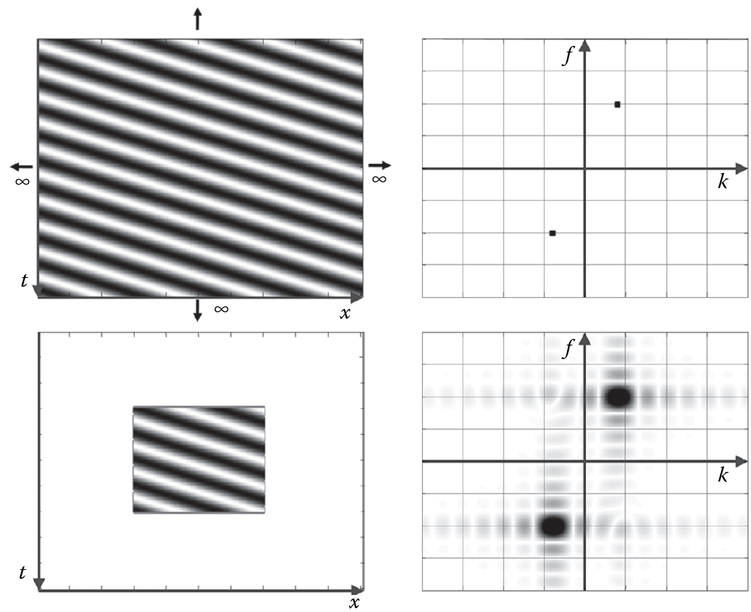

Figure 3.31 Effect of windowing on a harmonic plane wave—the spectral leakage transforms a Dirac's delta into a 2D sinc function.

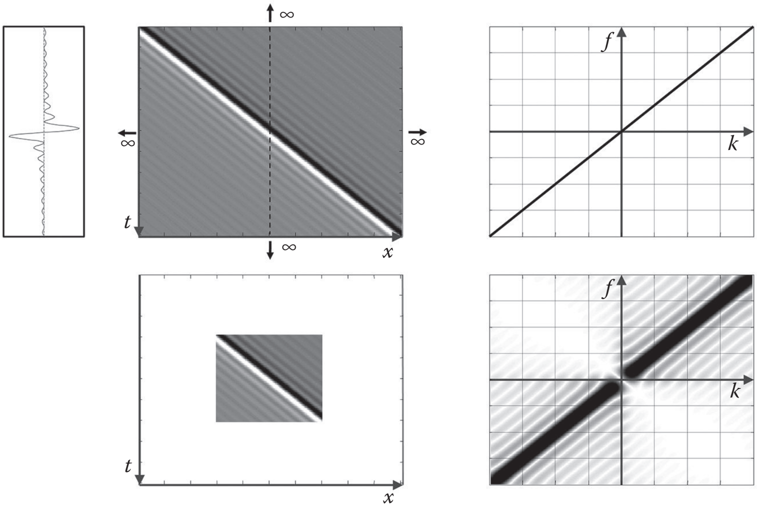

Figure 3.32 Effect of windowing on a broadband wave; the wavelet has a flat spectrum and is not dispersive. The spectral leakage creates a main lobe and side lobes.

points in space is necessarily smaller than in time. The abrupt truncation of the signal in time and space produces side lobes that have to be mitigated.

The sampling, with broadband surface waves, generates the described effects on each point of each mode. In Figure 3.32, the effect of the windowing is shown on a simple, nondispersive signal with a flat amplitude spectrum. The theoretical *f–k* spectrum is a Dirac's delta line passing through the origin of the *f–k* plane and with slope *V*. The finite sampling creates spectral leakage, as shown by the presence of side lobes. They appear in the spectrum of the finite signal as secondary maxima, which are parallel to the main event. Side lobes can be mitigated with tapering during the processing.

The sampling introduces a series of limitations that have important consequences on the inferred dispersion curves. In the following section, the limits introduced by the acquisition are discussed considering the final objective of designing the optimal acquisition to minimize the effects of the sampling over the extracted properties. The effects of the sampling in time and space will be discussed in terms of limitations of the frequency bandwidth, or modal resolution, and of accuracy of the extracted properties.

Windowing introduces a maximum wavenumber, a finite main lobe with leakage, and side lobes. The maximum wavenumber might limit the bandwidth and create spatial aliasing of the data. The spectral resolution affects the accuracy of the velocity estimation and the possibility of discriminating among different modes. A synthetic gather with only surface

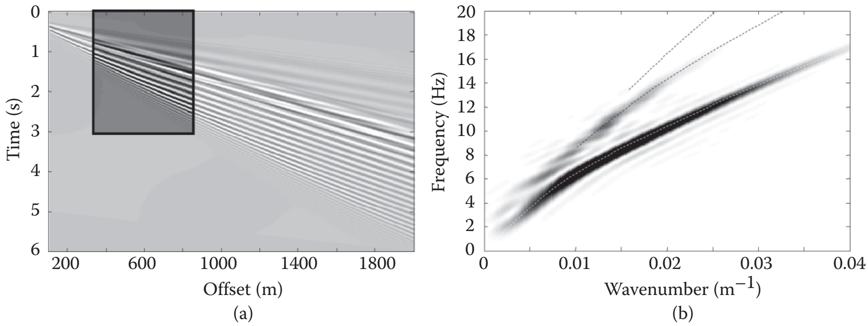

Figure 3.33 Finite sampling of a transient synthetic Rayleigh wave field: (a) shot gather and sampling window; (b) the three theoretical modes are represented in the *f*–*k* plane together with the computed spectral density. The truncation in time and offsets creates leakage and side lobes that affect the identification of higher modes.

waves is shown in Figure 3.33a. The spectral analysis of a limited time and space window is shown in Figure 3.33b together with the theoretical modal curves. The fundamental mode corresponds to the energy maximum, but the identification of higher modes is not straightforward.

These sampling issues can lead to quite irregular and misleading behavior of the experimental dispersion curve if they are not correctly assessed in the design of the testing array. Moreover, these limitations also have to be taken into account in the inversion.

Only the modal curves should be considered a characteristic of the site. The experimental dispersion curve is affected by mode superposition effects and also depends on the acquisition parameters. It can then be considered as an apparent dispersion curve (Tokimatsu 1995), which is not an intrinsic property of the site. To compare synthetic and experimental data within an inversion algorithm, it is then necessary to model the experimental test, taking into account all the acquisition parameters and the source position.

The limitations of the temporal sampling are less severe. It is normally possible to sample at a frequency high enough for the proper acquisition of the maximum propagating frequency, and the total duration of the signal can be normally acquired.

3.4.2.2 Maximum wavenumber and spatial aliasing

A broadband dispersion curve is needed for an accurate inversion. The maximum wavenumber that can be identified with a real array depends on the receiver spacing. When the spatial sampling is not sufficient, aliasing occurs, and the short wavelengths cannot be reconstructed properly. With 2D data, a true wavenumber higher than the Nyquist wavenumber $k_{true} > k_{nyquist}$ appears in the negative quadrant. The aliased replica falling in the spectrum has an apparent wavenumber of $k_{alias} = k_{true} - 2k_{nyquist}$. The apparent velocity is therefore negative.

The concept can be illustrated with monochromatic plane waves, with the same frequency and decreasing velocity, such as in Figure 3.34. The top row shows the *f*–*k* spectra, with a dot indicating the reconstructed position of the event; the bottom row shows the seismograms. The first two cases, with high V_1 and V_2, have a wavenumber smaller than the Nyquist wavenumber and are properly sampled. The apparent velocity in the seismic gather can be identified visually following a phase (e.g., a peak); it is positive with the wave traveling from left to right in the panel. The velocity V_3 corresponds to a wavenumber ($k_3 = f / V_3$) equal to the Nyquist wavenumber. The gather shows two opposite identical apparent velocities; the positive corresponds to the actual velocity.

The cases with V_4 and V_5 are aliased, with $k_{true} > k_{nyquist}$ and an apparent negative velocity. In the last case, V_6 has a wavenumber exactly equal to $2k_{nyquist}$ and the aliased as $k_{alias} = k_{true} - 2k_{nyquist} = 0$. The apparent velocity is therefore infinite, and the event appears flat.

The sign of the apparent velocity is an important element to consider. The event propagates with reference to the gathers in Figure 3.34, from left to right, and the sign of the velocity is known to be positive because of the acquisition geometry. When the acquisition is spatially aliased with a wavenumber smaller than $2k_{nyquist}$, it appears with a negative velocity (dashed lines in the figure).

In broadband data, all the components with $k_{true} > k_{nyquist}$ will be aliased in the negative quadrant (Figure 3.35). However, in end-off gathers, knowing that all the signal energy travels in the positive direction and is associated with positive wavenumbers, we can assume that only noise and aliased events are present in the negative quadrant. It is therefore possible to recover the aliased information in the range ($k_{nyquist} - 2k_{nyquist}$), between $-k_{nyquist}$ and 0, without introducing additional noise. The actual maximum wavenumber with off-end gathers is twice the Nyquist wavenumber; equivalently, the minimum wavelength is equal to the receiver spacing.

3.4.2.3 Spectral resolution and aperture

The actual length of the observation array affects the size of the main lobe of the window transform, which controls the spectral resolution. The leakage of the energy will be larger for a shorter array due to the scaling property of the Fourier transform. The accuracy of the identified maximum will be affected by the width of the main lobe, and the possibility of separating multiple events will be affected as well.

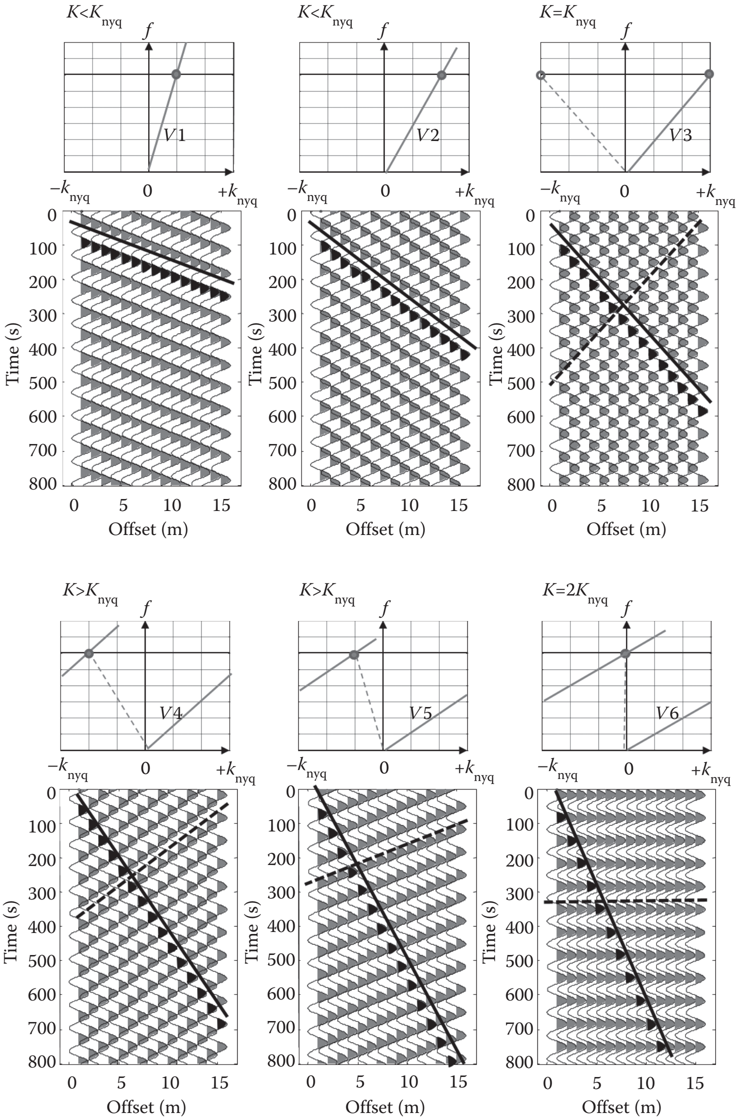

Figure 3.34 Spatial aliasing illustrated with a plane harmonic wave with decreasing velocity. The decrease of the velocity produces an increase of the wavenumber, which in the third panel reaches the Nyquist wavenumber for the adopted spatial sampling. Spatial aliasing occurs, and apparent velocities are negative in the *f*–*k* panels on the bottom. The continuous line in the seismic gather represents the actual phase velocity; the dotted line is the apparent phase velocity.

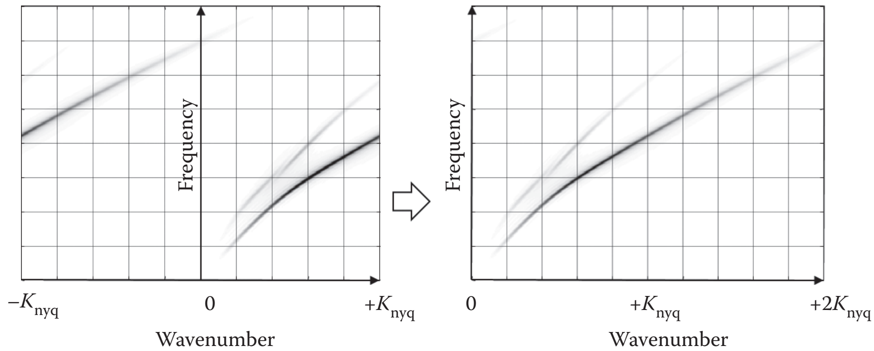

Figure 3.35 Unwrapping the negative quadrant of the *f*–*k* spectrum in off-end gathers to recover the correct representation of the propagation.

In theory, at a given frequency, the energy can propagate only at a discrete set of wavenumbers, one for each of the existing modes at the considered frequency. The wavenumber spectrum, at a constant frequency, should have spikes at the modal wavenumbers $k_1(f_1), k_2(f_1), \ldots, k_n(f)$ associated with the n existing modes. In reality, the wavenumber spectral resolution, derived from the spatial windowing of the acquisition (geophones spread length), allows only for smeared picks. This windowing effect may prevent an effective identification of modal curves. The ideal wavenumber spectrum and the observed spectrum for two different arrays are shown in Figure 3.36. The theoretical spectrum is shown in Figure 3.36a; the two events are resolved with an array of 96 m (Figure 3.36b) and are not resolved with an array of 12 m (Figure 3.36c).

As a result of the insufficient spectral resolution, multiple modes can be unresolvable, and the energy can appear as propagating at an intermediate velocity without corresponding to any of the real modes due to the mode superposition. What can be observed in the data depends on the true events and on the acquisition layout.

As an example, we show the picked energy maxima for a synthetic spectrum computed for the same layered model using two different arrays. The model has a velocity inversion, and higher modes dominate in the high-frequency range. The synthetic apparent dispersion curve for a short array (24 m) shows that modal superposition leads to a dispersion curve not corresponding to a modal curve because of insufficient spectral resolution. The absolute maximum of the spectrum is due to the superposition of the contributions of modes. The apparent dispersion curve shows a smooth passage to higher modes with increasing frequency (Figure 3.37b). When the array length is sufficient to separate modes (96 m in this example), the maximum follows the mode with the highest energy at each frequency (Figure 3.37a).

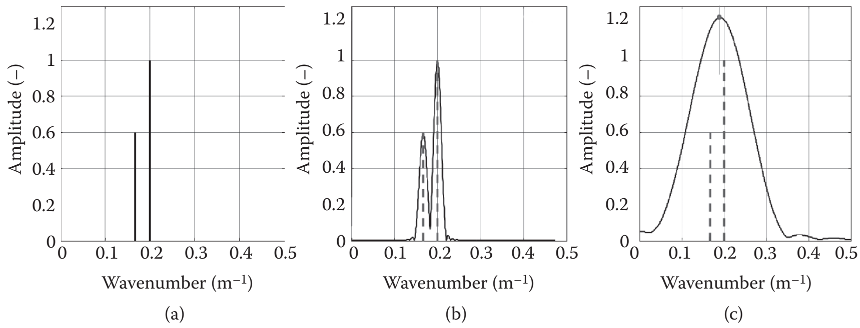

Figure 3.36 Spectral resolution at a constant frequency—possibility of resolving two modes with arrays of different lengths.

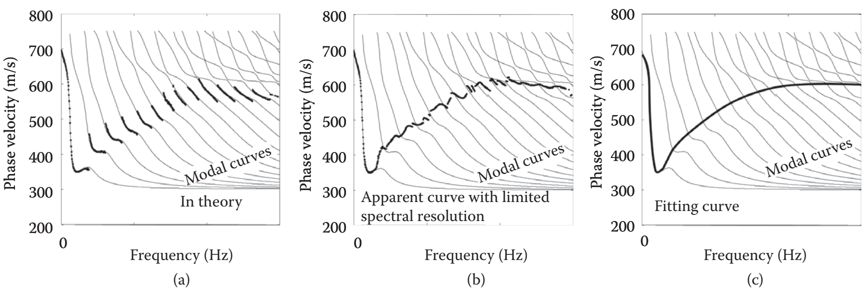

Figure 3.37 Example of modal superposition with two different arrays for the same synthetic case.

The example shows that the apparent velocity is affected by the modal superposition, depending on the array size. Similar results would be obtained with any of the processing techniques presented in Chapter 4 because the limitation is introduced by the acquisition and not by the processing.

Figure 3.38 represents the seismogram and picked maxima for an acquisition at a site composed of shallow, dry, overconsolidated clay over a clay sequence. The array length of 48 m used for the acquisition is not sufficient to separate the modes, and an apparent dispersion curve is observed in the range from 10 to 100 Hz.

Long arrays seem to be preferable because the longer the aperture of the spatial window (i.e., the array length), the higher the spectral resolution. On the contrary, with a short array there are less risks of lateral variations, insufficient *S*/*N* ratio, high-frequency attenuation, and spatial aliasing (for a given number of receivers).

The actual spectral resolution depends on the true array length. Yet, the computation of the spectrum requires evaluating the spectral density

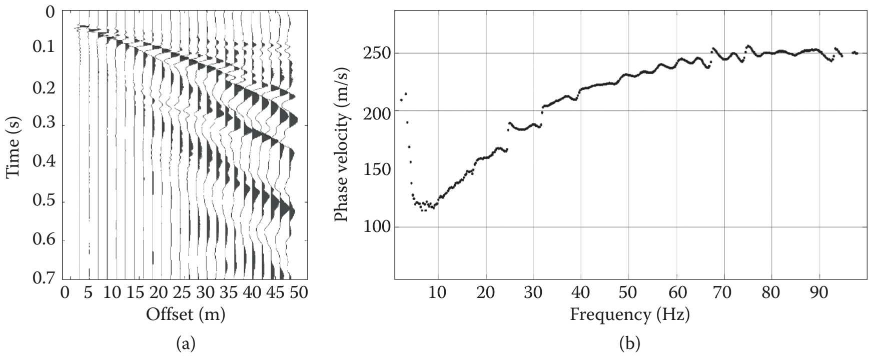

Figure 3.38 Example of modal superposition with real data. The picked curve from 10 to 100 Hz results from the superposition of multiple modes that are not resolved with the given acquisition parameters: (a) seismic gather; (b) apparent experimental dispersion curve.

on a number of frequency values much larger than the number of traces. This can be achieved in frequency–wavenumber analysis with the 2D Fourier transform via zero-padding the traces. The wavenumber spacing is crucial for an accurate picking of spectra but does not affect the actual spectral resolution and the size of the main lobe. Advanced techniques of array deconvolution, or synthetic aperture, can increase the spectral resolution.

3.4.2.4 Effects of side lobes

Spatial windowing introduces a fictitious spreading of energy in the *f*–*k* spectrum, and the side lobes might prevent higher modes from being identified and hiding the energy maxima associated with them (Linville and Laster 1966). The side lobes in the *k* domain might create apparent maxima that should not be confused with modes. This effect can be reduced using an appropriate tapering in the space domain, for example, a Hanning window. In Figure 3.39, the spectrum on the left is computed using a box window; the one on the right is computed using a Hamming window. The side lobes for a boxcar window can be computed as maxima of the sync function. The example on the right represents the set of maxima (including the first four side lobes on each side) of a 48 m array converted in velocity.

A generalized Gaussian window, without side lobes, helps in discriminating secondary maxima due to higher modes and those due to side lobes. Data are acquired without any special windowing, and this aspect refers more to processing (see Chapter 4).

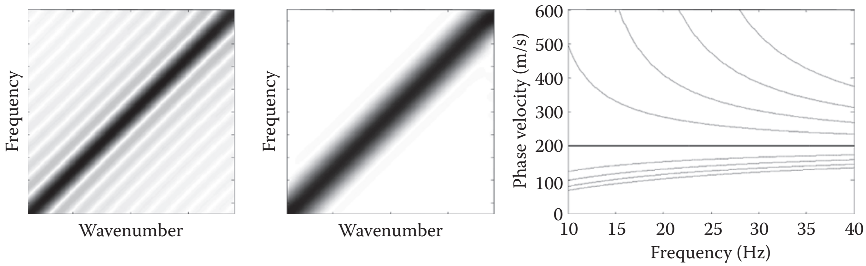

Figure 3.39 Effect of the side lobes without and with additional tapering. The side lobes can be misinterpreted as multiple modes.

3.4.3 Survey design

Considering the objectives of the surface wave test, the correct experimental design and the careful execution of the experiment are of primary importance for obtaining an optimal result. Many alternative approaches exist depending on the objective of the investigation and on the field of application. The spatial scale of surface wave testing can vary from millimeters to kilometers. Most of the acquisition parameters depend on the target depth and the desired resolution. In the following section, we discuss the acquisition geometries for active and passive tests at the intermediate scale of engineering applications for near-surface site characterization.

Designing a surface wave survey involves choosing the different elements of the experimental setup. The acquisition geometry is particularly important for its effects on the acquired data. The choice of the instruments (sources, receivers, and recording system) has a large impact as well. Often it imposes constraints because it is conditioned by the available instrumentation, by logistics, and by budget compatibility.

The term “acquisition geometry” usually indicates the space sampling of the wave field. It is characterized by the geometry of the receiver spread; the number and the position of the receivers define the total size of the array and the receiver spacing. In active tests, the position of the source, with minimum and maximum offset, is also a crucial parameter. The time sampling is defined by the sampling frequency and by the length of the records, or the time window.

In the next section, data acquisition on land is discussed. Surface waves at the seafloor in shallow water can be used with very similar approaches and methods. Some notes on the acquisition in this context will be given in Chapter 8.

3.4.3.1 Acquisition layout for active tests

An active surface wave test implies the use of a receiver array and of an active source. The most used scheme involves a linear array of evenly spaced receivers with in-line sources. The wave field is generated at the source location and is sampled along a radial line. The position and the direction, or azimuth, of the receiver line should be designed considering the local geology and trying to avoid known or likely lateral variations and dips, topographic changes, as well as surface or buried obstacles such as foundations or underground facilities. The acquired dataset will be a gather of traces with a constant spacing along the radius.

The acquisition parameters have to be designed depending on the aim of the test (i.e., the desired investigation depth and the resolution required for shallow layers). The parameters that have to be designed are the array length L and the receiver spacing ΔX, the source offset, and the time sampling parameters (Figure 3.40).

The *receiver spacing* is the spatial sampling frequency and affects the maximum wavenumber, which corresponds to the minimum wavelength. In turn, the shortest wavelength directly affects the possibility of resolving shallow layers, which can be important in some applications. As discussed in Section 3.3, the maximum wavenumber can be extended to $1/\Delta X$, which means that the minimum wavelength can be as low as the receiver spacing. In shallow applications, with spacing of a few meters, the limit often comes from the attenuation of the high frequencies and not from the sampling. Typically, values from 1 to 5 m are used for soil characterization.

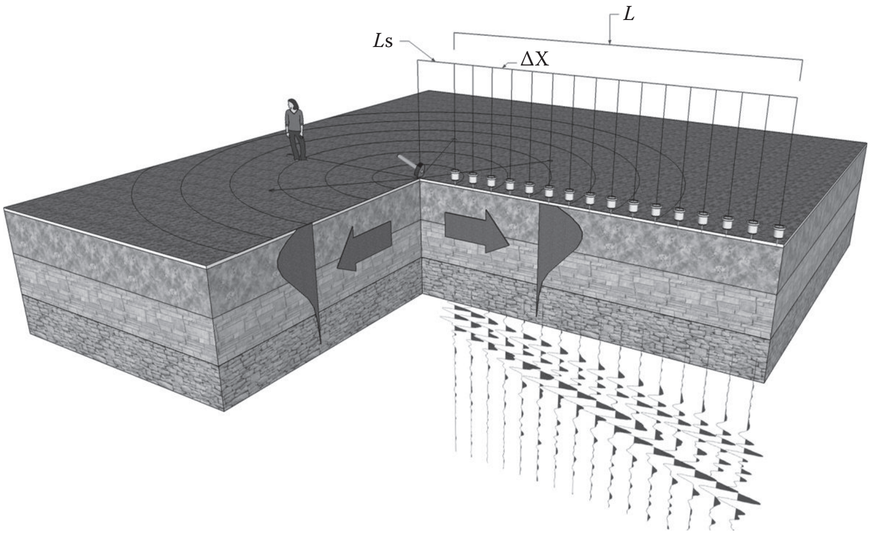

Figure 3.40 Schematic view of the active seismic acquisition—a linear array of receivers deployed radially from a surface source. The key geometrical parameters are indicated in the Figure.

The *array length* is important for many reasons. Provided that the signal is recorded with a sufficient SNR over the whole set of traces, a longer array provides a higher spectral resolution. Consequently, the mode separation possibility increases as does the accuracy of the identification of long wavelengths. If in theory there is no limitation to the maximum wavelength that can be observed with a given length, in practice the uncertainty increases as the ratio between the wavelength and the array length. It is important to stress that the effective length is limited by the SNR. If the source is not powerful enough, a long array can be counterproductive. Moreover, there is a trade-off between the improvement of the accuracy and the risk of lateral variations, which are more likely over a longer distance. Typical lengths are in the range from 20 to 100 m in shallow applications. The investigation depth is usually along the order of a half array length.

The *source offset* is the distance between the source and the closest receiver. The near-field effects create deviation from the linear wave hypothesis and can be considered as coherent noise. The distance affected by the near-field is a function of the wavelength, and empirical rules suggested in the literature indicate a half to one wavelength. Excluding near-offset traces will remove these effects but will also remove an offset range, which is valuable to estimate the high-frequency part of the wave propagation properties. The high frequencies might be too attenuated in the acquired traces and have too low a SNR to provide reliable information. Acquiring short offset data and removing the near offset from the low-frequency part are recommended, if necessary.

These parameters are also related and constrained by the hardware specifications. The number of available receivers relates to the spacing and total length of a single acquisition. The most common equipment for characterization at an engineering scale has 24 to 48 channels and uses light portable sources that can generate a wave field with sufficient energy up to distance of about 100 m.

The practical logistic limitations coming from the limited number of channels can be overcome using alternative schemes and geometries, but some processing approaches work only with evenly spaced arrays with end-off shots. Some acquisition procedures allow larger datasets to be collected by moving shots and/or receivers. Traces are subsequently sorted and assembled in a large seismic gather with some trace editing. Repeating a shot at the same location while moving the receivers allows one to acquire a virtually longer array (Gabriels et al. 1987).

If the source is repeatable and the receivers are moved along a line in two positions, as in Figure 3.37a, the final gather is equivalent to what would have been acquired with twice as many channels. The procedure can be repeated multiple times.

An alternative to varying the position of the receivers is the acquisition of a common receiver gather. A single receiver is deployed and left in place while the shot position is moved along a line at evenly spaced locations.

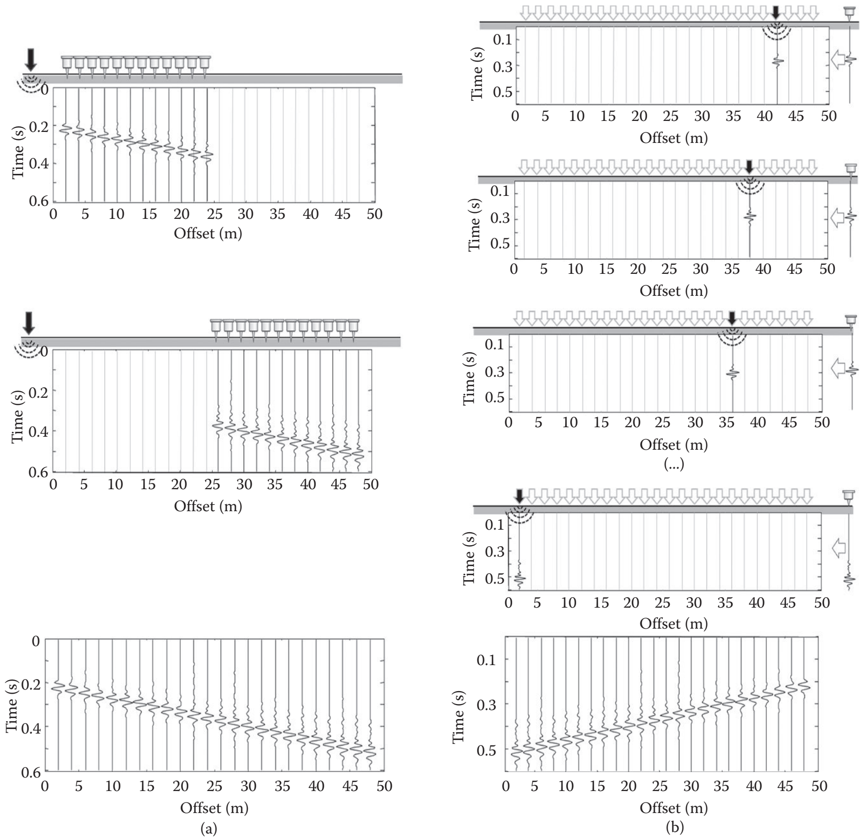

Figure 3.41 (a) Acquisition of a common shot gather merging two independent gathers, with the same shot position. (b) Acquisition of a common receiver gather with a single receiver.

The common receiver gather that is collected with this procedure (Figure 3.41b) for the reciprocity is equivalent to a common shot gather with the source at the receiver location. The repeatability of the source, and of the source phase in particular, is important for this type of acquisition.

Mixing multiple shot gathers, each with a plurality of receivers, could require more data processing. If the site does not present lateral variations, collecting multiple shot gathers with the same receiver array and sorting them with the offset could increase the aperture; this is the principle of the "walkaway test," as represented in Figure 3.42b. If the properties of the subsurface between the receiver array and the offset shot point are different from those under the receiver spread, the two gathers will not show a satisfactory phase continuity.

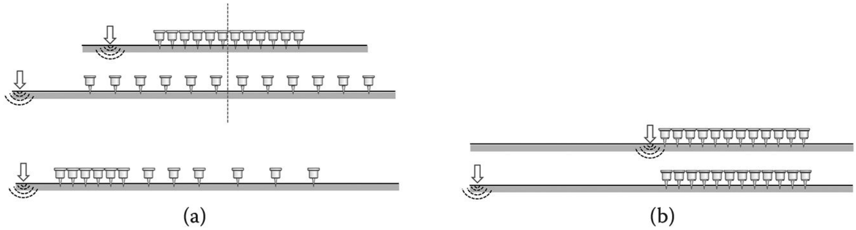

Figure 3.42 (a) Alternative acquisition schemes with irregular arrays, multiple arrays; and (b) mixing different common shot gathers.

Irregular spacing can finely sample the high frequencies in the near offset and can observe the low frequencies over a long distance. The processing of unevenly spaced arrays requires advanced signal processing tools (see Chapter 4). Merging arrays with different spacing and lengths requires more fieldwork and sometimes can be challenging at the data processing stage.

3.4.3.2 The two-station method

The "two-station" acquisition method is the standard approach to the measurement of surface waves in the so-called spectral analysis of surface waves (SASW) test (see Section 4.3). It can be seen as a particular case of an active multichannel test with only two channels. All the previously mentioned considerations for seismic acquisition also apply to this particular case, with spatial sampling issues even more relevant. Quite popular in the geotechnical community, the two-station method tries to mitigate the strong limitations of spatial sampling with complex field operation. A single pair of receivers is used, and it has to be moved during the test to sample different wavelengths.

The distance between the source and the closest receiver is usually equal to the receiver spacing. For a given spacing, only a limited range of frequencies can be obtained due to spatial aliasing, attenuation, near-field effects, and so on. The configuration is changed, often changing the source as well, to construct the dispersion over a wide frequency range. Short receiver spacings and light sources are used for the high frequency (short wavelength), while wide receiver spacings and heavier sources are used for the low frequency (long wavelength). The set of values of the receiver spacing is designed considering the need of an overlap among the different ranges. Different schemes have been proposed to change the configuration. The common receiver midpoint scheme (Figure 3.43) is considered the most reliable for sampling the region with forward and reverse shots more homogeneously, which mitigates the uncertainties related to noise

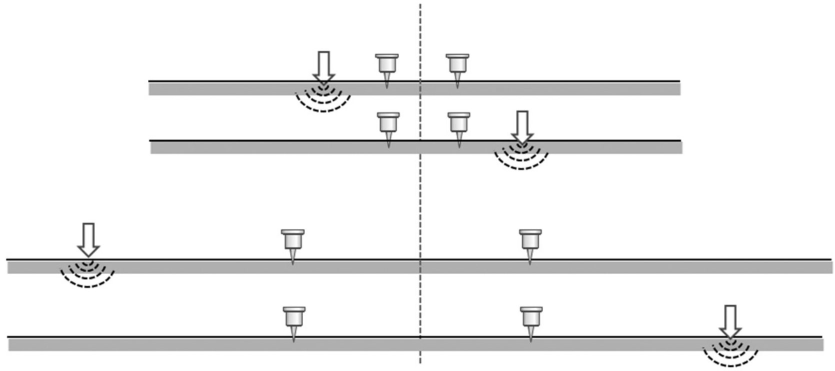

Figure 3.43 Two-station SASW acquisition with the common receiver midpoint scheme and forward and reverse shots.

and epistemic errors. A common source scheme in which only the receivers are moved is sometimes preferred for logistic considerations, especially when using heavy and difficult to deploy sources.

Reversing the source location compensates for phase distortions of the receivers and can help in identifying the effect of coherent noise. The effect of multiple modes, lateral variations, and other coherent noise types is more critical with two receivers than in multichannel shot gathers.

3.4.3.3 Acquisition of passive surface wave data

As discussed in the previous section, part of the ambient noise is a wave field generated by a number of sources of natural and human origin acting in the surroundings of the site (Bonnefoy-Claudet et al. 2006). In the low-frequency range, conventionally below 1 Hz, the noise sources are natural and mainly related to global geophysical events, and in particular to ocean waves. At a high frequency, conventionally above 1 Hz, the noise wave field is mainly generated by human activities (road traffic, industrial activities) even if atmospheric elements can largely contribute to the background vibrations and dominate the noise wave field in remote, quiet areas.

The ambient noises are often referred to as microtremors. The component of interest in shallow applications is the short period microtremor ($T < 1$ s, $f > 1$ Hz). The nature of this wave field makes its use particularly interesting in surface wave testing. Indeed the ambient noise wave field is generally dominated by surface wave components (Rayleigh and Love), with limited contributions from body wave propagation. The techniques based on the analysis of the ambient noise are called passive because of the absence of an active source.

At particularly noisy sites, such as in urban areas, the seismic background noise that degrades active seismic data can be seen as signal for the *passive* method.

In some frequency ranges, in particular at low frequency, the ambient noises can have stronger surface waves than those that can be generated using light sources at the surface. It is therefore interesting to extract these surface waves from the ambient noise and analyze them to identify their propagation properties. In near-surface applications, an efficient strategy is to combine active and passive data to increase the frequency range and therefore the investigation depth (see examples in Chapter 7).

The analysis techniques and the survey geometries used for active data cannot be directly applied, essentially because of the random component of the sources generating the wave field. In fact, the key difference in requirements affecting the geometry is related to the fact that the position of the source is not known and may change during the acquisition and for different frequency ranges. For a far source, the wave field is described by a value: the vector wavenumber k, with two components k_x and k_y. A linear array would detect the projection of the wavenumber along the array axis, obtaining an apparent velocity along the spread direction (see Section 4.6.3).

Hence, a 2D array is preferred, and the processing techniques have to be able to detect the wave parameters in two-dimensions. Details on the processing of passive data are presented in Chapter 4. Some considerations on the array are given in the following section. The performance of an array in enabling the estimation of the phase velocity depends on the array geometry and on the properties of the wave field. There is no universal agreement on the array design. The total array aperture, or diameter, is related to the maximum wavelength of interest; it is recommended to use an aperture from one-third of the maximum wavelength (Tokimatsu 1995) to a more conservative one wavelength (e.g., Asten and Henstridge 1984). In all directions, the station spacing should be designed considering the spatial aliasing— therefore being half of the minimum wavelength of interest.

The resolving power of an array depends on the maximum size and on the distribution of the receivers. The array performance can be evaluated using the theoretical array response function.

The array microtremor methods use a limited number of vertical receivers arranged in a 2D array. Common configurations are L-shaped and T-shaped, crosses, squares, triangles, and rectangular or hexagonal grids. The choice of the array also depends on the analysis method that is then used to process the data. A common choice for spatial autocorrelation (SPAC) processing is the triangle array with several nested equilateral triangles. This provides good results with a small number of geophones (Figure 3.44). The choice of the geometry is also constrained by the available equipment. Wireless receivers allow more complex and flexible geometries to be deployed.

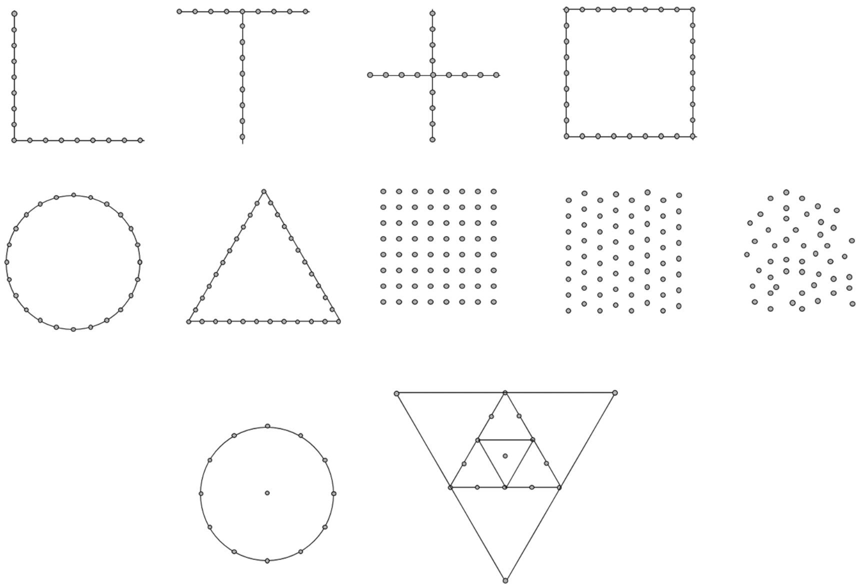

Figure 3.44 Examples of array configurations for passive surface wave tests.

The data acquisition is then fundamentally different from the active data acquisition. Ambient noise is recorded for a length of time sufficient to obtain a congruent number of data-segments. The acquisition parameters in time domain are set as a function of the desired frequency resolution and the maximum frequency of interest. The data segmentation is performed during the data processing, but often the maximum record length of the recording instrument will require recording multiple 30 s or 1 min gathers.

Interferometric techniques, with in-field data processing, can help in showing the improvement of the coherency of the events during the acquisition. The basic principles of passive surface wave interferometry are presented in Chapter 8.

The use of linear arrays for passive measurements has some appealing advantages from a logistic viewpoint. They allow using the same spread deployed for active tests, thus avoiding extra field operations, and allow the use of simple processing approaches. These are the main reasons for the popularity of the technique called refraction microtremor (ReMi), which indeed uses a linear array. It can be shown (see Section 4.6.3) that, in the case of a uniform and isotropic noise wave field, the response of the array can detect the real wavenumber.

However, the limitations induced by the linear array are severe; the linear array cannot identify the vector wavenumber but only its projection along the array direction. The method allows simple field operations but has limitations that can make the data of scarce value.

3.5 EQUIPMENT

The equipment for the acquisition of the surface waves has to record the wave field over the desired frequency range without distortions. The acquisition of seismic data requires three main functional elements: a source, a set of receivers, and an acquisition system. The positioning system is an important part of seismic acquisition. Knowing the exact location of sources and receivers is a crucial step in obtaining reliable velocity information. However, in near-surface surface wave testing, usually very simple surveying techniques are sufficient (e.g., measuring receiver positions with a tape measure).

The small acquisition devices for shallow applications are rather simple systems with a light source to input the energy in the ground. A set of receivers detects and transduces the ground motion into electric signals, which are transmitted as analog electric signals to a seismograph that digitizes and records the data. These are the key elements of the measurement chain, and their characteristics affect data quality.

Secondary components of the measurement chain, such as the data cable, trigger circuit, and control system, can also affect the data quality. They are important for the system's performance in terms of robustness and flexibility and for the efficiency of the field operations. A schematic view of the equipment for surface wave testing is represented in Figure 3.45. In the following section, we will discuss the main characteristics of sources, receivers, and seismographs.

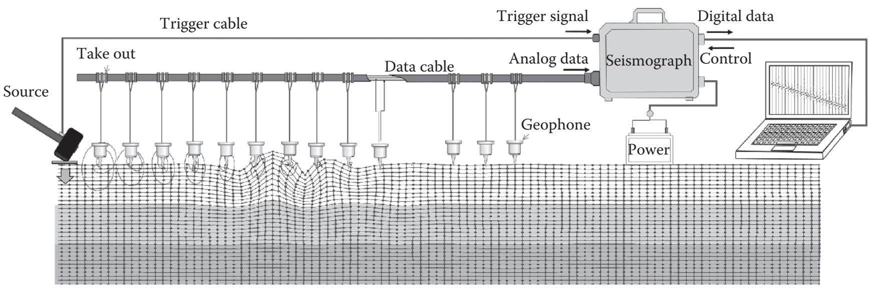

Figure 3.45 Schematic view of the field equipment. The essential elements are a source, receivers, and a digital acquisition device (usually a seismograph).

3.5.1 Sources

A seismic source is any device that releases energy into the Earth by generating seismic waves. In general, a seismic source should be able to generate high-amplitude signals without generating much coherent noise. For surface wave acquisition, the source should generate surface waves with enough energy to guarantee a sufficient SNR over the desired frequency range and offset range. In other seismic applications, a short compact wavelet is desirable. In surface wave acquisition, a wide frequency range is the objective. The challenge is often in having sufficient energy at the low-frequency end of the spectrum to increase investigation depth. The key parameters in the choice of the source are the energy and the frequency content. Logistical aspects can be very important—the site preparation, the cost, the source repeatability, the cycle time between shots, the environmental damage, the safety requirement, and the need for authorizations.

The comparison between sources is not straightforward because the source radiation depends upon environmental conditions at the surface and below the surface. A major factor to consider in surface wave data is the modal amplitude response. The amplitude spectrum of surface wave data largely depends on the subsurface. Even with a perfectly flat source spectrum in a lossless medium, the amplitude spectrum will show peaks and resonances and frequency ranges with high-amplitude attenuation.

Considering the data acquisition on land, sources can be classified as impulsive sources and vibrating sources. In shallow surveys, the most commonly used seismic sources are impulsive sources: hammers, sledgehammers, weight drops and accelerated weight drops, seismic guns, and explosives (Figure 3.46). Vibrating sources, standard in the exploration applications of the seismic methods, are used less in near-surface surveys. Vibrating sources include small controlled electromechanical vibrators, mini-vib, vibroseis, and Sosie™.

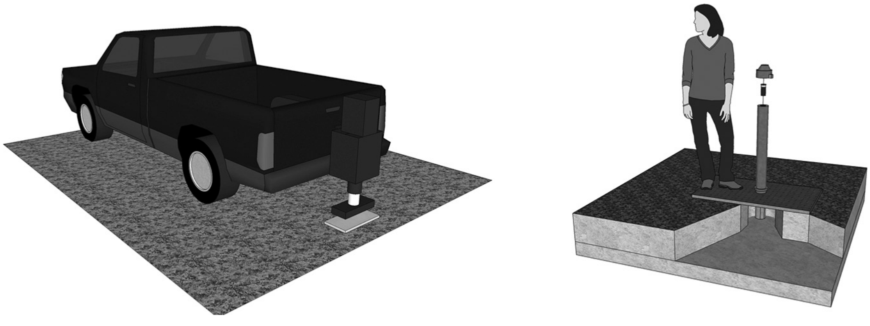

Figure 3.46 Weight drops, accelerated weight drops, and seismic guns are seismic sources used in surface wave data acquisition for near-surface characterization.

3.5.1.1 Impulsive sources

Impulsive sources input the energy into the ground with a short force application. The duration of the pulse directly affects the frequency content, as anticipated by Lamb (1904). Common impulsive sources can be chemical-based explosives (explosive, blasting caps, seismic guns, etc.) or mechanical (sledgehammer, weight drop, piezoelectric in holes).

Despite the number and variety of available seismic sources, probably the most used impact sources in shallow surveys are simple *sledgehammers* (weighing 1–15 kg) striking on metal or plastic plates. Indeed, a sledgehammer is easy to carry and use, and it has a low purchase and operational cost. It is easy to be triggered with an inertia switch on the hammer or with an electrical contact between hammer and metal plate. The triggering system is reliable and allows enhancement by in situ stacking (as has been shown earlier, the SNR increases with stacking with the square root of the number of shots).

The velocity at which a sledgehammer hits the plate can reach 15 m/s, depending on the hammer weight and on the force of the operator. Tests have shown that the impulsive dynamic force can exceed 20 kN. The weight of the hammer influences the frequency content of the generated pulse. A light hammer is preferable for the generation of high frequencies. The use of a metallic plate is recognized as increasing the frequency of the signal as well. On stiff surfaces, such as a rock outcropping or pavement, a very high frequency can be produced (more than 1000 Hz) with no need for striking plates. In most geological conditions and ambient noise levels, a sledgehammer can input sufficient energy for a 50–100 m array. Using a sledgehammer, there is virtually no site preparation or environmental damage.

Another popular source in shallow applications is the *weight drop* and the *accelerated weight drop*, also called thumper. It consists of a mass that is lifted using a winch or a piston and then released or accelerated to the ground. The mass can vary from a few kilograms to several tons. The height can vary as well, from less than a meter to tens of meters. Accelerated weight drops accelerate the falling mass to a larger velocity than that due to gravity by using different systems—springs, industrial elastic bands, compressed air, or gases. The velocity of the mass at impact can be increased without increasing the height, the dimension, and the portability of the systems and the cycle time between shots. The energy of a weight drop is related to the potential energy of the mass at the firing height; although for accelerated weight drops, the energy is increased by the work of the accelerating system during the fall of the mass. The use of a metallic baseplate usually increases the frequency content of the source.

Homemade systems can be triggered by circuit closing (when a metallic plate is used with an inertia or piezoelectric switch) or with a starter geophone. With homemade systems, the cycle time between shots can be slow, the movement in the field can be affected by logistic constraints, and the safety of the operation has to be considered.

In Figure 3.47, a single shot with a freefall weight drop of 130 kg from 3 m is compared with 20 stacked sledgehammer shots. Even if the overall SNR looks comparable, the spectral analysis reveals more energy at low frequencies in the weight-drop gather.

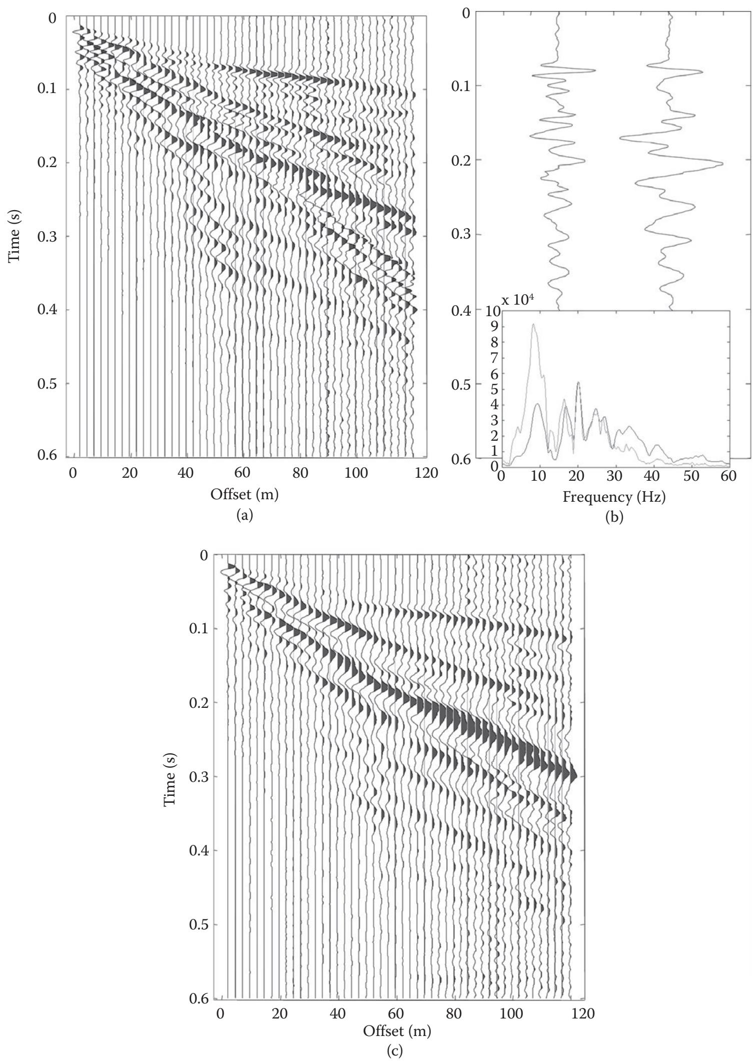

Figure 3.47 Comparison between (a) a vertically stacked sledgehammer acquisition and (c) a weight-drop acquisition. (b) Two traces at the same offset from the source are also compared in terms of spectral amplitude.

Other impulsive sources are *seismic guns*, ranging from modified rifles to iron water pipes with a coupler and nipple chamber on one end. Seismic guns typically input the energy by firing industrial cartridges into a shallow hole in the ground. The energy is greater than that of a sledgehammer, and the frequency content is similar. The use of seismic guns requires some preparation; after the purchase cost, the operation cost is essentially just that of the cartridges. The site preparation is quite simple, but a shallow hole has to be dug. Seismic gun operation requires an authorized operator and may cause noise and disturbance, especially in an urban context.

Explosives have been the primary source for land seismic surveys for decades, providing a scalable high-energy and broadband source. A small explosive charge can be placed in a shallow hole and detonated. Fast explosives are recommended for shorter pulses. The charge can be scaled according to the needs and to the local conditions and wave attenuation. The use of explosives is generally restricted and subject to specific licenses and is prohibited in many areas. The induced vibrations can create a nuisance and damages to structures and buried utility lines.

Besides the need for authorization and a licensed explosive handler, the necessity of drilling shot holes makes the speed of the operation slow and the cost high. Even in large-scale operations in desert areas, vibrators are used to replace explosives whenever possible. An interesting point regarding surface wave acquisition is that the depth of the source affects the energy distribution on the modes.

Figure 3.48 shows the seismograms and the spectra acquired at the same site with the same receiver array using two different sources. Figure 3.48a shows the seismogram acquired with a sledgehammer. Figure 3.48b shows the seismogram acquired with 100 g of explosives in a hole (1 m deep). The corresponding spectra for traces at a given offset from the source are shown in Figure 3.48c.

In large-scale surveys, several kg of explosives are placed in holes drilled to a depth of up to 20 m. Source studies show that one kg of explosive at a depth of 15–20 m produces roughly the same amplitude as a 450 kN vibrator with a 20 s sweep or a 20 ton weight dropped from 20 m. Exploration data can show surface waves up to very long offsets. Figure 3.49 shows a shot record with surface waves dominating up to an offset of 1500 m.

Marine sources will be briefly discussed in Chapter 8. In some cases of near-shore surveys, shots on land can be used with receiver arrays placed along the seafloor.

3.5.1.2 Vibrating sources

Impulsive sources input the energy into the ground with a short pulse of pressure lasting a few milliseconds. Therefore, the energy density in time (the source power) is high. An alternative approach is the use of longer controlled signals, using vibrating sources, with a low energy density.

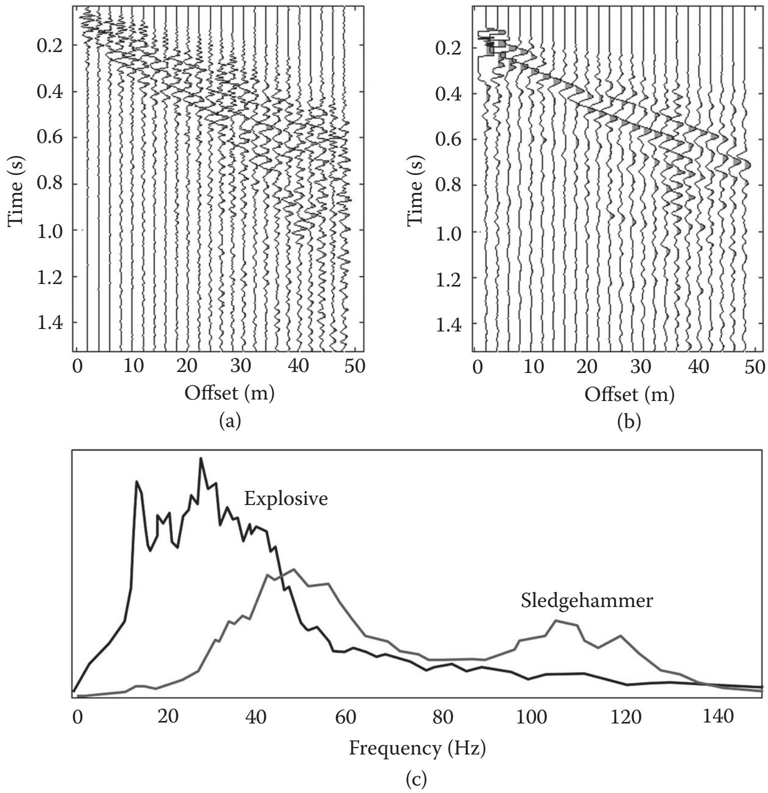

Figure 3.48 Seismogram comparison between a sledgehammer (a); an explosive in a shallow hole (b). The difference in frequency content is evident in the time signals and is confirmed by the corresponding spectra (c).

The typical example is the seismic vibrator, commonly used in the majority of land seismic reflection surveys. The energy is input using a controlled force signal lasting several seconds.

Nonimpulsive sources are called swept, controlled, or vibrating sources. They range from portable electromechanical shakers with a weight of less than 100 kg to large track-mounted hydraulic vibroseis weighting tens of tons. The principle of the vibrator is illustrated in Figure 3.50. A base plate is kept in contact with the ground by a hold down passive mass (the weight of the device or of the vehicle). An actuator imposes movement to a reaction mass, and the corresponding force is transmitted to the base plate. The schematic mechanical system is represented in Figure 3.50b.

The main mechanical parameters include the mass of the reaction mass M_r and of the base plate (M_b), the force of the actuator on the piston F_a,

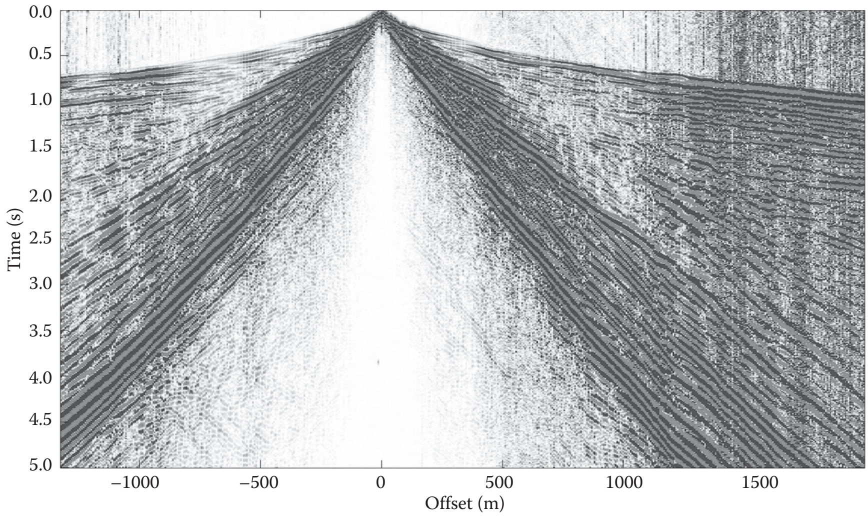

Figure 3.49 Explosive gather on a long spread.

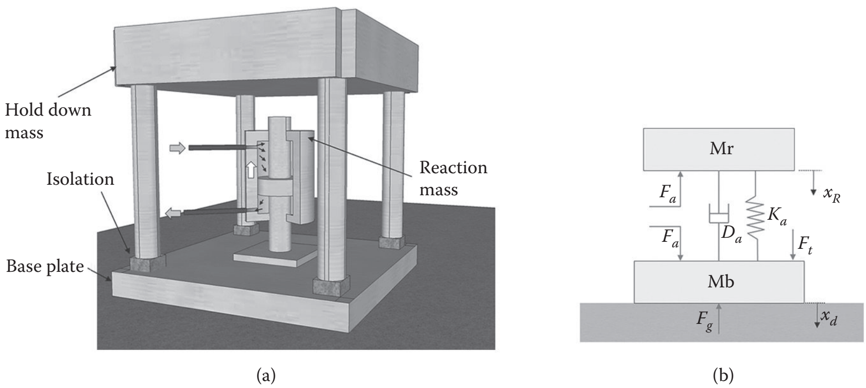

Figure 3.50 (a) Scheme of the vibrating source. (b) Representation of the mechanical system describing vibrating sources.

the force of the ground on the baseplate (and the opposite force of the baseplate on the ground), the force of the hold down mass F_t (often the frame of a vehicle) applied on the base plate through insulators, and the damping D_a and stiffness K_a of the actuator. The equation of motion for the two masses, considering the two masses as rigid, can be written as

$$-F_a = M_r\ddot{x}_r + D_a\left(\dot{x}_r - \dot{x}_b\right) + K_a(x_r - x_b) \tag{3.38}$$

$$F_a + F_t - F_g = M_b \ddot{x}_b + D_a(\dot{x}_r - \dot{x}_b) + K_a(x_r - x_b) \tag{3.39}$$

where x_r and x_b are the vertical displacements of the reaction mass and of the base plate, respectively. By summing Equations 3.38 and 3.39, it is possible to write

$$F_t - F_g = M_b \ddot{x}_b + M_r \ddot{x}_r \tag{3.40}$$

If the baseplate is isolated dynamically from the hold down mass, the dynamic force F_t is negligible and the force transmitted to the ground is equal to the weighted sum of inertia forces associated with the accelerations of the baseplate and of the reaction mass.

Small portable devices with a total weight of less than 100 kg can deliver forces up to 0.5 kN, while large-scale vibrators for seismic exploration have a peak force up to 400 kN. The maximum performance is actually a function of the frequency. Different factors limit the peak force in different frequency ranges. In the low-frequency range, there are the two factors limiting the energy that can be input into the ground: the peak-to-peak stroke of the reaction mass is the first limiting factor, at very low frequency, and the decoupling force. The boundary between the two depends on the magnitude of the reaction mass. A heavy vibrator with a high hold down weight, a large reaction mass, and a long stroke is best suited for high force at low frequency. In the high-frequency range, for hydraulic vibrators, the limit is usually related to the servo-valve bandwidth and the baseplate flexibility.

Quantification of the input source force is essential for some data processing techniques. For instance, the transfer function of a site can be obtained from the ratio in frequency domain between output (measured ground motion) and input (source force) (see Section 5.3). The source can be characterized using an accelerometer mounted on the frame.

Vibrating sources are used with different input signals. Long monochromatic signals guarantee a high SNR, but long operational times are required to sample the whole frequency range of interest. Sweeps and chirp signals are standard practice in reflection surveys. The advantage of the sweep for surface wave acquisition is that it covers the whole frequency range of interest in a single signal, and standard data processing can be applied.

3.5.1.3 Sweep signals

A sweep signal is a nonstationary function that can have the general form

$$A = A(t)\sin(2\pi f(t) + \varphi) \tag{3.41}$$

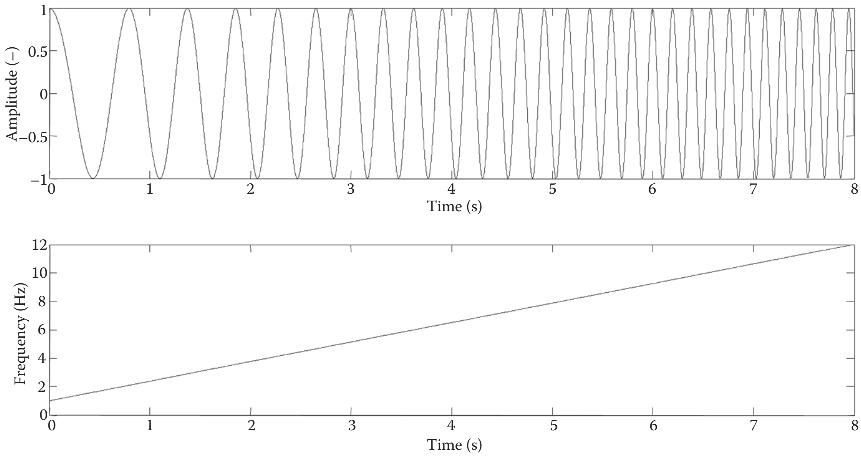

Figure 3.51 Example of a linear sweep and its time–frequency representation.

The frequency is therefore not constant but varies with time. Linear sweeps use a linear function for setting the frequency

$$f = c \cdot t + f_0 \tag{3.42}$$

The frequency can increase with time (upsweep) or can decrease (downsweep). Upsweeps, such as the example in Figure 3.51, are preferred in reflection surveying because the correlation ghosts appear at an earlier (often negative) time and have less interference with the primary reflections.

With such a frequency function, the different frequencies are differently sampled (i.e., a different number of periods are recorded for each of them). To better sample the low frequencies, logarithmic or quadratic sweeps are preferable. The use of a sweep generates a signal with a relatively low amplitude compared to the impulsive sources. The correlation can be used to create a compact signal. The random noise is not correlated and is therefore filtered out.

The principle of the correlation is illustrated in Figure 3.52, where an input sweep is reflected by three reflectors (impedance contrasts in the subsoil). The uncorrelated time signal shows interference among the different shifted and scaled copies of the sweep. The correlation process acts as deconvolution, producing a series of copies of the sweep autocorrelation at the position of the spikes.

The recording time should be equal to the sweep length plus the listen time. The correlation produces a record that has full overlap of the sweep onto the recorded signal only for the duration of the listen time. Longer lags and negative times have partial overlap and a limited frequency content. The total energy is proportional to the duration of the sweep. Long sweeps

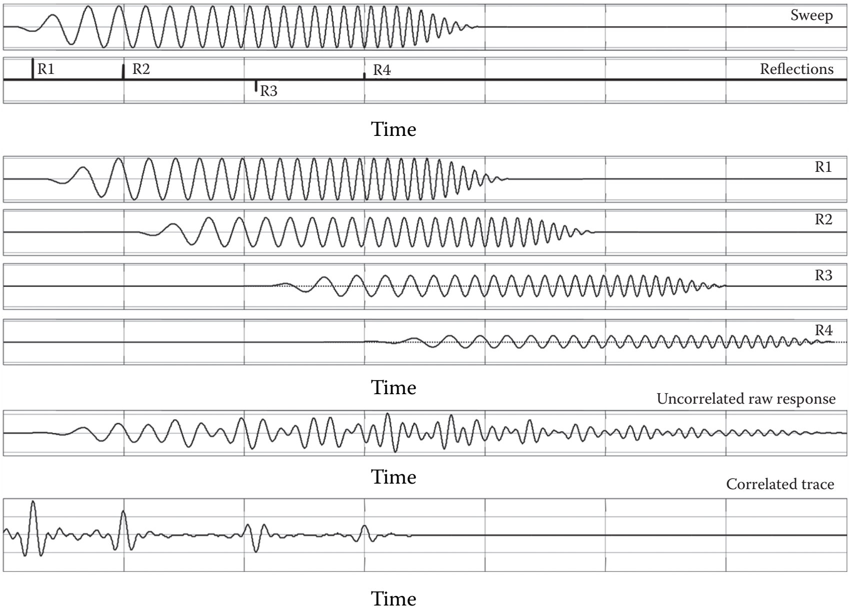

Figure 3.52 Principle of sweep correlation. The signal is the result of the convolution of the source signal with the Earth reflectivity series. It contains the superposition of shifted and scaled copies of the source signal. The correlation procedure decodes the recorded trace.

can be used for a higher SNR; the sweep rate in Hz/s affects the energy density in the different frequency ranges.

When acquiring data with controlled vibrating sources, the visual inspection of field records in time-offset domain can be challenging. The ambient noise over its full spectrum can create large amplitude vibrations, larger than those of the monochromatic signal. In the example shown in Figure 3.53, the raw gather is plotted on panel a and the filtered gather is plotted on panel b.

Alternatives to simple impulse and controlled sources are repetitive impulsive sources as the Sosie mini-Sosie, and Swept Impact Seismic Technique (SIST). These systems use impactors hitting the ground 5–15 times per second in a signal that can be 3–5 min long, with a pseudorandom series. A process similar to the correlation reconstructs a single impulse with high energy, compressing the series of pulses.

3.5.2 Receivers

The receivers are the first element of the recording chain. They transduce the ground motion into a measurable electric signal that is then transmitted, conditioned, and recorded by the acquisition device.

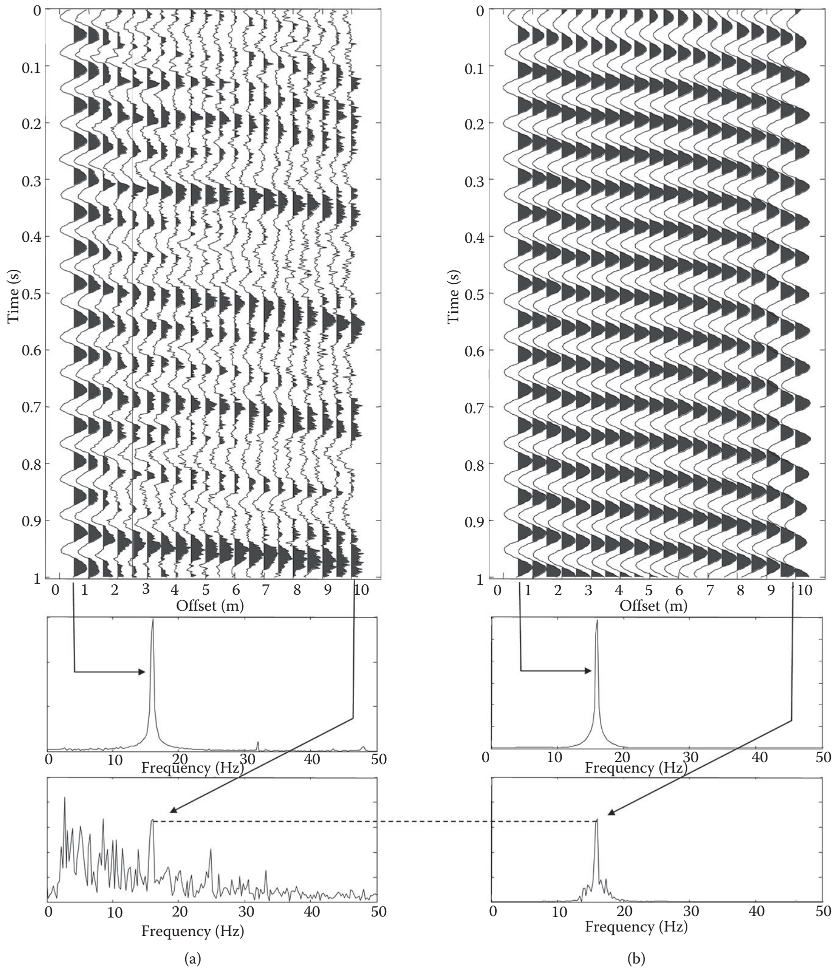

Figure 3.53 Harmonic multichannel record. (a) Raw data and (b) filtered data. In the far offset, the filtering enhances the coherency of the 16 Hz component.

The ground motion, on land, can be described in terms of particle displacement, velocity, or acceleration. The different sensing devices essentially can measure one or more components of the particles' velocity or acceleration, using a moving mass connected by a spring to the motion of the point to be measured. A soft spring allows the mass to stay in its initial position, providing a reference for measurement of displacement or velocity. A stiff spring causes the mass to move, with a residual displacement related to the acceleration. The mass motion, or the stress or strain, is measured using different physical principles.

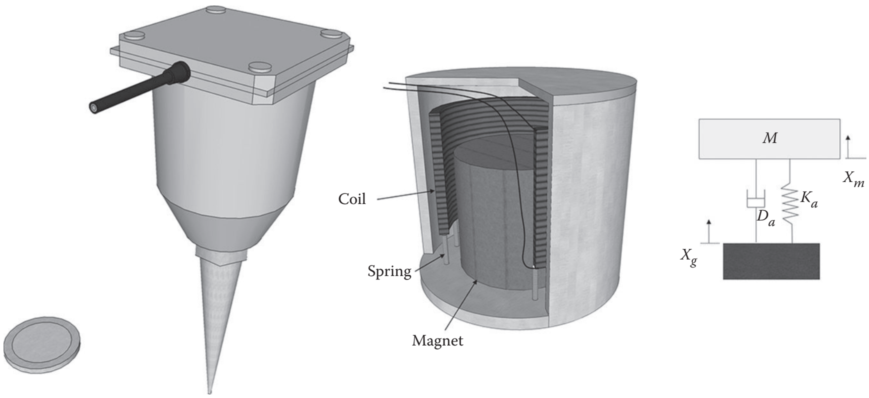

Figure 3.54 A geophone is an electrodynamic velocity transducer. Most geophones are of the moving coil type: a coil is suspended in a magnetic field and its oscillations generate a current.

In most shallow engineering surveys, the receivers are velocimeters called *geophones*. In some applications, especially with passive methods, the need to record a low frequency requires the use of very low-frequency geophones, called *seismometers*. In nondestructive testing, in pavement testing, and other applications where higher frequencies are needed, and often high-amplitude vibrations are generated, the use of *accelerometers* is preferred. Recently, the introduction of high-fidelity *microelectromechanical systems* (MEMS) digital accelerometers is also making accelerometers appealing for low-frequency applications (see Figure 3.54).

3.5.2.1 Geophones

Geophones are velocimeters (i.e., electrodynamic velocity transducers). Typically, they are of the moving coil type. A small coil is suspended by a spring in a magnetic field produced by a permanent magnet fastened to the casing. A similar system is obtained when the moving part is the magnet. The vibration of the soil causes a displacement of the geophone casing and magnet. Due to its inertia, the relative movement of the coil produces a small voltage proportional to the relative velocity. The axis of movement can be horizontal or vertical. When using a vertical geophone to transduce the vertical particle velocity, the tilt sensitivity has to be taken into account. Indeed the response is affected by the angle between the transducer and the vertical direction.

Ideally, the output of the geophone should be proportional to the input ground particle motion. In reality, a frequency-dependent transfer function, or response, is introduced by the geophone. It affects the transduced signal and the recorded seismic trace. The geophone can be considered as a single degree-of-freedom forced oscillator with mass m, a spring with stiffness k, and damping D. These mechanical parameters affect the oscillator response with the classical relationships of dynamics (see, for instance, Clough and Penzien 1993). The behavior of the oscillator can be described in terms of natural frequency and viscous damping ratio and by the complex receiver response function. The transfer response represents the ratio between the output and the input: it can be seen as a filter transforming the real particle motion signal into the recorded signal. The response function can be plotted as amplitude and phase as function of the normalized frequency (Figure 3.55).

Two main dynamic parameters have to be considered to understand the effect of the receiver on the recorded signals: the natural frequency and the damping. The value of the natural frequency, which is the resonance frequency of the oscillator, is important because it affects the minimum usable frequency of the transducer. The amplitude gets severely attenuated below the natural frequency. For example, a 40 Hz geophone will heavily attenuate the low frequencies used for surface wave measurements in shallow applications. For surface wave testing, geophones with a low natural frequency are typically used in order to obtain the desired

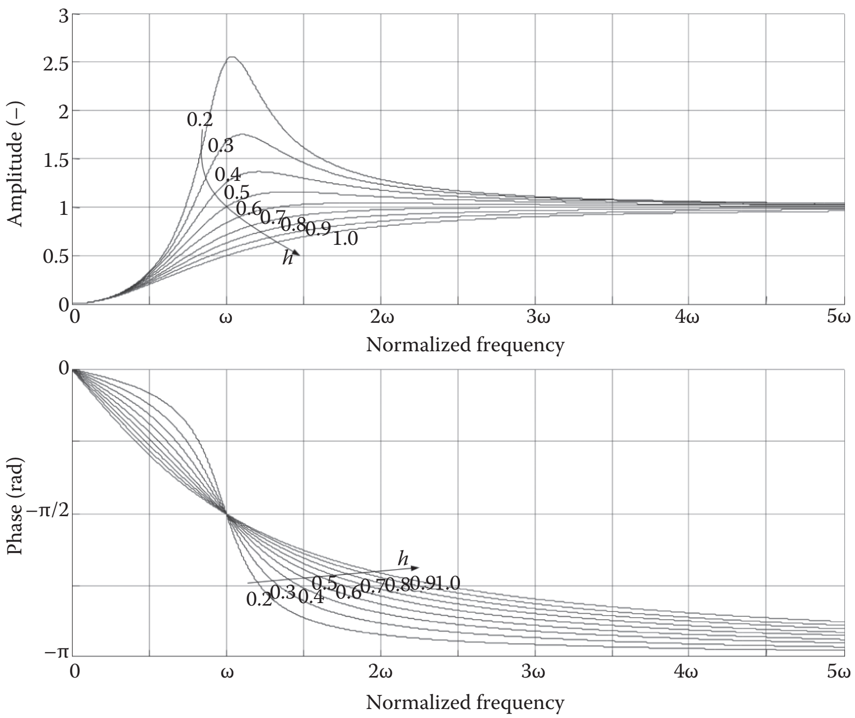

Figure 3.55 Amplitude and phase response of a geophone, as a function of the normalized frequency, for different values of damping.

investigation depth. Nevertheless, the trade-off between natural frequency and geophone roughness has to be considered. In order to obtain a low natural frequency, a large suspended mass has to be used. As a consequence, the instrument becomes heavier and less manageable in the field because the weight of the mass may easily damage the spring if the instrument is not carefully transported and deployed. Moreover, low-frequency geophones are more expensive. A compromise is represented by 4.5 or 2 Hz natural frequency geophones; 1 Hz geophones allow for deeper characterization but their cost is about one order of magnitude higher than 4.5 Hz geophones.

The effect of the damping on the response curve is important as well because it flattens the resonance peak at the natural frequency. It is recommended to tune the damping to get a flat response in the frequency band of interest. The damping of a geophone can easily be modified electrically by inserting a shunt resistor across the coil terminals. The current circulating in the coil passing through the shunt resistor produces a magnetic field opposing to the coil movement and damps it. The shunt resistor can be designed such that the damping factor is about equal to 0.6 to flatten the amplitude above the natural frequency as in Figure 3.55.

The sensitivity of the geophone can be expressed as the ratio of the output and the input in the flat-response frequency band. It is the output tension divided by the input particle velocity, and it is usually expressed in volt/mm/s.

In surface wave applications, the geophone response is critical for the amplitude response and the phase distortion.

In frequency domain, the recorded signal S_{rec} is the product of the complex spectrum of the true signal S_{true} by the receiver complex response function R

$$S_{rec}(f) = S_{true}(f) \cdot R(f) \tag{3.43}$$

Considering the amplitude A and the phase φ of the signal, the effect of the response of the receiver can be written as

$$A_{rec}(f) = A_{true}(f) \cdot A_R(f) \tag{3.44}$$

$$\varphi_{rec}(f) = \varphi_{true}(f) + \varphi_R(f) \tag{3.45}$$

The true amplitude is multiplied by the receiver response. This can attenuate the signal (and the noise) in certain frequency bands but does not influence the inferred phase velocity, which is essentially a function of the recorded phases. Geophones can also be used out of their flat band and below their natural frequency, obtaining attenuated signals. The recorded phase is affected by the receiver phase; the phase difference among multiple receivers, directly proportional to the wavenumber and affecting the velocity, is not affected by the receiver phase if the latter is constant on all traces. For this reason, using identical receivers is of fundamental importance to obtain good quality data for surface wave analysis.

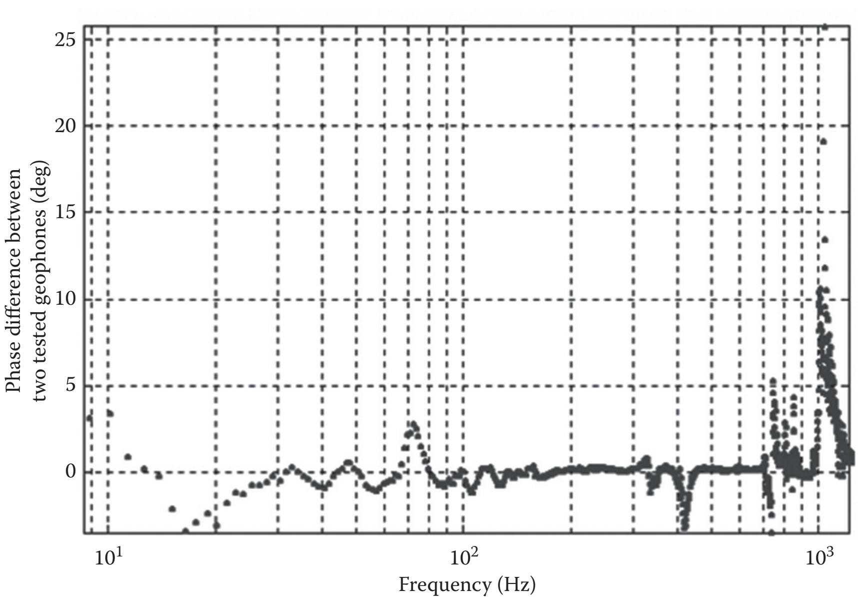

Figure 3.56 Real response of geophones measured on a shaking table: phase difference between two receivers.

The theoretical response curve describes the ideal response of the receiver. Real geophones have a different behavior at high frequency with spurious resonance noise due to undamped motion of the moving parts of the geophone along different directions. The real phase response is more relevant because the induced phase difference affects the estimated velocity. And the phase difference between two receivers can be large. Figure 3.56 shows the phase difference as a function of the frequency for two geophones tested on a shaking table (after Strobbia 2003). The phase distortion can become very large for weak excitation below half of the natural frequency.

A phase error of one degree can be relevant. The additional phase lag due to the receivers affects the phase velocity. For a two-receiver measurement, it can be computed as

$$V = \frac{2\pi}{\ } \ X \cdot f \tag{3.46}$$

where f is the frequency, $\Delta\varphi$ is the phase difference, and ΔX is the distance. The uncertainty on the velocity is

$$\sigma_V = \frac{2\pi \cdot \ X \cdot f}{\ ^{2}} \sigma \tag{3.47}$$

For example, a velocity of 600 m/s at 5 Hz and 5 m spacing would correspond to 15°. If the error is 5°, the obtained velocity is 900 m/s. Clearly, two-station measurements (see Sections 3.4.3.2 and 4.3) are the most sensitive to phase distortions, whereas the effects are mitigated and the estimate of phase velocity is robust in multichannel measurements.

The saturation can also be an important parameter. There is maximum input that clips the output for physical reasons. Indeed, the motion of the suspended mass is limited by the geometry of the casing. Saturation of the traces alters the frequency content and makes the experimental data useless.

3.5.2.2 Accelerometers and MEMS

In some applications, especially in nondestructive testing of pavement systems or structural elements, the high-frequency response is essential for estimating the surface wave components with very short wavelengths. Accelerometers can be more suitable for such applications. Conventional accelerometers are piezoelectric or piezoresistive. Piezoelectric accelerometers contain elements that are subjected to strain under acceleration. They rely on piezoceramic (such as lead zirkonate titanate) or single-crystal (such as quartz) piezoelectric elements to create a voltage. In general, they have an excellent high-frequency response. Piezoresistive accelerometers, often used in shock applications, are based on sensing flexure. They have a limited cost but typically a low sensitivity. Capacitive accelerometers can be used in servo mode and have a superior stability and linearity, but their cost is higher than the other accelerometers.

Recently MEMS sensors with the sensitivity, dynamic range, and low noise required for seismic acquisition have been developed. The term MEMS stands for microelectromechanical systems and indicates a combination of mechanical and electrical components built into very small devices, at the microscale. The first applications of MEMS were actually accelerometers. In principle, MEMS accelerometers could deliver a broader bandwidth and more accurate amplitude. One of the benefits is the possibility of recording the low frequency down to zero, hence also recording the gravity.

A typical MEMS accelerometer is composed of a moving proof mass attached through a mechanical suspension (e.g., a polysilicon spring) to a reference frame. The mass has radial fingers positioned between plates fixed to the frame. Acceleration causes deflection of the mass from its center position in one or more directions. The relative position of fingers and plates creates changes in the differential capacitance. The differential capacitance is measured electronically using modulation/demodulation techniques. A schematic representation of a MEMS accelerometer is reported in Figure 3.57.

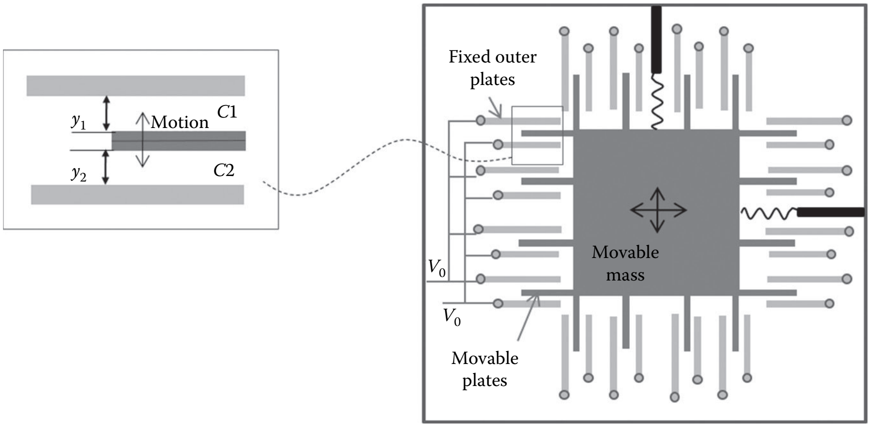

Figure 3.57 Scheme of a 2D MEMS accelerometer.

The mass of the moving part is as small as 1 μg, and the distance between the plates is of the order of 1 μm. Often, a single chip contains multicomponent MEMS accelerometers and measures the acceleration in the different directions. Besides the interest in multicomponent recording, it can also measure and correct the possible tilt of the sensor.

In acquisitions for deep exploration, the signals are digitized on the sensor or close to a group of sensors.

3.5.2.3 Receiver coupling and land streamers

The coupling of the receivers with the ground is a key element for obtaining good quality data. The receivers for land applications usually have steel spikes at the end of the casing. They are firmly planted in the ground or even planted and buried to minimize the local ambient noise. In some cases, base plates can be used, and the receiver's contact with the ground is assured only by its weight (for instance, on hard ground, where it is difficult to insert the spike).

An alternative to the use of receivers coupled individually to the ground is the land streamer (van der Veen et al. 2001; Vangkilde-Pedersen et al. 2006). The name comes from marine streamers, which are strings of hydrophones towed behind boats in marine seismic surveys. Land streamers consist of webbing supporting the data cable and the receivers, usually with plates for the coupling to the ground. Different designs, including wings (to avoid rotation), and different materials have been proposed. Compared to the marine acquisition, land data face more challenges related to the background noise, the coupling between receivers and ground, the cross-coupling between receivers, and the orientation of the receivers. Additional concerns can be the wear of the system and operational and logistic constraints. The coupling with the ground is less ideal than with planted or buried receivers, even if several comparisons show that similar data quality can be obtained in optimal conditions.

The streamer can be towed by a vehicle in a very efficient way in flat terrain or along roads. They can be very efficient in profiling, when the receiver array is moved by regular steps along a line, together with the source. Streamers for land applications can be short and have closely spaced geophones, offering a large increase in productivity.

3.5.2.4 Use of two-component receivers

The acquisition of multicomponent data can be necessary for some data-processing strategies. The polarization of the surface waves can be measured to estimate the contribution of different modes. It can also be used for joint or stand-alone inversion processes aimed at site characterization (see Chapter 8). In active surface wave acquisition, the motion components of interest are the vertical and the horizontal radial components. The polarization changes with the offset in real data. In Figure 3.58, the normalized particle velocity for nine two-component receivers and a harmonic source is plotted as a function of the offset.

With impulsive signals, the traces can show a more complicated hodogram, but often the ellipticity of the Rayleigh waves can be recognized. In Figure 3.59, the particle motion for a two-component acquisition with 20 evenly paced stations is plotted.

The particle motion can become complicated if multiple frequencies are present. Multiple frequencies, multiple modes, and the different excitability of vertical and horizontal components can create complex trajectories. The following example refers to a vibrating source generating a signal with a 7 Hz central frequency. The secondary peaks show different amplitudes on the vertical and horizontal components, and the combination of modes and frequencies creates complex but repeatable trajectories. In Figure 3.60, the hodograms represent cyclic signals following complex artistic paths.

3.5.2.5 Receivers for marine surveys

Geophones are usually sealed and can be used in moderately wet environments. When acquiring data in water, however, the use of hydrophones is needed. Hydrophones are normally piezoelectric devices, with a membrane sensing the pressure variations in the fluid. They have less coupling issues than the geophones and a wide frequency range in the response.

In seabed applications, sometimes geophones and hydrophones are combined in a cable (ocean bottom cable) or in nodes in which the digitalization is also performed (ocean bottom seismometers, ocean bottom nodes). In shallow applications, strings of hydrophones are the most common choice.

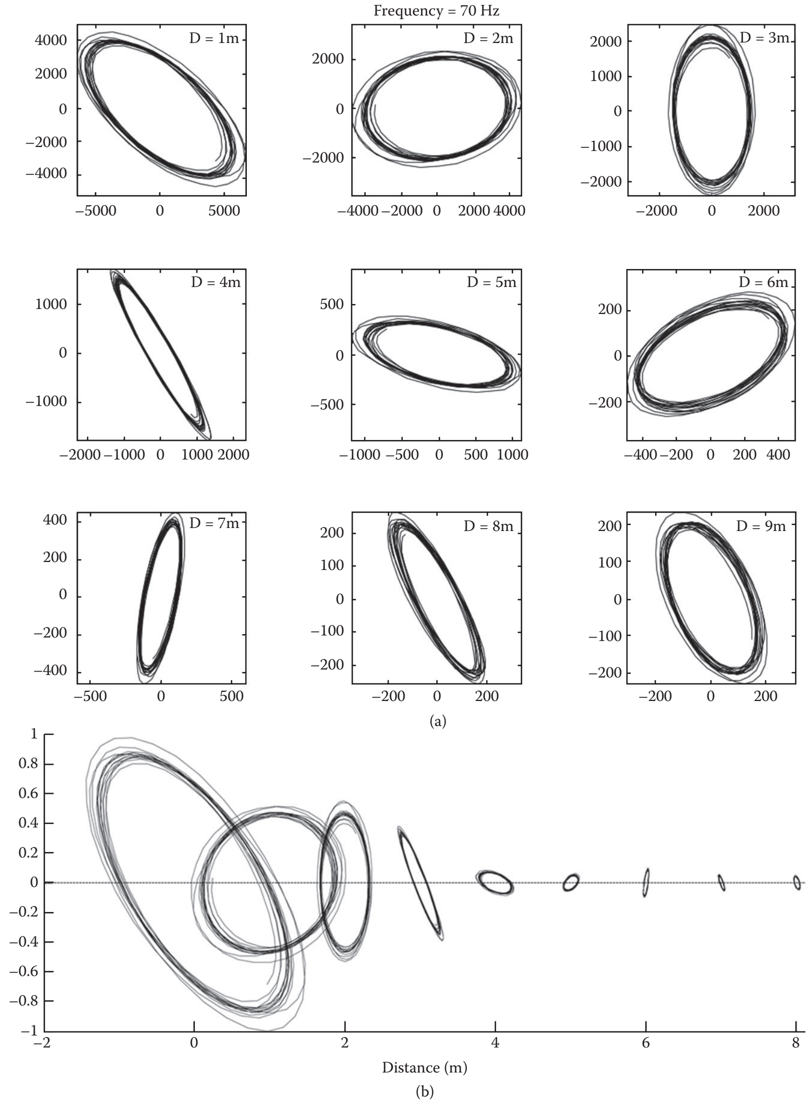

Figure 3.58 Particle velocity of the two components of in-line receivers': (a) normalized hodograms; (b) true amplitude hodograms as a function of source offset.

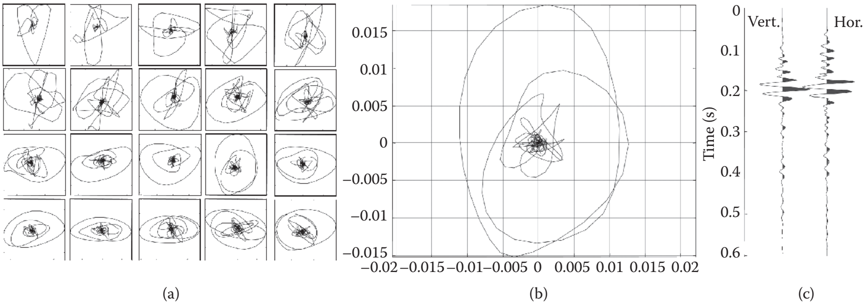

Figure 3.59 Hodograms of particle velocity for an impulsive source: (a) normalized hodograms; (b) example of hodogram at a given source offset; (c) vertical and horizontal velocity time histories.

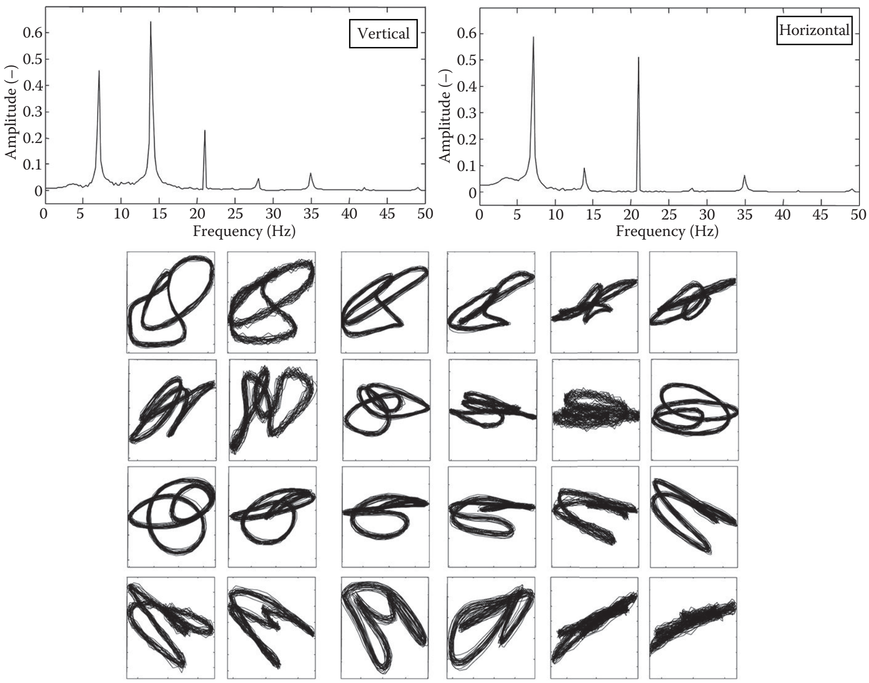

Figure 3.60 Amplitude spectrum of the horizontal and vertical component of the closest receiver (top). Normalised hodograms of the 24 recorder receivers (bottom).

3.5.3 Data acquisition systems

The main function of data acquisition systems is conditioning, sampling, and digitizing the signal generated by the transducers. This function can be performed by a central unit receiving the analog signals from the receiver spread or by different local units in a distributed system. In shallow applications, the first case is more common.

Different types of systems can be used in surface wave acquisitions. In two-station SASW field testing (Section 4.3), digital signal analyzers are typically used in the field, integrating the acquisition and the initial data processing. This choice may help in assessing data quality directly in the field, but usually signal analyzers are laboratory instruments, which are not designed for field operations. They require a power outlet and are not sturdy and waterproof. In multichannel acquisition, multichannel seismographs are the usual choice for field acquisition. These are multipurpose instruments specifically developed for seismic testing in the field; hence, they can be used in quite extreme conditions and are easily transported. Some of the preliminary data analyses can be performed in the field, for quality check purposes, although the processing is done at a later stage.

Seismographs are multichannel digital recorders for the acquisition and recording of the seismic signals generated by the transducers. The key parameters of a digital acquisition system are the number of channels, the fidelity of the digitization (the dynamic range, the distortion), and the allowed time sampling parameters (maximum sampling rate, the maximum record length).

Large-scale acquisition systems for deep seismic exploration have up to more than 100,000 channels. For near-surface applications, typical systems have 24 or 48 channels. Commercial systems are often modular and allow an easy expansion of the number of channels.

For passive surveys, a dedicated acquisition system is usually coupled to each receiver (usually a three-component geophone or seismometer). The instruments are then synchronized through the GPS signal or with other wireless systems. This option allows for an easy deployment of large aperture arrays, which are necessary for obtaining large exploration depths.

The key elements of an acquisition system for shallow seismic testing are those affecting the sampling, the digitization, and the data quality. The amplitude difference among Rayleigh waves at different frequencies and among different modes can be very high. The dynamic range of the seismograph affects the quality of data, the detectable frequency range, and the detectable modes. The dynamic range is a key characteristic of the analog-to-digital converter (A/D).

The A/D accuracy is measured in bits and describes the number of discrete values that can be produced within the input range. State-of-the-art instruments have 24-bit A/D converters, often based on sigma-delta oversampling, which corresponds to a 144 dB dynamic range. The resolution and quantization noise are discussed in Section 3.3.6.

The possible sample intervals are crucial in the choice of an acquisition system. They must allow the required sampling frequency and the maximum desired record length.

The minimum sampling interval can be as low as 10 microseconds on tens of channels. In surface wave acquisition for near-surface site characterization, usually a 1 or 2 ms sampling interval is sufficient because the analysis is performed in the frequency domain and the frequencies of interest are typically below 100 Hz (except for pavement testing where much higher frequencies are necessary). The maximum number of samples, combined with the largest sampling interval, gives the maximum recorded length, which is particularly important for passive measurements. The possibility of acquiring pretrigger data, based on circular buffering of the armed seismograph, enables the recording of data before the activation of the seismic source. This is necessary to evaluate the noise level at the moment of the acquisition and to enable synchronization of data in case of unreliable triggering. Moreover, a certain amount of pretrigger allows sampling windows different from the boxcar windows to be applied in order to mitigate leakage during the processing.

The analog filters are useful for some applications, but the built-in low-cut filter of some systems prevents them from acquiring low-frequency data (usually below 2 Hz). This feature is introduced in some systems to optimize the acquisition of seismic reflection survey, but it is unwanted for surface wave testing because it limits the investigation depth.

The bandwidth of the system indicates the frequency range where the system can record data reliably, and it obviously affects the measurable frequency range. Typical values range between 2 Hz and 20 kHz.

The system noise is measured in terms of system noise floor (crosstalk between channels). The system distortion usually is less than 10/million in the main seismic frequency range, and it is not critical for surface wave analysis.

Important parameters in coupling an acquisition system with receivers are the input signal range (in volts, as peak-to-peak maximum tension) and the available preamplifier gains. Indeed, the analog signals are usually amplified before their sampling and digitization. Automatic gains can affect the possibility of testing for surface waves at very short distances from the source and hence the possibility of obtaining very high resolutions close to the ground surface. This aspect may be critical for some applications, such as for pavement testing.

The trigger system and its accuracy are important properties for active data acquisition. In particular, high-accuracy triggering is necessary for stacking data in time domain. For surface wave analysis, stacking can be applied at a later stage of processing if single-shot data are recorded in the field. This strategy avoids the errors related to an asynchronous trigger. Accuracy of triggering is of paramount importance when multiple acquisitions with different source and/or receiver array positions are assembled to create a single large shot gather.

The possibility of recording an auxiliary channel is useful in controlled source acquisition. Indeed, monitoring of the input force is useful for the calculation of the experimental transfer function of the site (see Chapter 5).

The data transmission and data storage (internal or transmitted to the computer that is used to control the acquisition) can be important for the cycle time. The format of the saved data is usually one of the few standard formats for the seismic data (e.g., segy or seg2), which can then be converted in ASCII format for subsequent processing. Some systems use proprietary formats, although this choice is not recommended because it limits the possibility of sharing data.

Some software features can contribute to the efficiency of the field operations, for example, data acquisition and display filters (low-cut, high-cut, and notch) or software functions to test the system and the receiver spread. The possibility of visualizing the real-time output of the receivers or testing the receivers with a pulse test is useful for checking that all the receivers are working properly.

The system's robustness in various environmental conditions (operating temperature, waterproof rating, and so on) and the practical aspects of power consumption also have to be considered when selecting an acquisition system.